

Group sequential and adaptive designs for clinical trials

Christopher Jennison

Department of Mathematical Sciences,

University of Bath, UK

<http://people.bath.ac.uk/mascj>

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Phase III clinical trials

Phase III trials are conducted as the last stage in the drug development process.

Two positive studies are usually required to confirm that a new treatment is superior to the current standard treatment.

Regulators customarily require a hypothesis test to reach significance at the one-sided 2.5% level.

Studies may recruit hundreds, or even thousands, of subjects at a cost of as much as € 10k to € 50k per patient.

The time taken to reach a conclusion eats into the limited patent lifetime remaining to the company developing the drug.

Thus, there are strong incentives to reach an early conclusion for either a positive or negative decision.

Motivation of interim monitoring

Clinical trials methodology can also be applied to animal trials and epidemiological studies, where there is similar motivation from

Ethics

Administration (accrual, compliance, . . .)

Economics

to monitor the conduct of the trial and examine accumulating data.

Subjects should not be exposed to unsafe, ineffective or inferior treatments.

National and international guidelines for clinical trials call for interim analyses to be performed — and reported.

It is now standard practice for clinical trials to have a Data and Safety Monitoring Board (DSMB) to oversee the study and consider the option of early termination.

Care is needed when analysing data repeatedly during a study

Suppose θ is the difference in mean responses in a two-treatment comparison.

In a superiority trial, we wish to test $H_0: \theta \leq 0$ against $\theta > 0$.

If a test of H_0 is carried out at one-sided significance level $\alpha = 0.025$ on K occasions during the course of the trial, the **overall** type I error rate is:

Number of tests, K	Overall error rate	Number of tests, K	Overall error rate
1	0.025	10	0.097
2	0.042	20	0.124
3	0.054	100	0.190
5	0.071	∞	1.000

See Armitage et al. (*JRSS, A*, 1969).

Adaptation in clinical trials

Around the year 2000, there was a surge of interest in “adaptive” trials which allow a study design to be adjusted based on interim results.

An adaptive trial could:

- Route a larger share of patients to the treatment that seems to work best

- Drop treatments that don't seem to be effective

- Add more of the type of patients who react best to a particular treatment

- Merge two different phases of drug development into one trial

This represented a dramatic change from the existing philosophy of simple Phase III trials, designed to answer fully formulated questions through a pre-defined protocol and statistical analysis plan.

Time has shown just what can (and cannot) be achieved by such designs.

Outline of the course

Session 1. *Group sequential tests (GSTs)*

Group sequential testing and error spending designs

Group sequential tests for a delayed response

Session 2. *GSTs to test multiple hypotheses*

Inference on a secondary endpoint after a group sequential test

Multiple testing procedures and group sequential designs

Session 3. *Adaptive clinical trial designs I*

Using combination tests to construct adaptive designs

Multiple testing: Enrichment designs

Session 4. *Adaptive clinical trial designs II*

Seamless Phase 2/3 trials

Phase 3 trials with treatment selection

Session 1. Group sequential tests (GSTs) and delayed responses

1. Group sequential monitoring of clinical trials
2. Benefits of group sequential testing
3. Error spending GSTs
4. Group sequential testing for a delayed response
5. Optimising a delayed response GST
6. Error spending delayed response GSTs

1. Group sequential monitoring of clinical trials

Suppose a new treatment is being compared to a placebo or positive control in a Phase III trial.

The treatment effect θ represents the advantage of the new treatment over the control, with a positive value meaning that the new treatment is effective.

We wish to test the null hypothesis $H_0: \theta \leq 0$ against $\theta > 0$ with

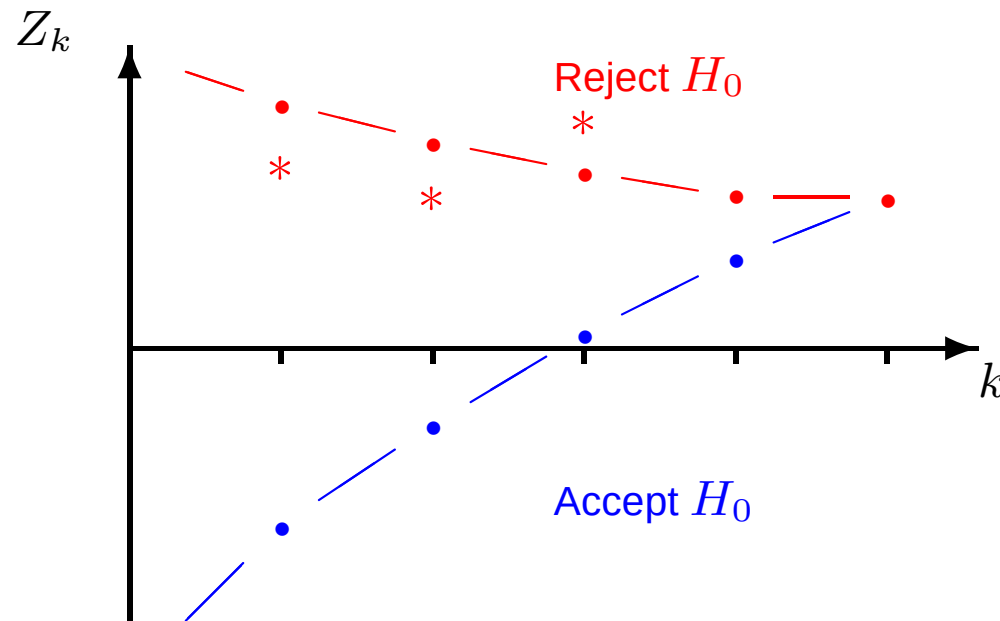
$$P_{\theta=0}\{\text{Reject } H_0\} = \alpha,$$

$$P_{\theta=\delta}\{\text{Reject } H_0\} = 1 - \beta.$$

Standardised test statistics $Z_1, Z_2, \dots$, are computed at interim analyses and these are used to define a stopping rule for the trial.

Group sequential tests

A typical boundary for a one-sided test has the form:



Crossing the upper boundary leads to early stopping for a positive outcome, rejecting H_0 in favour of $\theta > 0$.

Crossing the lower boundary implies stopping for “futility” with acceptance of H_0 .

Here, the trial stops to reject H_0 at the third of five analyses.

Joint distribution of parameter estimates

Reference: Sec. 3.5 and Ch. 11 of “*Group Sequential Methods with Applications to Clinical Trials*”, Jennison & Turnbull, 2000 (hereafter, JT).

Let $\hat{\theta}_k$ denote the estimate of θ based on data at analysis k .

The information for θ at analysis k is

$$\mathcal{I}_k = \{\text{Var}(\hat{\theta}_k)\}^{-1}, \quad k = 1, \dots, K.$$

Canonical joint distribution of $\hat{\theta}_1, \dots, \hat{\theta}_K$

In many situations, $\hat{\theta}_1, \dots, \hat{\theta}_K$ are approximately multivariate normal,

$$\hat{\theta}_k \sim N(\theta, \{\mathcal{I}_k\}^{-1}), \quad k = 1, \dots, K,$$

and

$$\text{Cov}(\hat{\theta}_{k_1}, \hat{\theta}_{k_2}) = \text{Var}(\hat{\theta}_{k_2}) = \{\mathcal{I}_{k_2}\}^{-1} \quad \text{for } k_1 < k_2.$$

Sequential distribution theory

The joint distribution of $\hat{\theta}_1, \dots, \hat{\theta}_K$ can be demonstrated directly for:

θ a single normal mean,

$\theta = \mu_A - \mu_B$, comparing two normal means.

The canonical distribution also applies when θ is a parameter in:

a general normal linear model,

a general model fitted by maximum likelihood (large sample theory).

Thus, theory supports general comparisons, including:

crossover studies,

analysis of longitudinal data,

comparisons adjusted for covariates.

Canonical joint distribution of $\hat{\theta}_1, \dots, \hat{\theta}_K$

A single normal mean

Suppose $X_1, X_2, \dots$ are independent $N(\theta, \sigma^2)$ responses.

For $n_1 < n_2$, define

$$\hat{\theta}_1 = \frac{X_1 + \dots + X_{n_1}}{n_1} \quad \text{and} \quad \hat{\theta}_2 = \frac{X_1 + \dots + X_{n_1} + \dots + X_{n_2}}{n_2}.$$

The joint distribution of $\hat{\theta}_1$ and $\hat{\theta}_2$ is bivariate normal.

Marginally

$$\hat{\theta}_1 \sim N(\theta, \mathcal{I}_1^{-1}) \quad \text{and} \quad \hat{\theta}_2 \sim N(\theta, \mathcal{I}_2^{-1}),$$

where

$$\mathcal{I}_1 = \frac{n_1}{\sigma^2} \quad \text{and} \quad \mathcal{I}_2 = \frac{n_2}{\sigma^2}.$$

Canonical joint distribution of $\hat{\theta}_1, \dots, \hat{\theta}_K$

It remains to check the covariance:

$$\begin{aligned}\text{Cov}(\hat{\theta}_1, \hat{\theta}_2) &= \text{Cov}\left(\frac{X_1 + \dots + X_{n_1}}{n_1}, \frac{X_1 + \dots + X_{n_1} + \dots + X_{n_2}}{n_2}\right) \\&= \text{Cov}\left(\frac{X_1 + \dots + X_{n_1}}{n_1}, \frac{X_1 + \dots + X_{n_1}}{n_2}\right) \\&= \frac{1}{n_1 n_2} \text{var}(X_1 + \dots + X_{n_1}) \\&= \frac{\sigma^2}{n_2} = \{\mathcal{I}_2\}^{-1} \\&= \text{Var}(\hat{\theta}_2).\end{aligned}$$

Canonical joint distribution of $\hat{\theta}_1, \dots, \hat{\theta}_K$

A two-treatment comparison

Suppose observations on treatments A and B, respectively, are

$$X_{Ai} \sim N(\mu_A, \sigma^2) \quad \text{and} \quad X_{Bi} \sim N(\mu_B, \sigma^2),$$

and $\theta = \mu_A - \mu_B$.

At analysis k , with n_{Ak} observations on treatment A and n_{Bk} on treatment B,

$$\hat{\theta}_k = \hat{\mu}_{A,k} - \hat{\mu}_{B,k} = \frac{1}{n_{Ak}} \sum_{i=1}^{n_{Ak}} X_{Ai} - \frac{1}{n_{Bk}} \sum_{i=1}^{n_{Bk}} X_{Bi}.$$

Exercise:

Show that $\hat{\theta}_1, \dots, \hat{\theta}_K$ have the canonical joint distribution, i.e., they are multivariate normal,

$$\hat{\theta}_k \sim N(\theta, \mathcal{I}_k^{-1})$$

for appropriately defined $\mathcal{I}_k$, and $\text{Cov}(\hat{\theta}_{k_1}, \hat{\theta}_{k_2}) = \text{Var}(\hat{\theta}_{k_2})$ for $k_1 < k_2$.

Canonical joint distribution of $\hat{\theta}_1, \dots, \hat{\theta}_K$

In general: A sequence of efficient estimators has the canonical covariance

So, if $\hat{\theta}_1$ at analysis 1 and $\hat{\theta}_2$ at analysis 2 are efficient, unbiased estimators of θ , then, $\text{Cov}(\hat{\theta}_1, \hat{\theta}_2) = \text{Var}(\hat{\theta}_2)$.

Proof

Suppose $\text{Cov}(\hat{\theta}_1, \hat{\theta}_2) \neq \text{Var}(\hat{\theta}_2)$, which implies that $\text{Cov}(\hat{\theta}_1 - \hat{\theta}_2, \hat{\theta}_2) \neq 0$.

Consider unbiased estimators of the form $\hat{\theta}_2^* = \hat{\theta}_2 + \epsilon(\hat{\theta}_1 - \hat{\theta}_2)$.

For ϵ small and of the opposite sign to $\text{Cov}(\hat{\theta}_1 - \hat{\theta}_2, \hat{\theta}_2) \neq 0$, we have

$$\text{Var}(\hat{\theta}_2^*) < \text{Var}(\hat{\theta}_2),$$

contradicting the assumption that $\hat{\theta}_2$ is an efficient, estimator of θ .

Thus, if $\hat{\theta}_2$ is efficient, we *must have* $\text{Cov}(\hat{\theta}_1, \hat{\theta}_2) = \text{Var}(\hat{\theta}_2)$.

Canonical joint distribution of z -statistics

In testing $H_0: \theta = 0$, the *standardised statistic* at analysis k is

$$Z_k = \frac{\hat{\theta}_k}{\sqrt{\text{Var}(\hat{\theta}_k)}} = \hat{\theta}_k \sqrt{\mathcal{I}_k}.$$

For this,

$(Z_1, \dots, Z_K)$ is multivariate normal,

$$Z_k \sim N(\theta \sqrt{\mathcal{I}_k}, 1), \quad k = 1, \dots, K,$$

$$\text{Cov}(Z_{k_1}, Z_{k_2}) = \sqrt{\mathcal{I}_{k_1} / \mathcal{I}_{k_2}} \quad \text{for } k_1 < k_2.$$

Canonical joint distribution of score statistics

The score statistics, $S_k = Z_k \sqrt{\mathcal{I}_k}$, are also multivariate normal with

$$S_k \sim N(\theta \mathcal{I}_k, \mathcal{I}_k), \quad k = 1, \dots, K.$$

The score statistics possess the “independent increments” property,

$$\text{Cov}(S_k - S_{k-1}, S_{k'} - S_{k'-1}) = 0 \quad \text{for } k \neq k'.$$

It can be helpful to know that the score statistics behave as Brownian motion with drift θ observed at times $\mathcal{I}_1, \dots, \mathcal{I}_K$.

Survival data

The canonical joint distributions also arise for

- a) estimates of a parameter in Cox's proportional hazards regression model
- b) log-rank statistics (score statistics) for comparing two survival curves

— and to Z -statistics formed from these.

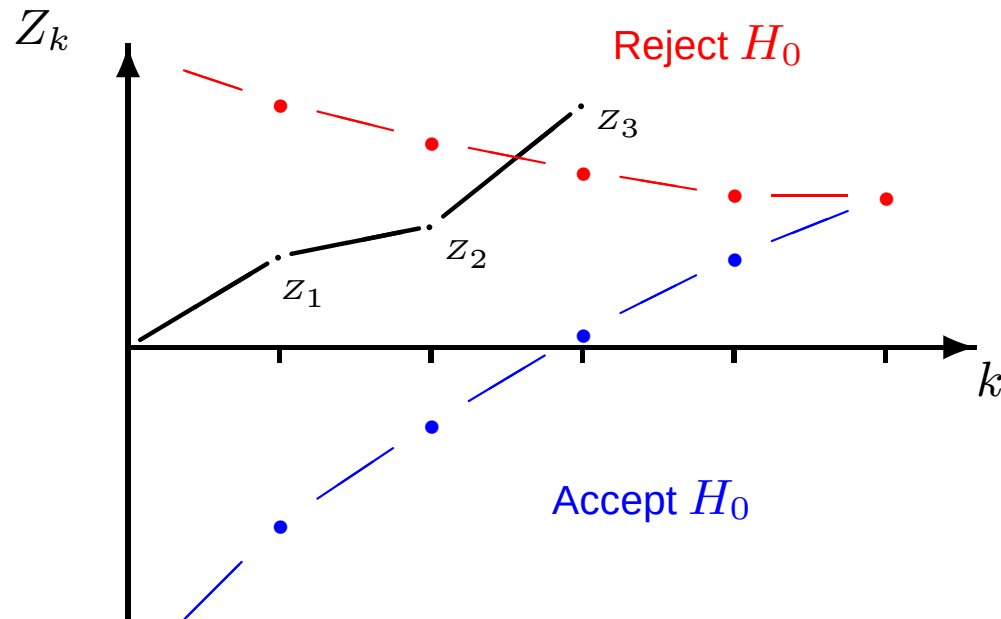
For survival data, observed information is roughly proportional to the number of failures.

Special types of group sequential test are needed to handle unpredictable and unevenly spaced information levels: see *error spending tests*.

Reference:

“Group-sequential analysis incorporating covariate information”, Jennison and Turnbull (*J. American Statistical Association*, 1997).

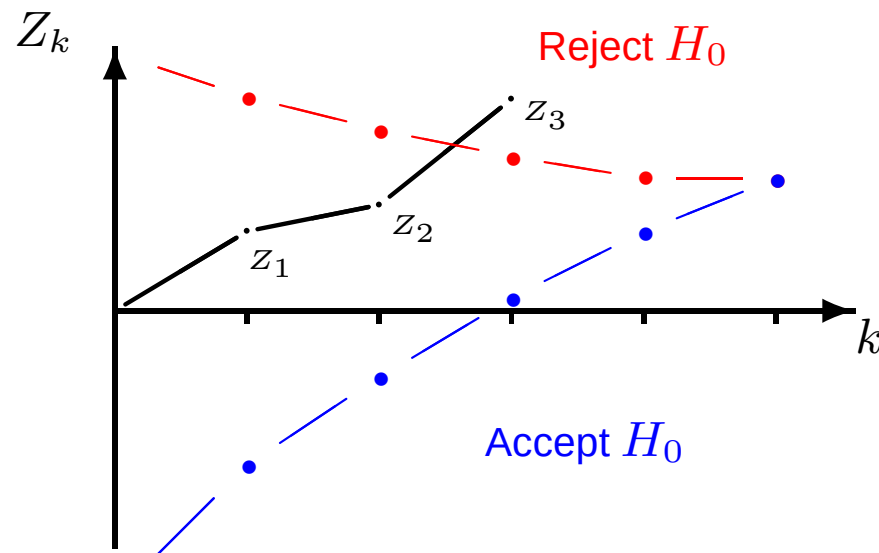
Computations for group sequential tests



In order to find $P_\theta\{\text{Reject } H_0\}$, etc., we need to calculate the probabilities of basic events such as

$$a_1 < Z_1 < b_1, \quad a_2 < Z_2 < b_2, \quad Z_3 > b_3.$$

Computations for group sequential tests



Probabilities such as $P_{\theta}\{a_1 < Z_1 < b_1, a_2 < Z_2 < b_2, Z_3 > b_3\}$ can be computed by repeated numerical integration (see JT, Ch. 19).

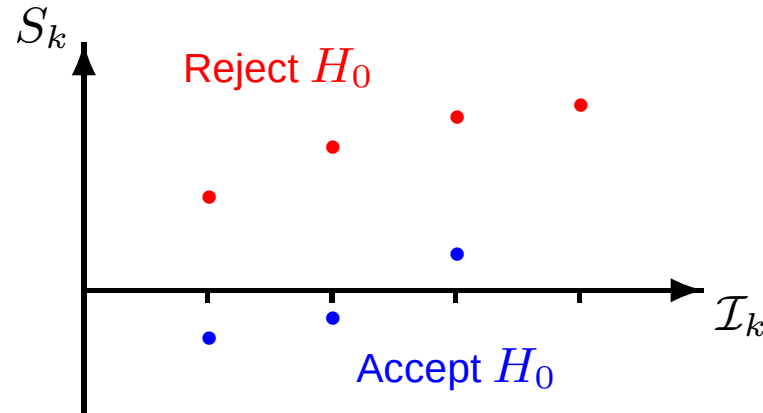
Combining such probabilities yields properties of a group sequential boundary.

Constants and group sizes can be chosen to define a test with a specific type I error probability and power.

Example of one-sided tests: The Pampallona & Tsiatis family

Pampallona & Tsiatis (*J. Statistical Planning and Inference*, 1994).

To test $H_0: \theta \leq 0$ against the *one-sided* alternative $\theta > 0$ with type I error probability α and power $1 - \beta$ at $\theta = \delta$.



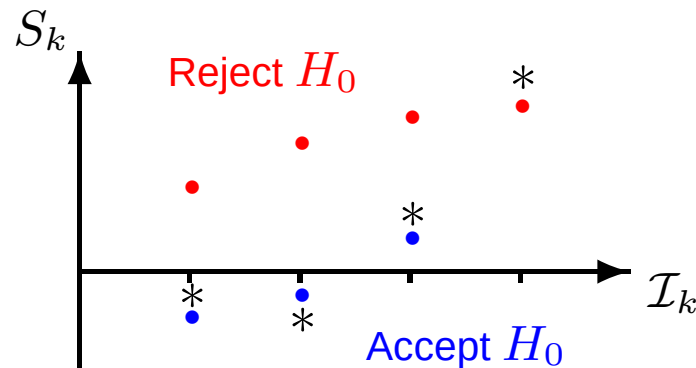
The computational methods just described can be used to define tests with parametric stopping boundaries meeting the design criteria.

For the P & T test with parameter Δ , boundaries on the score statistic scale are

$$a_k = \mathcal{I}_k \delta - C_2 \mathcal{I}_k^\Delta, \quad b_k = C_1 \mathcal{I}_k^\Delta.$$

One-sided tests with a non-binding futility boundary

Regulators are not always convinced a trial monitoring committee will abide by the stopping boundary specified in the study protocol.



The sample path shown above leads to rejection of H_0 . Since such paths are not included in type I error calculations, the true type I error rate is under-estimated.

If a futility boundary is deemed to be *non-binding*, the type I error rate should be computed ignoring the futility boundary.

For planning purposes, power and expected sample size should be computed assuming the futility boundary will be obeyed.

Constants can be computed in this way for, say, a Pampallona & Tsiatis test.

2. Benefits of group sequential testing

In order to test $H_0: \theta \leq 0$ against $\theta > 0$ with type I error probability α and power $1 - \beta$ at $\theta = \delta$, a fixed sample size test needs information

$$\mathcal{I}_{fix} = \frac{\{\Phi^{-1}(1 - \alpha) + \Phi^{-1}(1 - \beta)\}^2}{\delta^2}.$$

Information is (roughly) proportional to sample size in many clinical trial settings.

A group sequential test with K analyses will need to be able to continue to a maximum information level $\mathcal{I}_K$ which is greater than $\mathcal{I}_{fix}$.

The benefit is that, on average, the sequential test can stop earlier than this and expected information on termination, $\mathbb{E}_\theta(\mathcal{I})$, will be considerably less than $\mathcal{I}_{fix}$, especially under extreme values of θ .

We term the ratio $R = \mathcal{I}_K / \mathcal{I}_{fix}$ the “inflation factor” for a group sequential design.

Benefits of group sequential testing

In specifying a group sequential test's boundary, one can aim to minimise the expected information $\mathbb{E}_\theta(\mathcal{I})$ under effect sizes of θ of most interest, subject to a fixed number of analyses K and inflation factor R .

Eales & Jennison (*Biometrika*, 1992) and Barber & Jennison (*Biometrika*, 2002) report on designs optimised for criteria of the form $\sum_i w_i \mathbb{E}_{\theta_i}(\mathcal{I})$ or

$$\int f(\theta) \mathbb{E}_\theta(\mathcal{I}) d\theta,$$

where f is a normal density.

These optimal group sequential designs can be used in their own right.

They also serve as benchmarks for other methods which may have additional useful features.

Computing optimal GSTs

In optimising a group sequential test, we create a Bayes sequential decision problem, placing a prior on θ and defining costs for sampling and for making incorrect decisions.

Such a problem can be solved rapidly by dynamic programming.

We then search for the combination of prior and costs such that the solution to the (unconstrained) Bayes decision problem has the specified frequentist error rates α at $\theta = 0$ and β at $\theta = \delta$.

The resulting design solves both the Bayes decision problem and the original frequentist problem.

Note: Although the Bayes decision problem is introduced as a computational device, this derivation demonstrates that an efficient frequentist procedure should also be good from a Bayesian perspective.

Benefits of group sequential testing

One-sided tests with binding futility boundaries, minimising $\{\mathbb{E}_0(\mathcal{I}) + \mathbb{E}_\delta(\mathcal{I})\}/2$ for equal group sizes, $\alpha = 0.025$, $1 - \beta = 0.9$, K analyses, $\mathcal{I}_{max} = R\mathcal{I}_{fix}$.

Minimum values of $\{\mathbb{E}_0(\mathcal{I}) + \mathbb{E}_\delta(\mathcal{I})\}/2$, as a percentage of $\mathcal{I}_{fix}$

K	R					Minimum over R
	1.01	1.05	1.1	1.2	1.3	
2	80.8	74.7	73.2	73.7	75.8	73.0 at $R=1.13$
3	76.2	69.3	66.6	65.1	65.2	65.0 at $R=1.23$
5	72.2	65.2	62.2	59.8	59.0	58.8 at $R=1.38$
10	69.2	62.2	59.0	56.3	55.1	54.2 at $R=1.6$
20	67.8	60.6	57.5	54.6	53.3	51.7 at $R=1.8$

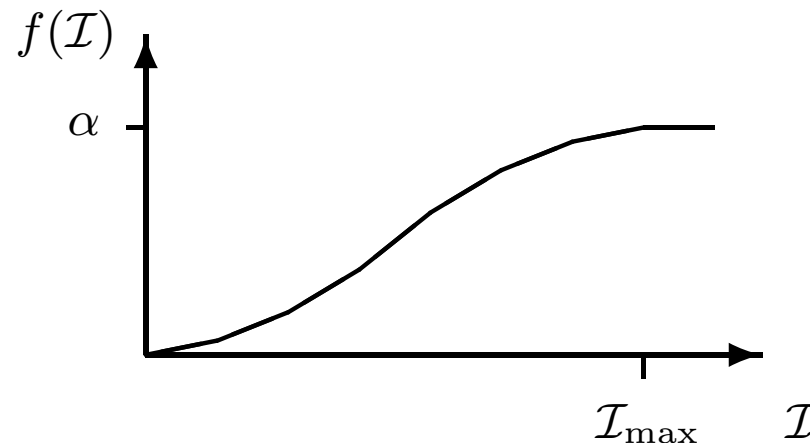
Note: $\mathbb{E}(\mathcal{I}) \searrow$ as $K \nearrow$ but with diminishing returns,
 $\mathbb{E}(\mathcal{I}) \searrow$ as $R \nearrow$ up to a point.

3. Error spending tests

Since the sequence $\mathcal{I}_1, \mathcal{I}_2, \dots$ is often unpredictable, it is good to have a group sequential design that can adapt to the observed information levels.

Lan & DeMets (*Biometrika*, 1983) presented two-sided tests of $H_0: \theta = 0$ against $\theta \neq 0$ which “spend” type I error as a function of observed information.

Maximum information design with error spending function $f(\mathcal{I})$:



The boundary at analysis k is set to give cumulative type I error probability $f(\mathcal{I}_k)$.

The null hypothesis, H_0 , is accepted if $\mathcal{I}_{\max}$ is reached without rejecting H_0 .

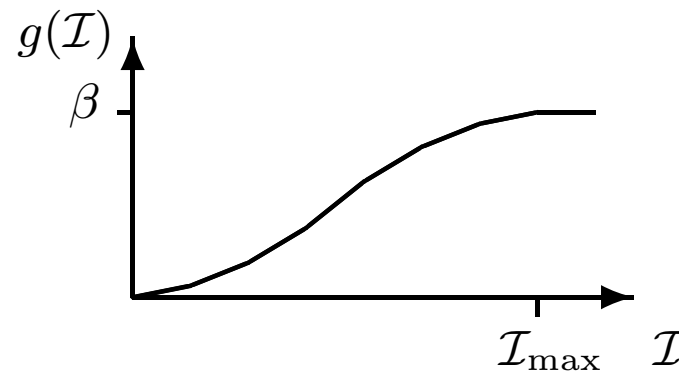
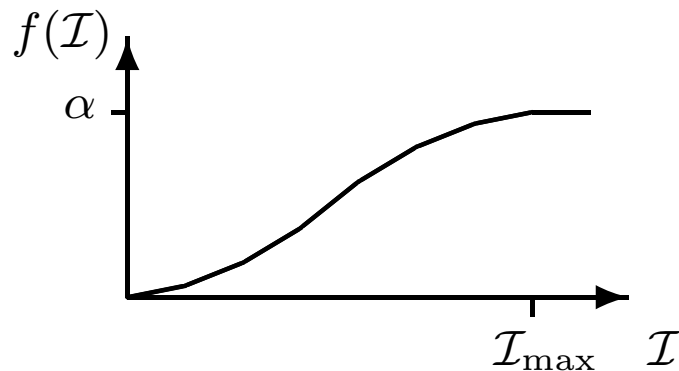
One-sided error spending tests

For a one-sided test of $H_0: \theta \leq 0$ against $\theta > 0$ with

Type I error probability α at $\theta = 0$,

Type II error probability β at $\theta = \delta$,

we need two error spending functions.



Type I error probability α is spent according to the function $f(\mathcal{I})$, and type II error probability β according to $g(\mathcal{I})$.

One-sided error-spending tests

Analysis 1:

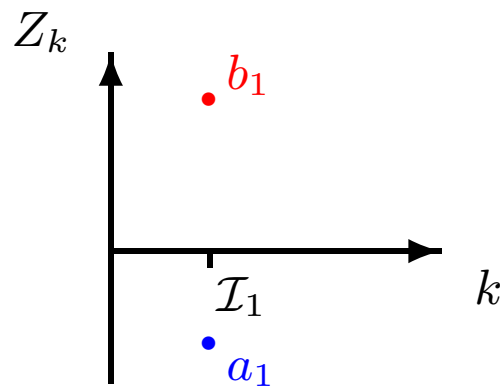
Observed information $\mathcal{I}_1$.

Reject H_0 if $Z_1 > b_1$, where

$$P_{\theta=0}\{Z_1 > b_1\} = f(\mathcal{I}_1).$$

Accept H_0 if $Z_1 < a_1$, where

$$P_{\theta=\delta}\{Z_1 < a_1\} = g(\mathcal{I}_1).$$



One-sided error-spending tests

Analysis 2: Observed information $\mathcal{I}_2$

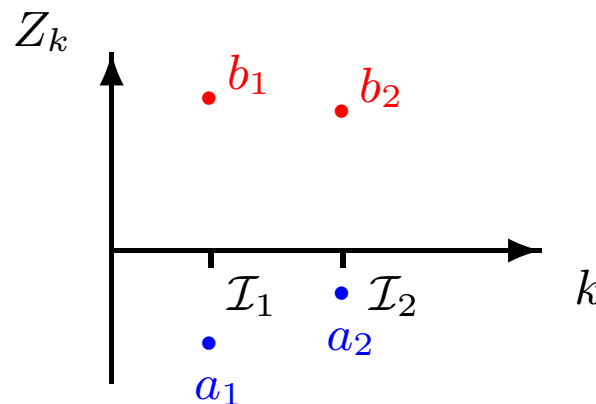
Reject H_0 if $Z_2 > b_2$, where

$$P_{\theta=0}\{a_1 < Z_1 < b_1, Z_2 > b_2\} = f(\mathcal{I}_2) - f(\mathcal{I}_1)$$

— note that, for now, we assume the futility boundary is binding.

Accept H_0 if $Z_2 < a_2$, where

$$P_{\theta=\delta}\{a_1 < Z_1 < b_1, Z_2 < a_2\} = g(\mathcal{I}_2) - g(\mathcal{I}_1).$$



One-sided error-spending tests

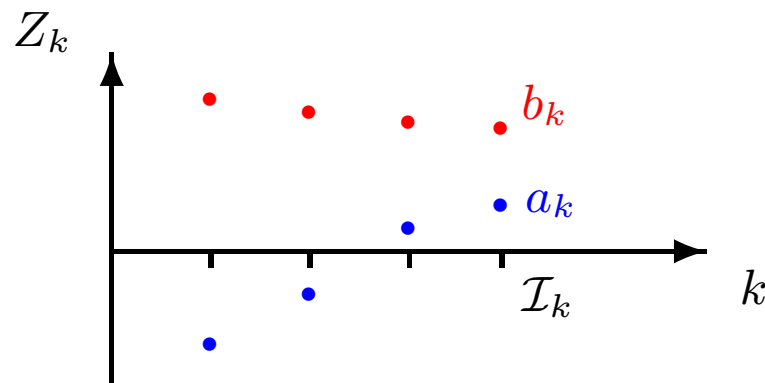
Analysis k: Observed information $\mathcal{I}_k$

Find a_k and b_k to satisfy

$$P_{\theta=0}\{a_1 < Z_1 < b_1, \dots, a_{k-1} < Z_{k-1} < b_{k-1}, Z_k > b_k\} = f(\mathcal{I}_k) - f(\mathcal{I}_{k-1}),$$

and

$$P_{\theta=\delta}\{a_1 < Z_1 < b_1, \dots, a_{k-1} < Z_{k-1} < b_{k-1}, Z_k < a_k\} = g(\mathcal{I}_k) - g(\mathcal{I}_{k-1}).$$

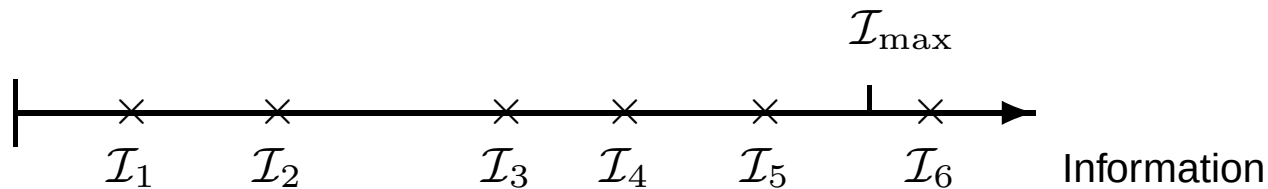


Remarks on error spending tests

1. Computation of (a_k, b_k) does **not** depend on future information levels, $\mathcal{I}_{k+1}, \mathcal{I}_{k+2}, \dots$.

2. A “maximum information design” continues until a boundary is crossed or an analysis with $\mathcal{I}_k \geq \mathcal{I}_{\max}$ is reached.

If necessary, patient accrual can be extended to reach $\mathcal{I}_{\max}$.



If a maximum of K analyses is specified, the study terminates at analysis K with $f(\mathcal{I}_K)$ defined to be α . Then, a_K is chosen to give cumulative type I error probability α and we set $b_K = a_K$.

Remarks on error spending tests

3. The value of $\mathcal{I}_{\max}$ can be chosen so that boundaries converge at the final analysis under a typical sequence of information levels, e.g.,

$$\mathcal{I}_k = (k/K) \mathcal{I}_{\max}, \quad k = 1, \dots, K.$$

4. The ρ -family provides a convenient choice of error spending functions.

In the case of one-sided tests, type I error probability is spent as

$$f(\mathcal{I}) = \alpha \min \{1, (\mathcal{I}/\mathcal{I}_{\max})^\rho\}$$

and type II error probability as

$$g(\mathcal{I}) = \beta \min \{1, (\mathcal{I}/\mathcal{I}_{\max})^\rho\}.$$

The value of ρ determines the inflation factor $R = \mathcal{I}_{\max}/\mathcal{I}_{fix}$.

Barber & Jennison (*Biometrika*, 2002) show ρ -family tests have excellent efficiency properties when compared with designs for the same number of analyses K and inflation factor R .

One-sided error-spending tests: Non-binding futility boundary

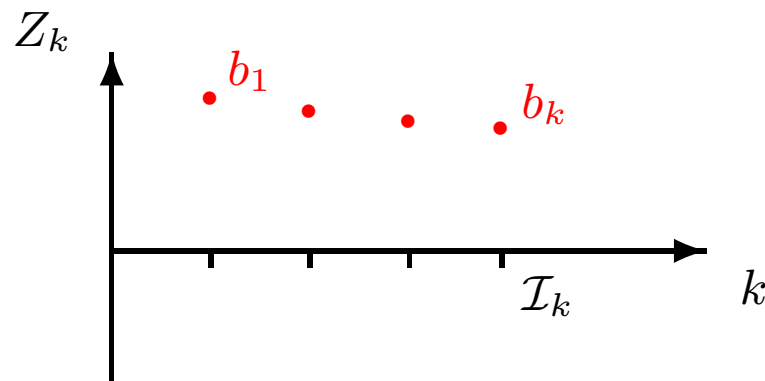
If the futility boundary is treated as non-binding, computation of the error-spending efficacy boundary only involves the type I error spending function $f(\mathcal{I})$.

Boundary values, $b_1, b_2, \dots$, are calculated one by one as the trial proceeds.

Analysis k : Observed information $\mathcal{I}_k$

Reject H_0 if $Z_k > b_k$, where

$$P_{\theta=0}\{Z_1 < b_1, \dots, Z_{k-1} < b_{k-1}, Z_k > b_k\} = f(\mathcal{I}_k) - f(\mathcal{I}_{k-1}).$$



One-sided error-spending tests: Non-binding futility boundary

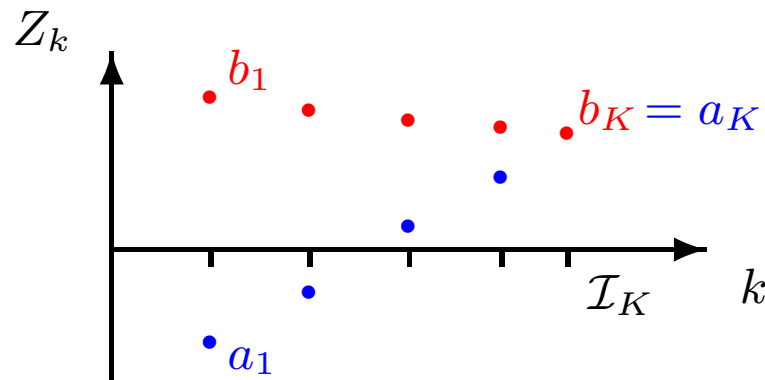
The futility boundary can be added through a type II error spending function $g(\mathcal{I})$.

For $k = 1, \dots, K - 1$:

At analysis k with observed information $\mathcal{I}_k$, set a_k to satisfy

$$P_{\theta=\delta}\{a_1 < Z_1 < b_1, \dots, a_{k-1} < Z_{k-1} < b_{k-1}, Z_k < a_k\} = g(\mathcal{I}_k) - g(\mathcal{I}_{k-1}).$$

For $k = K$: Set $a_K = b_K$.



4. Group sequential testing for a delayed response

Reference: Hampson & Jennison (HJ), (*JRSS B*, 2013)

Example: Cholesterol reduction after 4 weeks of treatment

In their Example A, HJ describe a group sequential trial where there is a delay of four weeks between the start of treatment and observation of the primary endpoint.

The recruitment rate is around 4 patients per week, so at each interim analysis we expect about 16 subjects to have started treatment but not yet given a response.

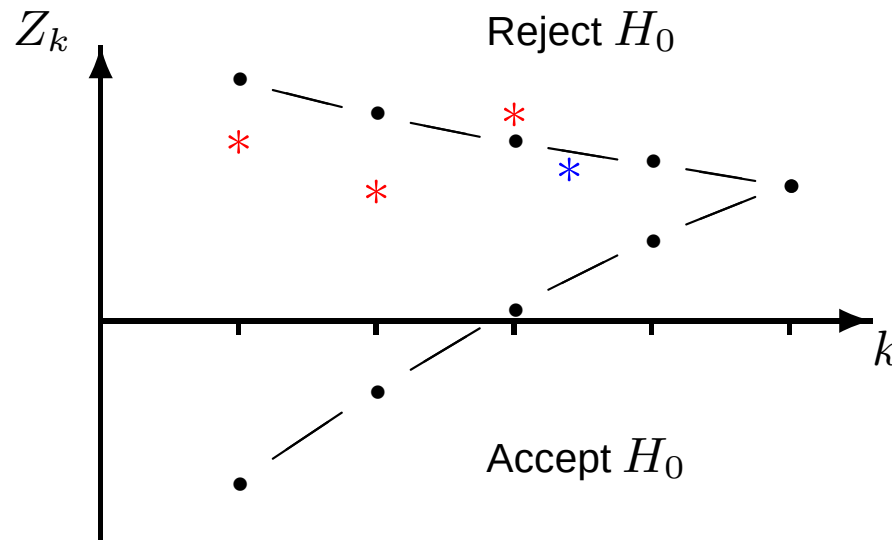
We refer to these as patients as being “in the pipeline”.

If a group sequential test reaches its conclusion at an interim analysis, we still expect investigators to follow up pipeline subjects and observe their responses.

How should these data be analysed?

The problem of delayed responses

A possible outcome for the cholesterol reduction trial



Suppose $Z_3 = 2.4$, exceeding the boundary value of 2.3.

The trial stops but, with the pipeline data included, $Z = 2.1$.

Can the investigators claim significance at level α ?

Short term information on “pipeline” subjects

Example: Prevention of fracture in postmenopausal women

In their Example D, HJ consider a study where the primary endpoint is occurrence of a fracture within five years.

Changes in bone mineral density (BMD) are measured after one year.

It is expected that these two variables are correlated.

How might we use the BMD data to gain information from subjects who have been followed for between one and five years?

Would fitting a Kaplan-Meier curve for time to first fracture also help — remember that inference is about the binary outcome defined at five years?

Incorporating delayed observations after a GST terminates

1. Whitehead (*Controlled Clinical Trials*, 1992) proposed a “deletion” method.

The analysis k at which termination occurs is “deleted” and one behaves as if analysis k originally had information $\tilde{\mathcal{I}}_k$, appropriate to the final set of responses.

A boundary value $\tilde{b}_k$ is computed and H_0 rejected if the final statistic $\tilde{Z}_k \geq \tilde{b}_k$.

2. Hall & Ding (*Univ. Rochester, Technical Report*, 2002) applied a combination test (Bauer & Köhne, *Biometrics*, 1994) to the two sets of data obtained before and after the GST terminates.

Sorriyarachchi et al. (*Biometrics*, 2003) investigated these methods and found they perform poorly with respect to power:

The deletion method is conservative and can lead to lower power than a GST which ignores the additional data,

With a moderate number of pipeline subjects, Hall & Ding’s method leads to greater loss of power than the deletion method.

Incorporating delayed observations after a GST terminates

The methods of Whitehead (1992) and Hall & Ding (2002) are based on applying a GST as if response were immediate, then trying to deal with additional pipeline data once this GST has terminated.

A more systematic approach is to recognise that there will be pipeline data when designing the trial.

Interestingly, T. W. Anderson (*JASA*, 1964) recognised this issue, well before the advent of modern group sequential methods.

The methods of Hampson & Jennison (*JRSS, B*, 2013) follow the same basic structure as proposed by Anderson:

With delayed response data, a trial comes to an end in two stages

1. Stop recruitment of any more subjects,
2. After responses have been observed for all recruited subjects, make a decision to accept or reject H_0 .

Defining a group sequential test with delayed responses

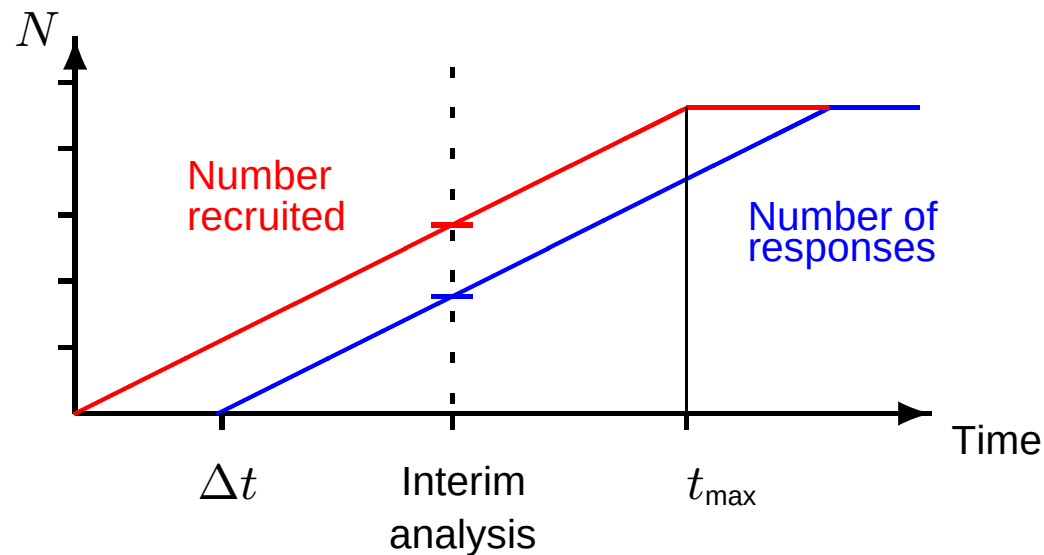
We assume:

The primary endpoint is measured a fixed time after treatment commences,

The endpoint will be known (eventually) for all treated subjects,

If recruitment is stopped, it cannot be re-started.

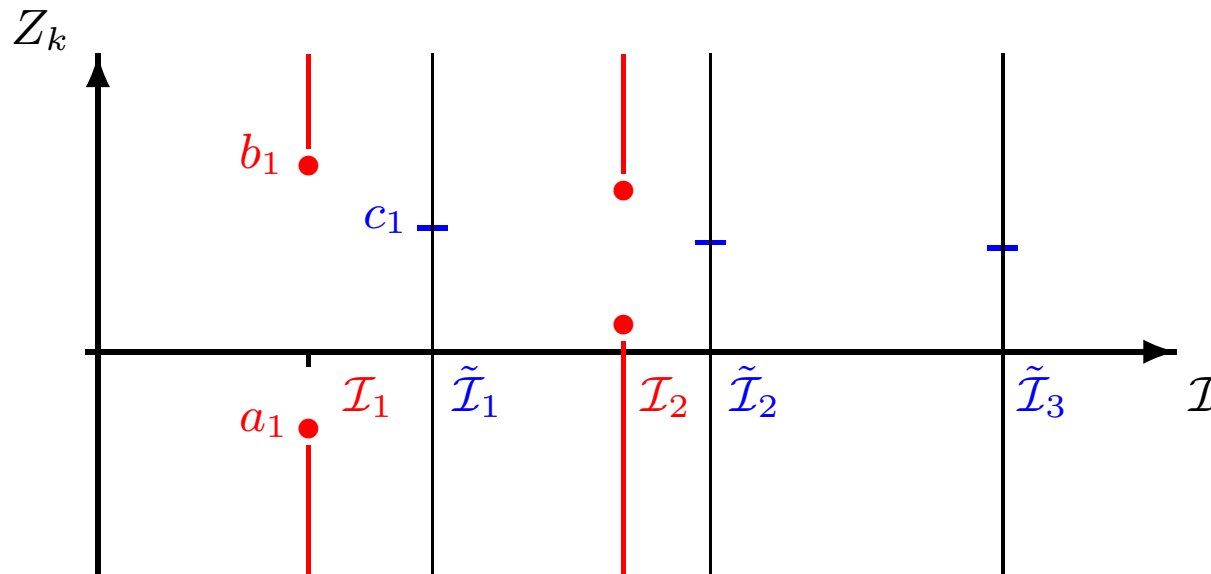
Consider a trial where responses are observed time Δ_t after treatment.



At each analysis, patients arriving in the last Δ_t units of time are “in the pipeline”.

Boundaries for a Delayed Response GST

At interim analysis k , the observed information level is $\mathcal{I}_k = \{\text{Var}(\hat{\theta}_k)\}^{-1}$.



If $Z_k > b_k$ or $Z_k < a_k$ at analysis k , we cease enrolment of patients and follow-up all recruited subjects.

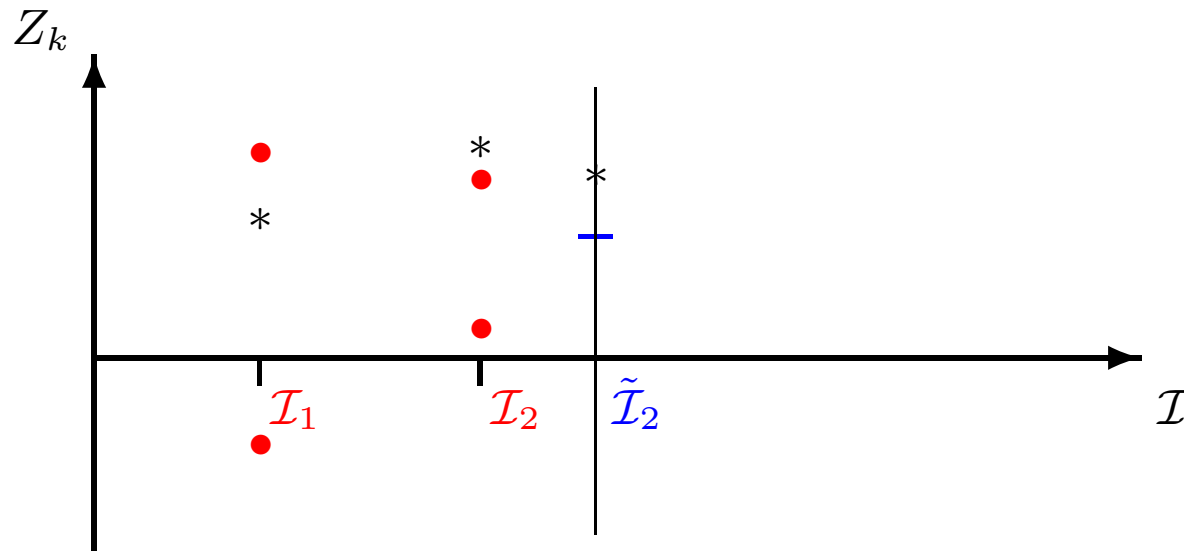
At the subsequent decision analysis, denote the observed information by $\tilde{\mathcal{I}}_k$ and reject H_0 if $\tilde{Z}_k > c_k$.

Delayed Response Group Sequential Tests (DR GSTs)

For a particular sequence of observed responses, we apply boundary points at a sequence of information levels of the form

$$\mathcal{I}_1, \dots, \mathcal{I}_k, \tilde{\mathcal{I}}_k.$$

In the example below, recruitment ceases at the second analysis and the final decision is made with extra “pipeline” data bringing the information up to $\tilde{\mathcal{I}}_2$.



Calculations for a Delayed Response GST

The type I error rate, power and expected sample size of a Delayed Response GST depend on joint distributions of test statistic sequences:

$$\{Z_1, \dots, Z_k, \tilde{Z}_k\}, \quad k = 1, \dots, K - 1,$$

and

$$\{Z_1, \dots, Z_{K-1}, \tilde{Z}_K\}.$$

Each sequence is based on accumulating data sets.

Given $\{\mathcal{I}_1, \dots, \mathcal{I}_k, \tilde{\mathcal{I}}_k\}$, the sequence $\{Z_1, \dots, Z_k, \tilde{Z}_k\}$ follows the canonical distribution we saw earlier for the sequence of Z -statistics in a GST with immediate responses (JT, Ch. 11).

Thus, properties of Delayed Response GSTs can be calculated using numerical routines devised for standard group sequential designs.

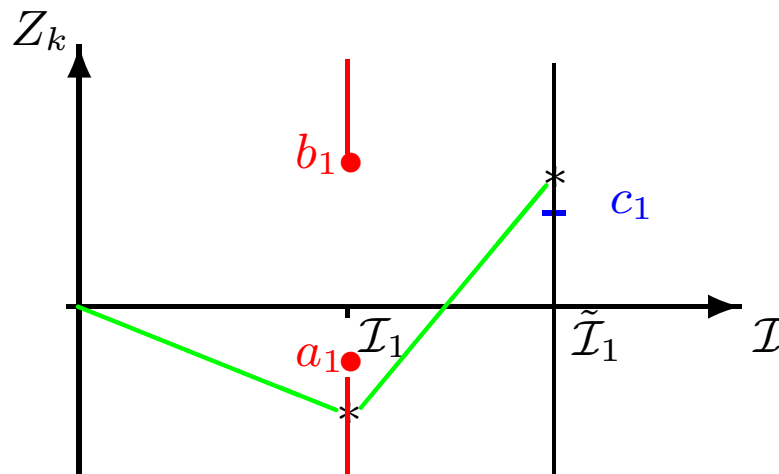
The value of information from pipeline subjects

When recruitment is terminated at interim analysis k with $Z_k > b_k$ or $Z_k < a_k$, current data suggest the likely final decision.

However, the pipeline data provide further information to use in this decision.

The pipeline data will occasionally produce a “reversal”, with the final decision differing from that anticipated when recruitment was terminated.

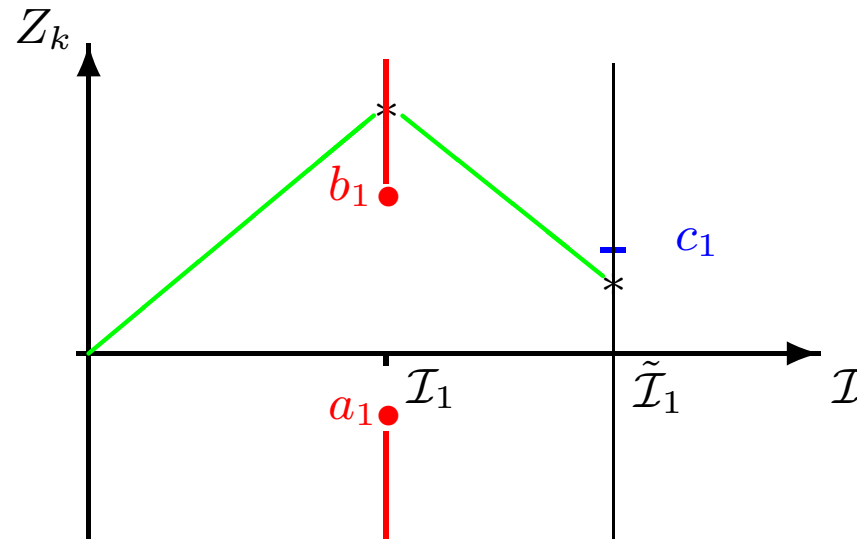
We could observe:



Here, accrual stops at analysis 1 because of unpromising results, but H_0 is rejected when the pipeline data are observed.

The value of information from pipeline subjects

Or, recruitment may cease with promising data only for H_0 to be accepted.



Note that there is no option of “banking” the good evidence at analysis 1 — we are assuming all pipeline subjects will eventually be observed.

Decisions based on more data ought to be more accurate: perhaps the pipeline data have helped to avoid a false positive conclusion here.

An optimised design will place boundary points to achieve high power for the permitted type I error rate, α .

5. Optimising a Delayed Response GST

We specify the type I error rate α and power $1 - \beta$ to be attained at $\theta = \delta$.

We set maximum sample size n_{max} , number of stages K , and analysis schedule.

Let r be the fraction of n_{max} in the pipeline at each interim analysis.

Let N denote the total number of subjects recruited.

Objective:

Given $\alpha, \beta, \delta, n_{max}, K$ and r , we find the Delayed Response GST minimising

$$F = \int \mathbb{E}_{\theta}(N) f(\theta) d\theta$$

where $f(\theta)$ is the density of a $N(\delta/2, (\delta/2)^2)$ distribution.

Other weighted combinations of $\mathbb{E}_{\theta}(N)$ can also be used.

Computing optimal Delayed Response GSTs

We follow the same approach as for optimising a GST with immediate response.

We create a Bayes sequential decision problem, placing a prior on θ and defining costs for sampling and for making incorrect decisions.

This problem can be solved rapidly by dynamic programming.

We then search for the combination of prior and costs such that the solution to the (unconstrained) Bayes decision problem has the specified frequentist error rates α at $\theta = 0$ and β at $\theta = \delta$.

The resulting design solves both the Bayes decision problem and the original frequentist problem.

Again, the Bayes decision problem is introduced as a computational device, but the derivation demonstrates the relationship between admissible frequentist designs and Bayes procedures.

An optimal design for the cholesterol treatment example

In the cholesterol treatment trial, the primary endpoint is reduction in serum cholesterol after 4 weeks of treatment.

Responses are assumed normally distributed with variance $\sigma^2 = 2$.

The treatment effect θ is the difference in mean response between the new treatment and control.

An effect $\theta = 1$ is regarded as clinically significant.

It is required to test $H_0: \theta \leq 0$ against $\theta > 0$ with

Type I error rate $\alpha = 0.025$,

Power 0.9 at $\theta = 1$.

A fixed sample test needs $n_{fix} = 85$ subjects over the two treatments.

An optimal design for the cholesterol treatment example

We consider designs with a maximum sample size of 96.

We assume a recruitment rate of 4 per week:

Data start to accrue after 4 weeks,

At each interim analysis, there will be $4 \times 4 = 16$ pipeline subjects,

so the “pipeline fraction” is $r = 16/96 = 0.17$.

Recruitment will close after 24 weeks.

Interim analyses are planned after $n_1 = 28$ and $n_2 = 54$ observed responses and the final decision is based on:

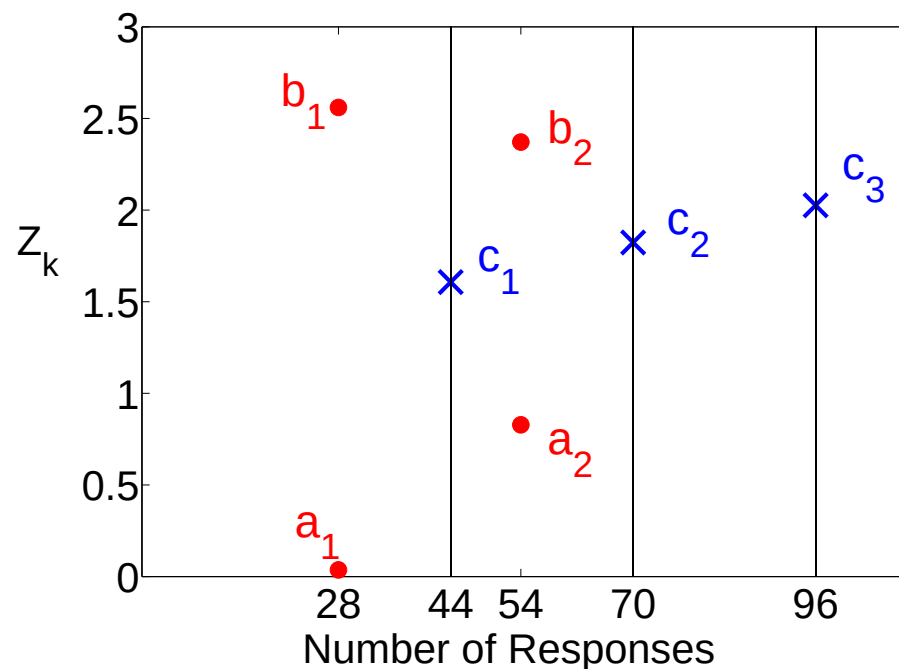
$\tilde{n}_1 = 44$ responses if recruitment stops at interim analysis 1,

$\tilde{n}_2 = 70$ responses if recruitment stops at interim analysis 2,

$\tilde{n}_3 = 96$ responses if there is no early stopping.

An optimal design for the cholesterol treatment example

The following Delayed Response GST minimises $F = \int \mathbb{E}_\theta(N) f(\theta) d\theta$, where $f(\theta)$ is the density of a $N(0.5, 0.5^2)$ distribution.



Both c_1 and c_2 are less than 1.96. If desired, these can be raised to 1.96 with little change to the design's power curve.

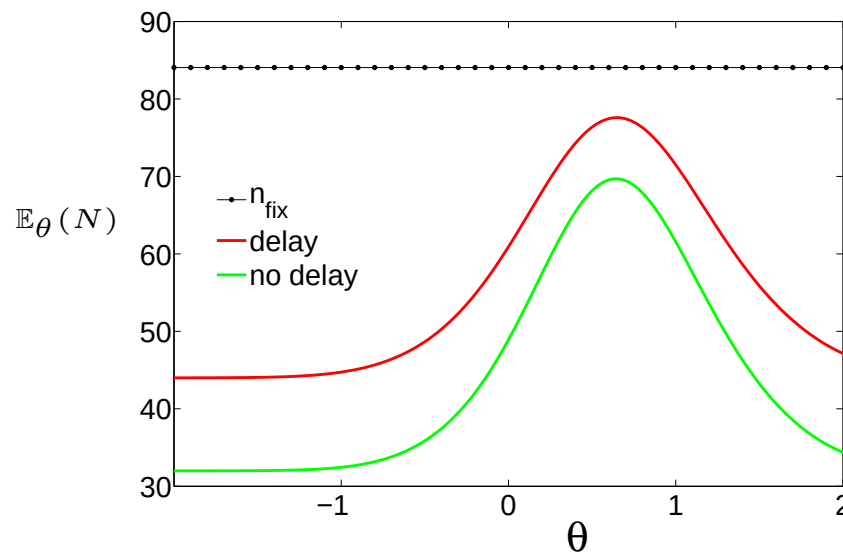
An optimal design for the cholesterol treatment example

The figure shows expected sample size curves for

The fixed sample test with $n_{fix} = 85$ patients,

The Delayed Response GST minimising F ,

The GST for immediate responses with analyses after 32, 64 and 96 responses, also minimising F .



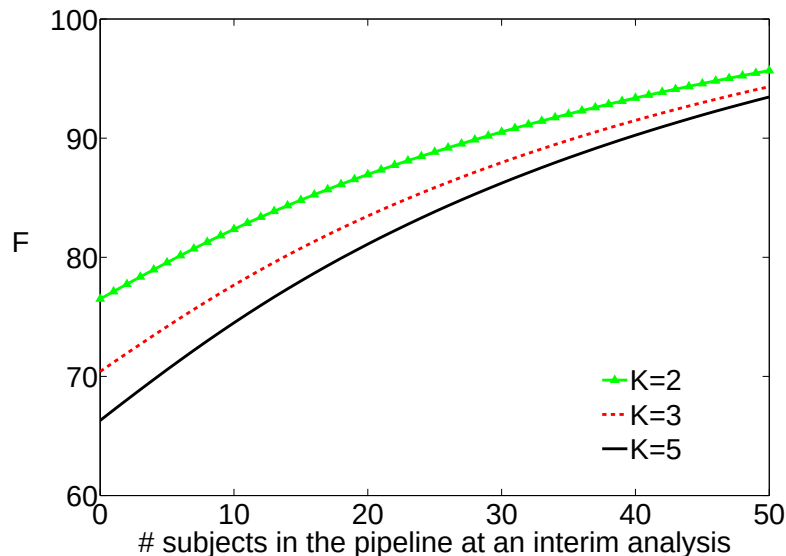
Efficiency loss when there is a delay in response

In general, a delay in response erodes the benefits of sequential testing.

Consider tests with $\alpha = 0.025$, power 0.9 and response variance, σ^2 , such that the fixed sample test needs $n_{fix} = 100$ subjects.

Suppose a group sequential design has $n_{max} = 1.1 n_{fix} = 110$.

The figure shows the minima of $F = \int \mathbb{E}_\theta(N) f(\theta) d\theta$, attained by optimal Delayed Response GSTs with K analyses for a range of “pipeline” sizes.



The reduction in average $\mathbb{E}_\theta(N)$ when the pipeline size is 25 patients is around half the reduction achieved by a GST when response is observed immediately.

Using a short term endpoint to improve efficiency

Suppose a second endpoint, correlated with the primary endpoint, is available soon after treatment.

For patient i on treatment $T = A$ or B , let

$Y_{T,i}$ = *The short term endpoint,*

$X_{T,i}$ = *The long term endpoint.*

Assume that we have a parametric model for the joint distribution of $(Y_{T,i}, X_{T,i})$ in which

$$\mathbb{E}(X_{A,i}) = \mu_A, \quad \mathbb{E}(X_{B,i}) = \mu_B \quad \text{and} \quad \theta = \mu_A - \mu_B.$$

We analyse all the available data at each interim analysis.

Using a short term endpoint to improve efficiency

At an interim analysis, subjects are

- *Unobserved,*
- *Partially observed (with just $Y_{T,i}$ available),*
- *Fully observed (both $Y_{T,i}$ and $X_{T,i}$ available).*

We fit the full model to all the data available at analysis k , then extract

$$\hat{\theta}_k \quad \text{and} \quad \mathcal{I}_k = \{\text{Var}(\hat{\theta}_k)\}^{-1}.$$

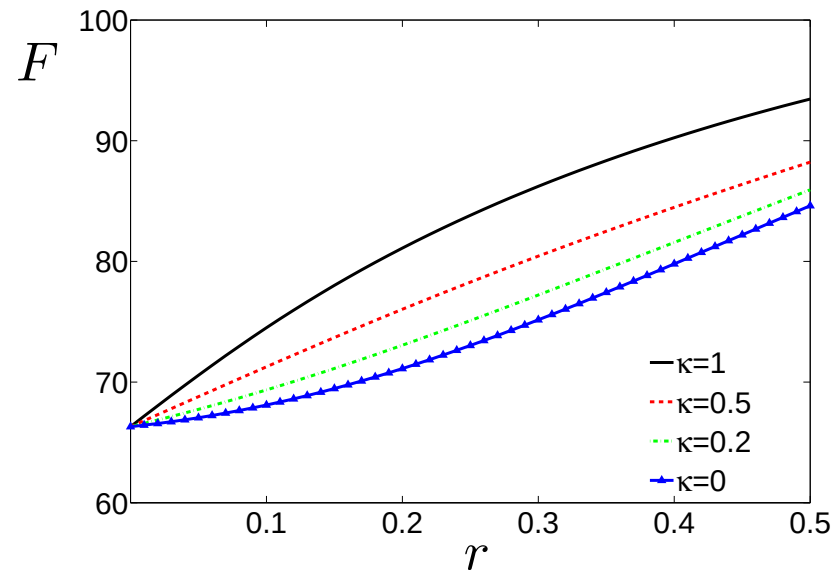
Including the short term endpoint in the model increases the information, $\mathcal{I}_k$, for the long term endpoint.

The sequence of estimates $\{\hat{\theta}_k\}$ follows the standard joint distribution for a group sequential trial with observed information levels $\{\mathcal{I}_k\}$.

Thus, we can design a Delayed Response GST in the usual way.

Using a short term endpoint to improve efficiency

Values of F achieved using a second, short-term endpoint



Results are for the previous testing problem with $K = 5$ analyses.

The endpoints $Y_{T,i}$ and $X_{T,i}$ are bivariate normal with correlation 0.9.

The parameter κ is the ratio of time to recording the short-term and long-term endpoints, so $\kappa = 1$ equates to having no short-term endpoint.

Using a short term endpoint to improve efficiency

Note: Although the short-term endpoint may itself be of clinical interest, the final inference is about the primary endpoint alone.

The same approach can be used with repeated measurements as follow-up continues for each patient.

Nuisance parameters, such as variances and the correlation between short-term and long-term endpoints, can be estimated within the trial.

In HJ's Example D, prevention of fracture in postmenopausal women, we could:

Fit a joint model for bone mineral density measured at one year and incidence of fracture within five years,

Use censored time-to-event data on the fracture endpoint for subjects with less than five years of follow-up.

6. Error spending Delayed Response GSTs

In practice, information levels at interim analyses and decision analyses are unpredictable.

In the error spending approach, the type I error probability to be spent by stage k is defined through a function $f(\mathcal{I}_k)$.

Similarly, the type II probability to be spent by stage k is specified as $g(\mathcal{I}_k)$.

A target information level $\mathcal{I}_{max}$ is defined and recruitment stops when this is reached (or will be reached with the responses from pipeline subjects).

HJ show how to construct error spending Delayed Response GSTs that protect type I error rate exactly.

The attained power is close to its specified level as long as the information levels take values similar to those assumed in planning the trial.

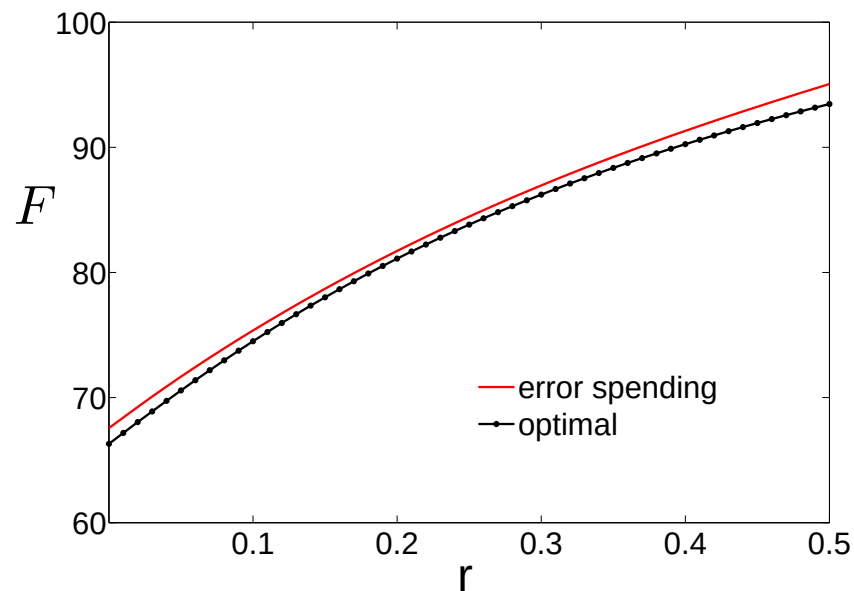
The ρ -family of error spending functions

HJ recommend error spending functions of the form

$$f(\mathcal{I}) = \alpha \min\{1, (\mathcal{I}/\mathcal{I}_{\max})^\rho\}, \quad g(\mathcal{I}) = \beta \min\{1, (\mathcal{I}/\mathcal{I}_{\max})^\rho\}.$$

The efficiency of the resulting designs can be seen in our example with $\alpha = 0.025$, power 0.9, $K = 5$ stages, $n_{fix} = 100$ and $n_{max} = 110$.

Values of F achieved by ρ -family error spending designs



Further development of Delayed Response GSTs

A variety of optimality criteria

HJ show how designs can be optimised for criteria involving both the number of subjects recruited and the time to a final decision.

The nature of a specific clinical trial will determine which approaches may be possible, depending on whether:

- All pipeline subjects must be followed to the response time,

- Investigators may decide whether to wait and observe pipeline subjects,

- Data from (some) pipeline subjects will not be “valid” and cannot be used.

Discussants of the HJ paper commented on the nature of “pipeline” data and HJ categorised possible types of situation in their response.

Further development of Delayed Response GSTs

Inference on termination

HJ explain how to construct p-values and confidence intervals, with the usual frequentist properties, on termination of a Delayed Response GST.

These methods can also provide median unbiased point estimates.

The bias of maximum likelihood estimates can be reduced following the approach which Whitehead (*Biometrika*, 1986) introduced for standard GSTs.

Non-binding futility boundaries

It is commonly required that a group sequential design should protect the type I error rate, even if the trial may continue after crossing the “futility” boundary.

We are currently working to extend our error spending methods to the “non-binding” case.

Further development of Delayed Response GSTs

Adaptive choice of group sizes in a Delayed Response GST

There have been many proposals for “sample size re-estimation” in response to interim treatment effect estimates.

With an immediate response, these designs can be regarded as GSTs with the added feature that the size of each group is data-dependent.

Faldum & Hommel (*J. Biopharm. Statistics*, 2007) and Mehta & Pocock (*Statistics in Medicine*, 2011) present 2-group designs with sample size re-estimation and a delayed response.

Jennison & Turnbull (*Statistics in Medicine*, 2015) pursue new developments in the spirit of Mehta & Pocock’s “promising zone” methods and arrive at designs which can be regarded as adaptive versions of HJ’s Delayed Response GSTs.

Whether response is immediate or delayed, only minor efficiency gains are achieved by adapting group sizes to treatment effect estimates.

Recapitulation: Group sequential tests

- It is natural to monitor clinical trials with a view to possible early stopping.
- Distribution theory and computational methods support a variety of group sequential designs which control the type I error rate.
- These designs can be optimised for a stated objective.
- Error spending designs offer efficient, flexible monitoring of a variety of response types, including survival data.
- Information monitoring allows changes to be made to ensure the trial has enough subjects or failure events to achieve desired statistical power.
- Group sequential tests can be extended to accommodate a delayed response.
- These designs offer (nearly) all the usual features of GSTs for an immediate response and they can be optimised for particular objectives.
- The impact on efficiency of a delayed response, can be mitigated by using short term end-points to capture interim information from pipeline subjects.

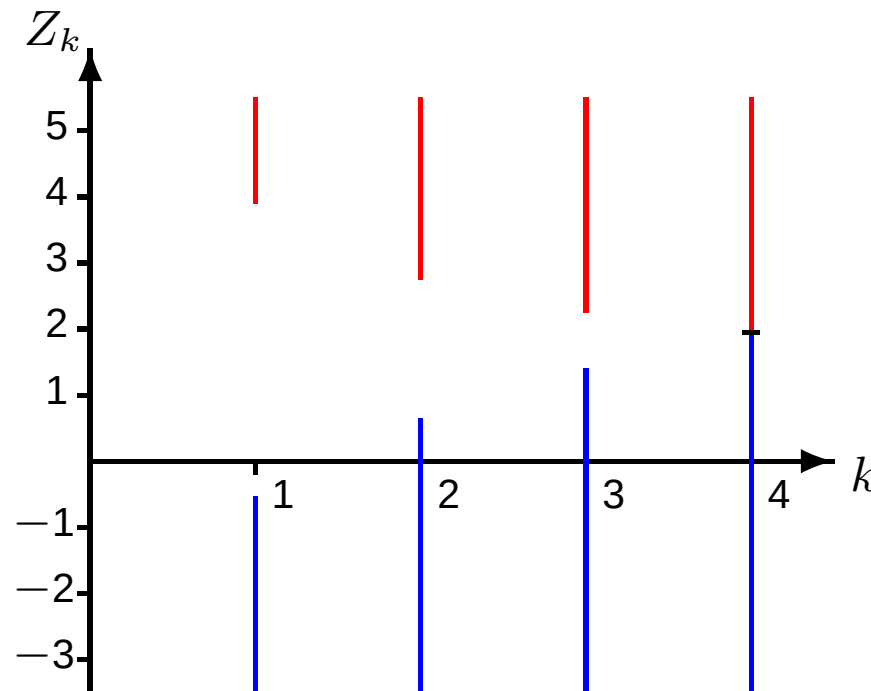
Session 2. Multiple hypothesis testing and GSTs

7. Analysis on termination of a group sequential test
8. Inference on a secondary endpoint after a group sequential test
9. Testing multiple endpoints in a single trial
10. Multiple testing procedures and group sequential designs
11. Further reading on GSTs and multiple hypothesis testing
12. GSTs and co-primary endpoints

7. Analysis on termination of a group sequential test (JT, Ch. 8)

A group sequential test's sample space is all possible pairs (k, Z_k) on termination.

The figure shows this sample space for a Pampallona & Tsiatis test of $H_0: \theta \leq 0$ vs $\theta > 0$ with $\Delta = 0$, $K = 4$ analyses, type I error rate $\alpha = 0.025$ and power $1 - \beta = 0.8$ at $\theta = 1$.



Frequentist inference is based on probabilities over the *sample space of the study*.

The need for special methods

Suppose our 4 stage study with a Pampallona & Tsiatis boundary ends at stage 3 with $Z_3 = 2.6$. It is tempting to quote a 1-sided P-value of

$$P\{N(0, 1) > 2.60\} = 0.0047.$$

But using this definition, we would also get a P-value ≤ 0.0047 by

stopping at stage 1 with $Z_1 > 3.90$,

stopping at stage 2 with $Z_2 > 2.76$,

stopping at stage 3 with $Z_3 > 2.60$,

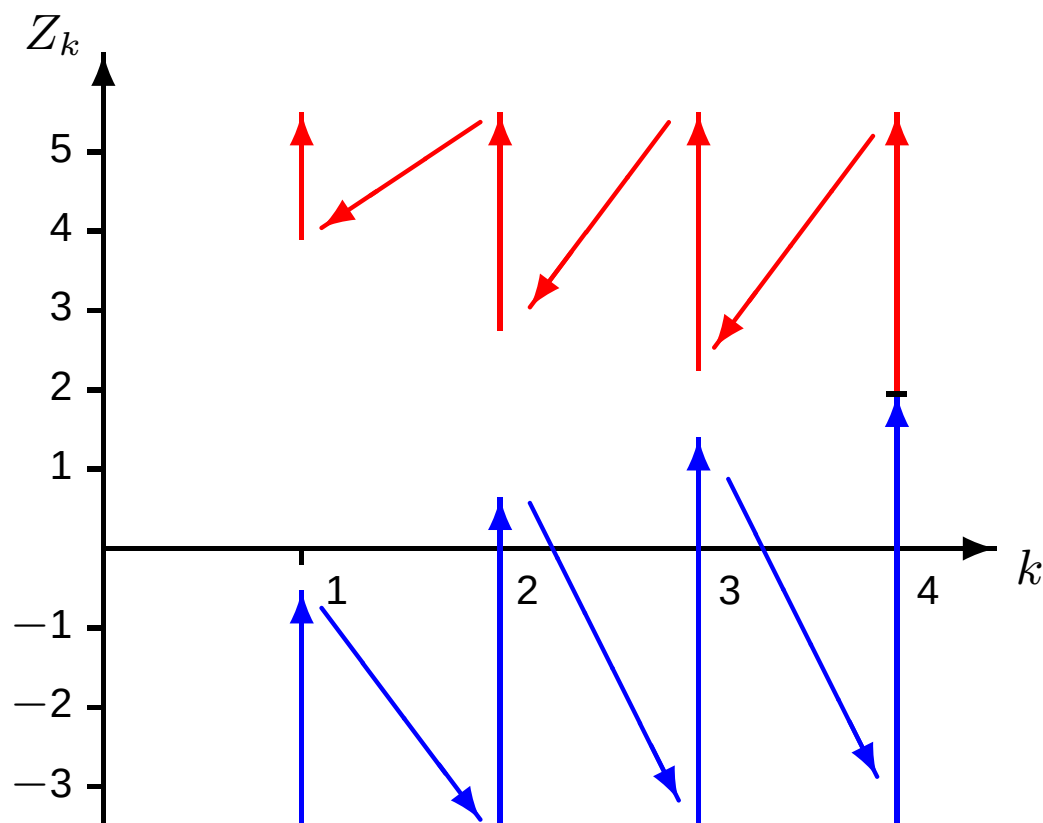
stopping at stage 4 with $Z_4 > 2.60$,

and the total probability under $\theta = 0$ of a “P-value” ≤ 0.0047 would be 0.0076.

So, this “P-value” is *not* distributed as $U(0, 1)$.

Analysis on termination

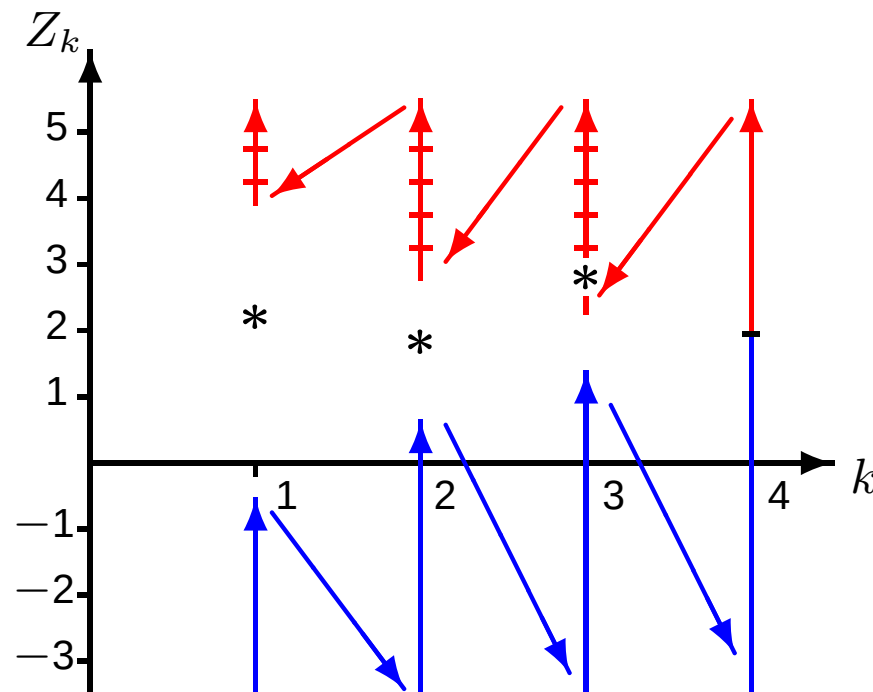
For proper frequentist inference, we first order the sample space.



We shall define P-values and confidence intervals with respect to this ordering.

A P-value on termination

The *P-value* for $H_0: \theta = 0$ is the probability under H_0 of seeing an outcome as extreme as that observed.



So, on stopping at analysis 3 with $Z_3 = 2.60$, the 1-sided P-value for $H_0: \theta \leq 0$ is

$$P_{\theta=0}\{\text{Terminate with } Z_1 \geq 3.90 \text{ or } Z_2 \geq 2.76 \text{ or } Z_3 \geq 2.60\} = 0.0063.$$

A P-value on termination

With the above definition, based on a specific ordering of the sample space:

The P-value has a $U(0, 1)$ distribution under H_0 .

If the group sequential test has one-sided type I error probability α , the P-value is $\leq \alpha$ precisely when the test stops with rejection of H_0 , i.e., in the part of the sample space coloured red.

The P-value will tend to take low values when the parameter θ is large and positive.

A confidence interval on termination

Suppose the test terminates at analysis k^* with $Z_{k^*} = Z^*$.

A $100(1 - 2\alpha)\%$, equal-tailed confidence interval for θ contains precisely those values θ for which the observed outcome (k^*, Z^*) is in the middle $(1 - 2\alpha)$ of the probability distribution of outcomes under θ .

This can be seen to be the interval (θ_1, θ_2) where

$$P_{\theta=\theta_1}\{\text{An outcome above } (k^*, Z^*)\} = \alpha$$

and

$$P_{\theta=\theta_2}\{\text{An outcome below } (k^*, Z^*)\} = \alpha.$$

This follows from the relation between a $100(1 - 2\alpha)\%$ lower confidence limit for θ and the family of level 2α , two-sided tests of hypotheses $H: \theta = \tilde{\theta}$.

A confidence interval on termination

Example:

If the trial stops at analysis 3 with $Z_3 = 2.6$, the 95% confidence interval for θ is

$$(0.22, 1.77)$$

using our specified ordering.

In contrast:

The “naive” fixed sample CI would be $(0.25, 1.78)$.

However, it is not appropriate to use this fixed sample interval as this fails to take account of the sequential stopping rule.

Consequently, the coverage probability of this fixed sample interval is *not* $1 - 2\alpha$.

Consistency of hypothesis testing and CI on termination

Suppose a group sequential study is run to test $H_0: \theta \leq 0$ vs $\theta > 0$ with one-sided type I error probability α .

Then, a $1 - 2\alpha$, equal-tailed confidence interval on termination should lie completely above $\theta = 0$ if and only if H_0 is rejected.

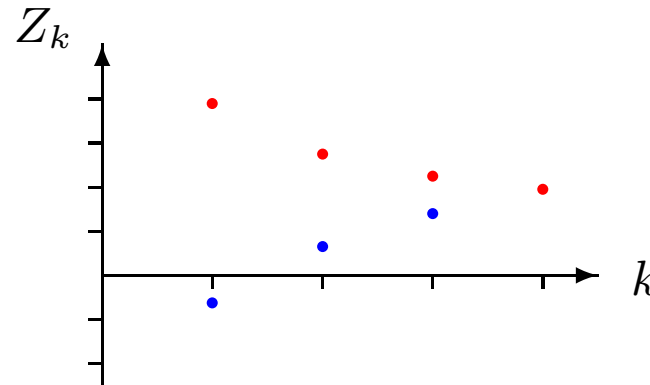
This happens automatically if outcomes for which we reject H_0 are at the top end of the sample space ordering — and any sensible ordering does this.

Why the naive approach does not work

Note that a naive $1 - 2\alpha$ level CI on termination lies completely above $\theta = 0$ if an *unadjusted* α level, one-sided significance test rejects H_0 .

Because of the multiple testing effect, the probability of such an outcome is liable to be greater than the desired level α .

Estimating θ after a group sequential test



In a balanced two-treatment comparison, the maximum likelihood estimate (MLE) of θ on termination of the trial is

$$\hat{\theta}_M = \sum_{i=1}^{n_k} (X_{Ai} - Y_{Bi}) / n_k.$$

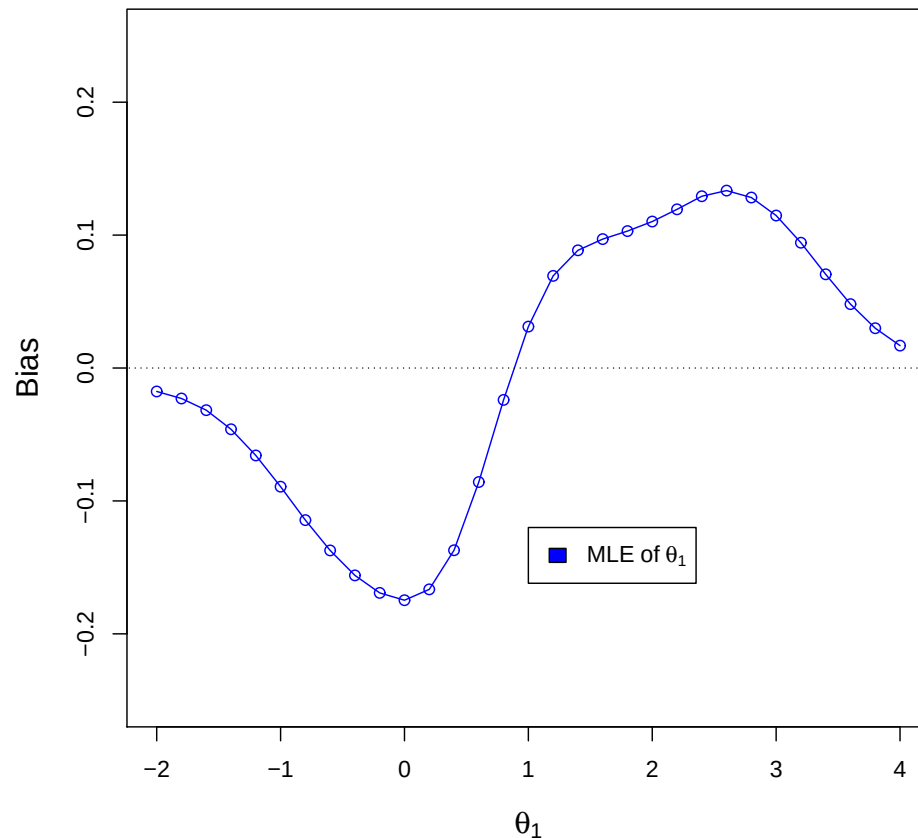
For large, positive values of θ , high values of $\hat{\theta}$ lead to early stopping, while lower values lead to collection of more data and the chance for $\hat{\theta}$ to increase.

This results in an upward bias of the MLE, so $E_{\theta}(\hat{\theta}_M) > \theta$ for larger values of θ .

Similarly, $E_{\theta}(\hat{\theta}_M) < \theta$ for lower values of θ .

Bias of the MLE of θ after a 4 group Pampallona & Tsiatis test

The bias of the MLE can be calculated as a function of the true effect size, θ .



In our example, sample size is chosen to give power 0.8 for detecting a treatment effect of 1, and the bias of the MLE is around 0.1 at values of θ just above 1.

Correcting the bias of the MLE

Denote the bias function of the MLE by

$$b(\theta) = E_{\theta}(\hat{\theta}_M) - \theta.$$

Whitehead (*Biometrika*, 1986) suggested correcting the MLE by subtracting an estimate of its bias.

Since the true θ is unknown, the bias of the MLE is estimated by $b(\hat{\theta}_M)$.

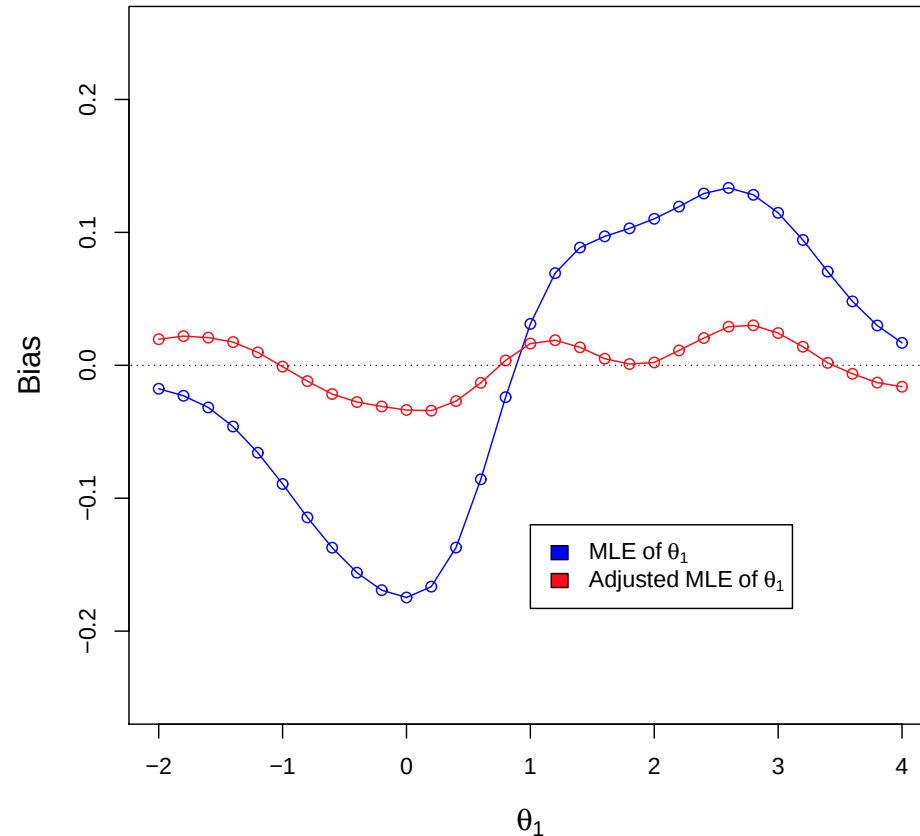
The adjusted estimator is then

$$\hat{\theta}_{adj} = \hat{\theta}_M - b(\hat{\theta}_M).$$

Bias of the MLE of θ after a 4 group Pampallona & Tsiatis test

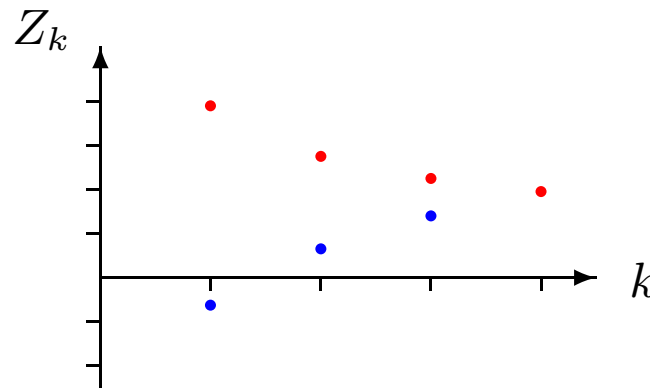
Simulation results show that Whitehead's adjusted estimator has much smaller bias than the MLE on which it is based.

For our example:



8. Inference on a secondary endpoint after a group sequential test

Stopping boundary for the primary endpoint



Denote the treatment effect on the primary endpoint by θ_1 .

Suppose the trial terminates with rejection of $H_0: \theta_1 \leq 0$ in favour of $\theta_1 > 0$.

On stopping, data on a secondary endpoint are analysed to estimate the treatment effect, θ_2 , on this endpoint.

For an individual subject, the primary and secondary responses are correlated.

The group sequential design leads to bias in the MLE $\hat{\theta}_1$ — and the correlation in responses implies that bias is passed on to the MLE $\hat{\theta}_2$.

Estimation for a secondary endpoint after a group sequential test

Suppose responses for an individual subject are bivariate normal with correlation ρ .

For a patient on Treatment A,

$$\text{Primary endpoint} \quad X_1 \sim N(\mu_{A1}, \sigma_1^2),$$

$$\text{Secondary endpoint} \quad X_2 \sim N(\mu_{A2}, \sigma_2^2).$$

Similarly, for a patient on Treatment B,

$$\text{Primary endpoint} \quad X_1 \sim N(\mu_{B1}, \sigma_1^2),$$

$$\text{Secondary endpoint} \quad X_2 \sim N(\mu_{B2}, \sigma_2^2).$$

The primary treatment effect is

$$\theta_1 = \mu_{A1} - \mu_{B1}$$

and the secondary treatment effect is

$$\theta_2 = \mu_{A2} - \mu_{B2}.$$

Estimation for a secondary endpoint after a group sequential test

Consider a group sequential design where the bias in the MLE $\hat{\theta}_1$ is

$$b_1(\theta) = E_{\theta}(\hat{\theta}_1) - \theta_1$$

when the true treatment effects are $\theta = (\theta_1, \theta_2)$.

Note that $E_{\theta}(\hat{\theta}_1)$ depends on θ_1 and not on θ_2 .

Whitehead (*Biometrics*, 1986) shows that the MLE, $\hat{\theta}_2$, has bias

$$b_2(\theta) = E_{\theta}(\hat{\theta}_2) - \theta_2 = \rho \sqrt{\frac{\sigma_2^2}{\sigma_1^2}} b_1(\theta)$$

when the true treatment effects are $\theta = (\theta_1, \theta_2)$.

Since this bias is a multiple of $b_1(\theta)$, it depends on θ_1 — and not on θ_2 .

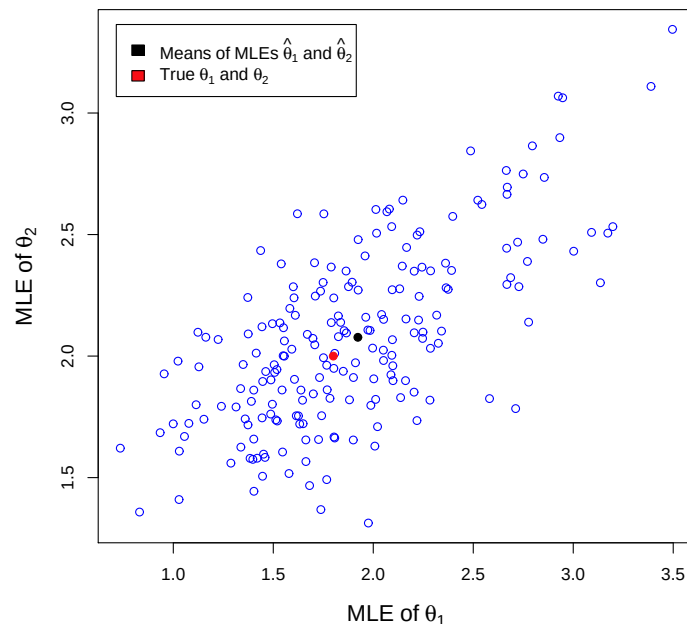
We can follow the same approach as for the primary endpoint and adjust the MLE, $\hat{\theta}_2$, by subtracting an estimate of its bias, $(\rho \sigma_2 / \sigma_1) b_1(\hat{\theta})$.

Estimation for a secondary endpoint: Example

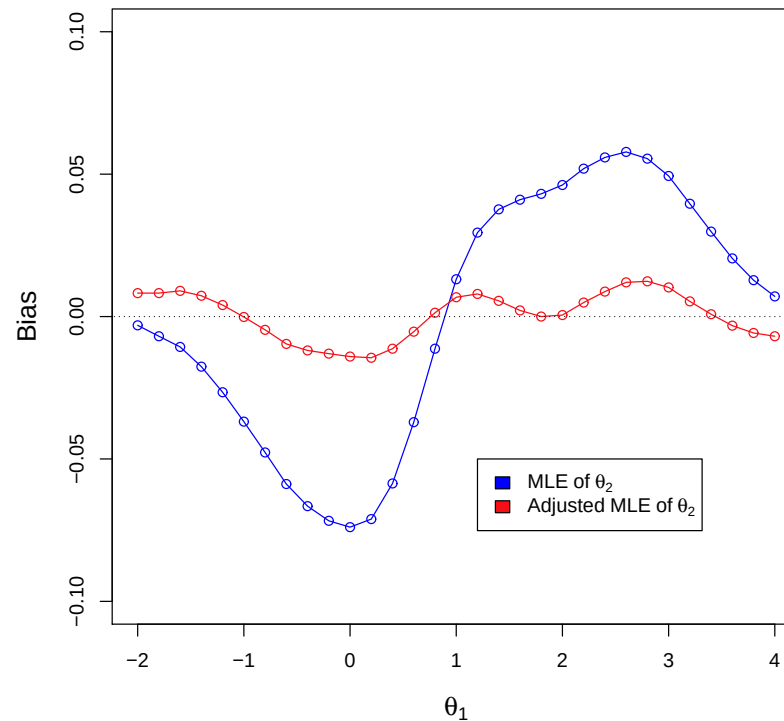
As previously, suppose a trial is designed for the primary endpoint using a Pampallona & Tsiatis test with $\Delta = 0$, 4 analyses, type I error rate $\alpha = 0.025$ and power 0.8 at $\theta_1 = 1$.

Assume responses are bivariate normal with correlation $\rho = 0.6$ and $\sigma_1^2/\sigma_2^2 = 2$.

The plot, for the case $\theta_1 = 1.8$ and $\theta_2 = 2$, shows the correlation between the MLEs, $\hat{\theta}_1$ and $\hat{\theta}_2$, on termination of the Pampallona & Tsiatis test.



Estimation for a secondary endpoint: Example



The plot shows the bias in the MLE $\hat{\theta}_2$ is largely eliminated in the adjusted estimator

$$\hat{\theta}_2 - \rho \sqrt{\frac{\sigma_2^2}{\sigma_1^2}} b_1(\hat{\theta}).$$

9. Testing multiple endpoints in a single trial

Consider a trial comparing treatments A and B where the treatment effect for the primary endpoint is θ_1 .

The trial has a group sequential design with a Pampallona & Tsiatis test with $\Delta = 0$, 4 analyses, $\alpha = 0.025$ and power 0.8 at $\theta_1 = 1$.

If the trial has a positive outcome, rejecting $H_1: \theta_1 \leq 0$ in favour of $\theta_1 > 0$, a secondary endpoint with treatment effect θ_2 is analysed.

The investigators believe it is appropriate to carry out a fixed sample size, level α test of $H_2: \theta_2 \leq 0$ against $\theta_2 > 0$.

Suppose that for an individual patient, the primary and secondary responses are bivariate normal with correlation ρ .

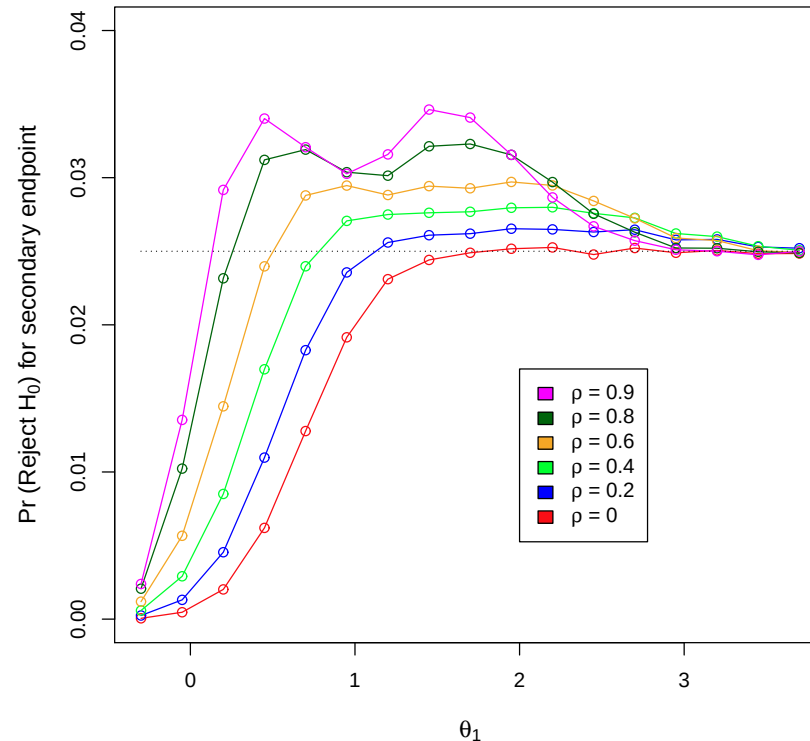
Is this approach to testing the two endpoints valid?

Hung, Wang and O'Neill (*J. Biopharm. Statis.*, 2007) explain why it is not valid.

Testing a secondary endpoint after a group sequential test

The plot shows the overall probability of rejecting $H_2: \theta_2 \leq 0$ (which requires rejection of H_1 first), when $\theta_2 = 0$. Values of θ_1 range from below 0 to above 3.

As ρ increases, the type I error rate for testing H_2 exceeds the nominal 0.025.



The type I error rate for the test of H_2 is inflated for the same reason that the MLE of θ_2 is biased upon conclusion of a group sequential test of θ_1 .

Testing multiple hypotheses

A clinical trial may involve

Co-primary endpoints

Positive outcomes required for at least one endpoint

Positive outcomes required on all endpoints

Secondary endpoints, tertiary endpoints, . . .

The trial may have

Multiple treatments,

Pre-defined sub-populations of patients.

The trial design may be

Of fixed sample size,

Group sequential.

The familywise error rate

Suppose we have h null hypotheses, $H_i: \theta_i \leq 0$ for $i = 1, \dots, h$.

A testing procedure yields a decision to accept or reject each of the h hypotheses.

The procedure's **familywise error rate** under a set of values $(\theta_1, \dots, \theta_h)$ is

$$Pr\{\text{Reject } H_i \text{ for some } i \text{ with } \theta_i \leq 0\} = Pr\{\text{Reject any true } H_i\}.$$

The familywise error rate is controlled **strongly** at level α if this error rate is at most α for all possible combinations of θ_i values. Then

$$Pr\{\text{Reject any true } H_i\} \leq \alpha \quad \text{for all } (\theta_1, \dots, \theta_h).$$

Using such a procedure, we can choose to focus on a parameter θ_{i^*} and claim significance for a test of the null hypothesis H_{i^*} , without having to worry that this choice of hypothesis was based on observed data.

Bonferroni adjustment

Carlo Bonferroni (1892–1960) is associated with a simple, but very useful, result.

Suppose we have h null hypotheses and test each one at significance level α/h .

Then, if all h null hypotheses are true,

$$\begin{aligned} & Pr\{\text{Reject at least one of } H_1 \dots H_h\} \\ & \leq Pr\{\text{Reject } H_1\} + \dots + Pr\{\text{Reject } H_h\} = h \frac{\alpha}{h} = \alpha. \end{aligned}$$

If only some of the h null hypotheses are true,

$$Pr\{\text{Reject at least one true } H_i\} < \alpha.$$

So we have **strong control** of the **familywise error rate**.

We start by considering applications in fixed sample size study designs . . .

A Bonferroni test for co-primary endpoints

Example: Co-primary endpoints (non-sequential trial)

A trial compares a new treatment against control with respect to two endpoints

Endpoint 1: Core MACE (*Major Adverse Cardiac Event* — CV-related death, nonfatal stroke, or nonfatal myocardial infarction)

Endpoint 2: Expanded MACE (Core MACE plus hospitalization for unstable angina or coronary revascularization).

One-sided type I error probability $\alpha = 0.025$ is divided between the endpoints.

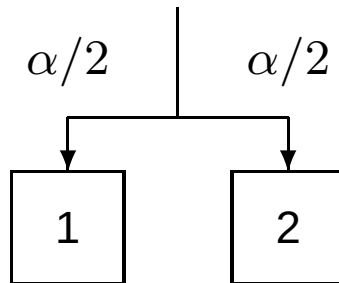
With Z -statistics Z_1 and Z_2 for endpoints 1 and 2,

An effect on Core MACE is declared if $Z_1 > \Phi^{-1}(1 - \alpha/2) = 2.24$,

An effect on Expanded MACE is declared if $Z_2 > \Phi^{-1}(1 - \alpha/2) = 2.24$.

Example: Co-primary endpoints

This Bonferroni procedure can be represented graphically as:



Familywise type I error is protected conservatively as there is a positive correlation between the two tests, due to the common aspects of the two endpoints.

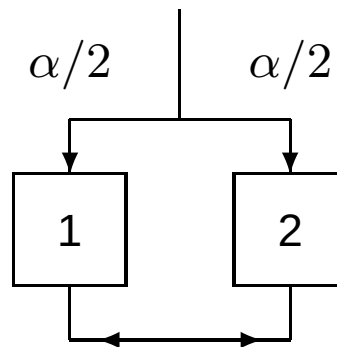
Suppose we have rejected H_1 , might it be permissible to test H_2 at significance level α rather than $\alpha/2$?

If H_1 is false, we only need to worry about type I errors concerning H_2 .

If H_1 is true, we have already made a type I error (a rare event given H_1 is true) and making another will not increase the familywise error rate!

Bonferroni procedure with recycling (the Holm procedure)

We can represent a new version of the Bonferroni procedure, which “recycles” error probability after rejecting H_1 or H_2 , as:

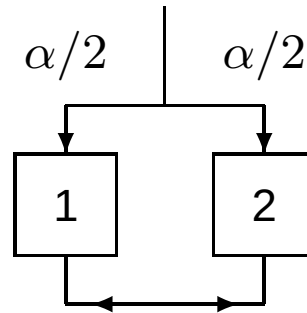


Proof that FWER is protected

If H_1 and H_2 are both true,

$$\begin{aligned}\text{FWER} &= Pr\{\text{Reject } H_1 \text{ or } H_2\} \\ &\leq Pr\{Z_1 > \Phi^{-1}(1 - \alpha/2)\} + Pr\{Z_2 > \Phi^{-1}(1 - \alpha/2)\} \\ &\leq \alpha/2 + \alpha/2 = \alpha.\end{aligned}$$

Bonferroni procedure with recycling (the Holm procedure)



Proof that FWER is protected (continued)

If H_1 is true and H_2 is false,

$$\text{FWER} = \Pr\{\text{Reject } H_1\} \leq \Pr\{Z_1 > \Phi^{-1}(1 - \alpha)\} = \alpha.$$

Similarly, if H_2 is true and H_1 is false,

$$\text{FWER} = \Pr\{\text{Reject } H_2\} \leq \Pr\{Z_2 > \Phi^{-1}(1 - \alpha)\} = \alpha.$$

If H_1 and H_2 are both false,

A type I error cannot be made so $\text{FWER} = 0$.

A hierarchical testing or “gatekeeping” procedure

Example: Primary and secondary endpoints

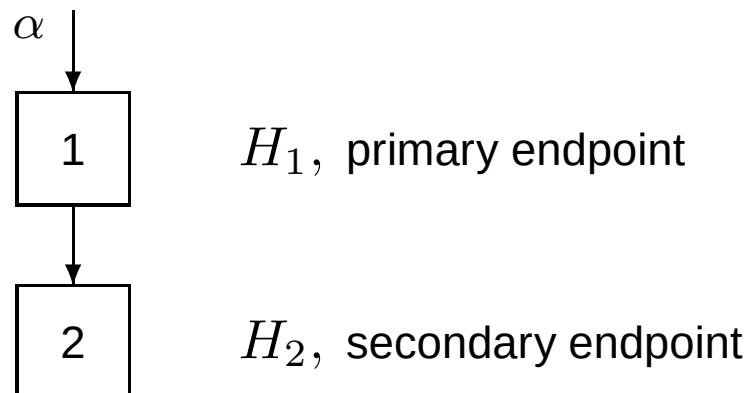
Consider a trial where

The null hypothesis H_1 concerns the primary endpoint,

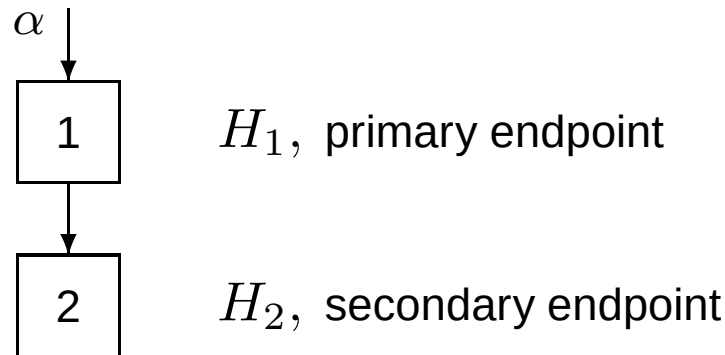
The null hypothesis H_2 relates to a secondary endpoint.

Suppose H_2 will only be tested if H_1 has already been rejected — O'Neill (*Controlled Clinical Trials*, 1997) states this is the only time one should test H_2 .

We test H_1 first at significance level α . If H_1 is rejected, we continue to test H_2 at significance level α .



Example: Primary and secondary endpoints



Proof that FWER is protected

If H_1 is true, a family-wise error occurs if H_1 is rejected (regardless of H_2),

$$\text{FWER} = \Pr\{\text{Reject } H_1\} = \Pr\{Z_1 > \Phi^{-1}(1 - \alpha)\} = \alpha.$$

If H_1 is false and H_2 is true,

$$\begin{aligned} \text{FWER} &= \Pr\{\text{Reject } H_1 \text{ and then reject } H_2\} \\ &\leq \Pr\{Z_2 > \Phi^{-1}(1 - \alpha)\} = \alpha. \end{aligned}$$

If H_1 and H_2 are both false, a type I error cannot be made and $\text{FWER} = 0$.

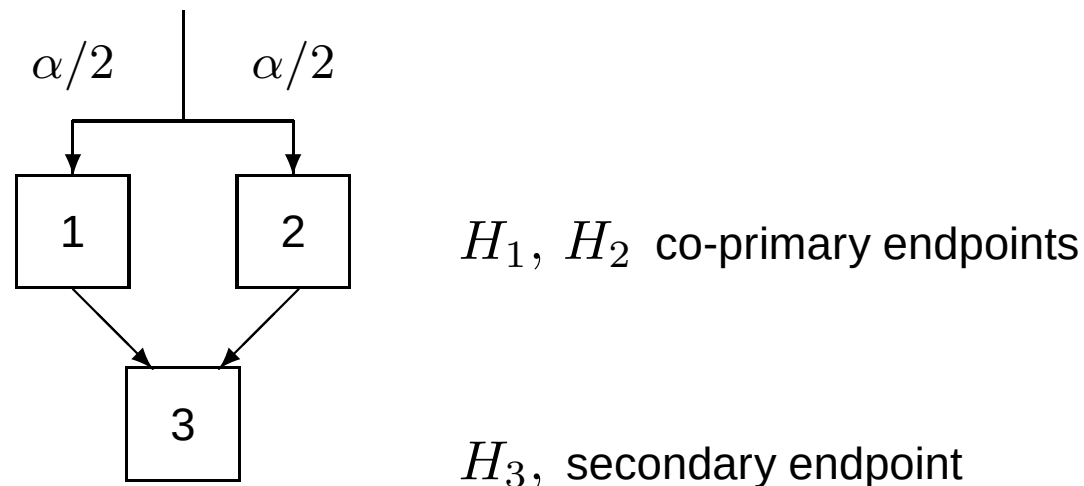
Co-primary and secondary endpoints

Suppose we wish to test a secondary endpoint if a positive result is obtained on either primary endpoint.

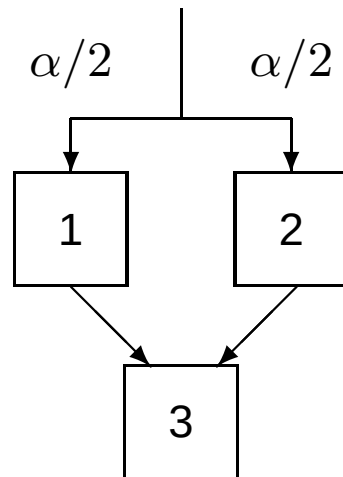
To do this, we recycle family wise error probability from the primary endpoints.

The secondary endpoint is tested at significance level $\alpha/2$ if just one of H_1 and H_2 is rejected, and at level α if both H_1 and H_2 are rejected.

We can represent this testing procedure as:



Co-primary and secondary endpoints



H_1, H_2 co-primary endpoints

H_3 , secondary endpoint

There are eight different combinations of true and false values for H_1 , H_2 and H_3 .

Taking these eight cases in turn, it is quite easy to prove that FWER is at most α , whichever set of null hypotheses is true.

Questions?

1. Can we add more “recycling” to reduce conservatism and increase power?
2. Can we opt to recycle error between H_1 and H_2 before testing H_3 at all?

Closed testing procedures

In constructing and validating more elaborate forms of multiple testing, we can make use of “closed testing procedures” (Marcus et al, *Biometrika*, 1976).

The closed testing procedure

Suppose we have h null hypotheses, $H_i: \theta_i \leq 0$ for $i = 1, \dots, h$.

For each subset I of $\{1, \dots, h\}$, define the intersection hypothesis

$$H_I = \cap_{i \in I} H_i$$

which states that all hypotheses H_i are true, for $i \in I$.

Construct a level α test of each intersection hypothesis H_I , i.e., a test which rejects H_I with probability at most α whenever all hypotheses specified in H_I are true.

The simple hypothesis $H_j: \theta_j \leq 0$ is rejected overall if, and only if, H_I is rejected for every set I containing index j .

Closed testing procedures

Proof that a closed testing procedure provides strong control of the FWER at level α

Let $\tilde{I}$ be the set of indices of all true hypotheses H_i .

For a familywise error to be committed, $H_{\tilde{I}}$ must be rejected.

Since $H_{\tilde{I}}$ is true,

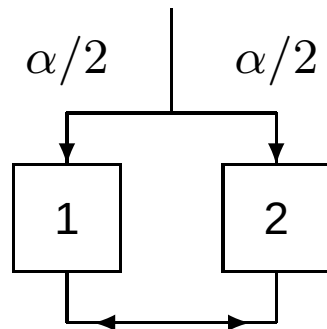
$$Pr\{\text{Reject } H_{\tilde{I}}\} = \alpha.$$

Thus,

$$Pr\{\text{Reject } H_j \text{ for at least one } j \in \tilde{I}\} \leq Pr\{\text{Reject } H_{\tilde{I}}\} = \alpha,$$

so the probability of a familywise error is no greater than α .

The Bonferroni test with recycling as a closed testing procedure



Let P_1 and P_2 denote P-values for simple tests of H_1 and H_2 .

Write $H_{1,2}$ for the intersection hypothesis, $H_1 \cap H_2$.

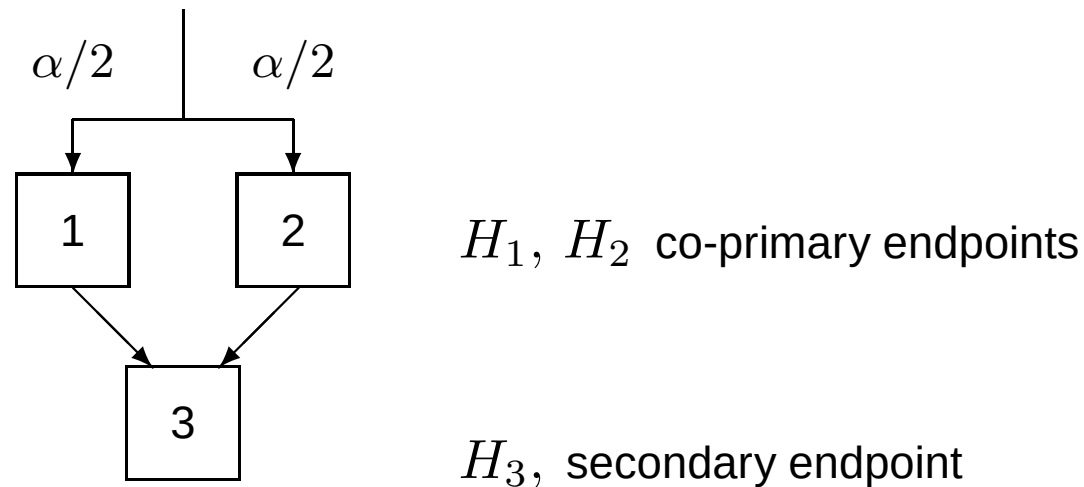
Using the closed testing procedure with the following set of tests is equivalent to the Bonferroni test with recycling.

<i>Hypothesis</i>	<i>Reject if</i>
H_1	$P_1 \leq \alpha$
H_2	$P_2 \leq \alpha$
$H_{1,2}$	$\min(P_1, P_2) \leq \alpha/2$

E.g., to reject H_1 overall we need individual tests to reject both H_1 and $H_{1,2}$, i.e.,

$$P_1 \leq \alpha \quad \text{and} \quad \min(P_1, P_2) \leq \alpha/2.$$

Testing co-primary and secondary endpoints as a closed testing procedure



Let P_1 , P_2 and P_3 denote P-values for simple tests of H_1 , H_2 and H_3 .

The procedure depicted above is equivalent to a closed testing procedure with suitably defined tests of H_1 , H_2 and H_3 , and the related intersection hypotheses.

Co-primary and secondary endpoints: Closed testing procedure

Tests of intersection hypotheses are:

<i>Hypothesis</i>	<i>Reject if</i>
H_1	$P_1 \leq \alpha/2$
H_2	$P_2 \leq \alpha/2$
H_3	$P_3 \leq \alpha$
$H_{1,2}$	$\min(P_1, P_2) \leq \alpha/2$
$H_{1,3}$	$\min(P_1, P_3) \leq \alpha/2$
$H_{2,3}$	$\min(P_2, P_3) \leq \alpha/2$
$H_{1,2,3}$	$\min(P_1, P_2) \leq \alpha/2$

E.g., to reject H_3 overall needs rejection of H_3 , $H_{1,3}$, $H_{2,3}$ and $H_{1,2,3}$, i.e.,

$$P_3 \leq \alpha, \min(P_1, P_3) \leq \alpha/2, \min(P_2, P_3) \leq \alpha/2, \min(P_1, P_2) \leq \alpha/2,$$

which can be seen to agree with the procedure described earlier.

Answer to Question 1: Can we add recycling to gain power?

A closed testing procedure with additional recycling

The tests of intersection hypotheses include:

<i>Hypothesis</i>	<i>Reject if</i>
H_1	$P_1 \leq \alpha/2$
H_2	$P_2 \leq \alpha/2$

This indicates conservatism. We can replace these tests by

<i>Hypothesis</i>	<i>Reject if</i>
H_1	$P_1 \leq \alpha$
H_2	$P_2 \leq \alpha$

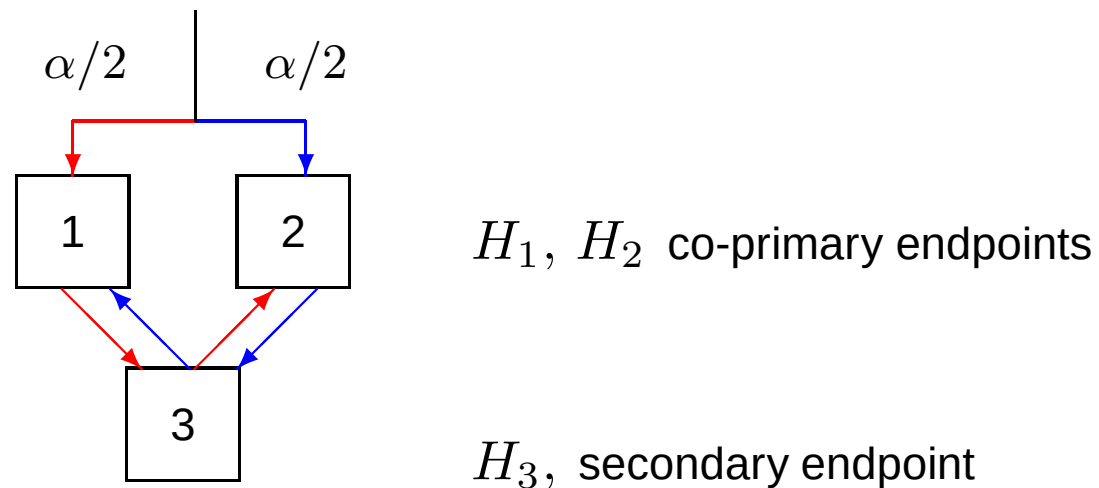
and their type I error rates will still be at most α .

This modification corresponds to recycling error probability from the test of H_3 back to whichever of H_1 and H_2 has not been rejected at level $\alpha/2$.

The extra opportunities to reject H_1 and H_2 give greater power.

Co-primary and secondary endpoints: A closed testing procedure with additional recycling

We can represent the testing procedure with additional recycling graphically.



The additional lines in the graph indicate that:

If $P_1 \leq \alpha/2$ and $P_3 \leq \alpha/2$, then H_2 is tested at level α ,

If $P_2 \leq \alpha/2$ and $P_3 \leq \alpha/2$, then H_1 is tested at level α .

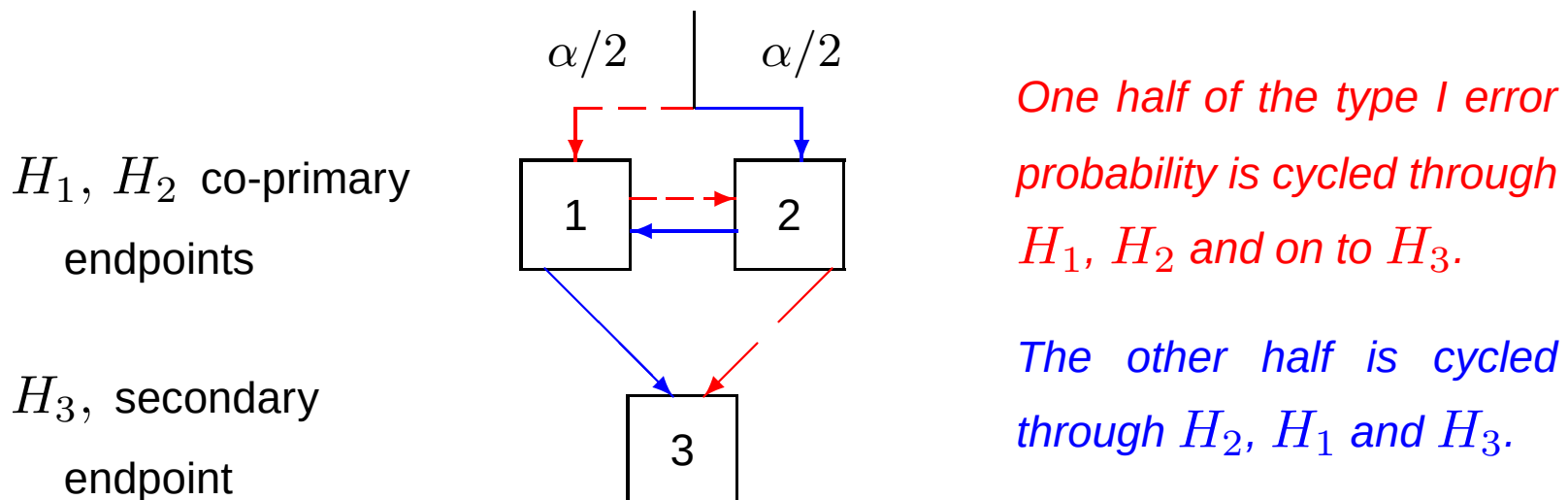
Answer to Question 2: Can we opt to recycle error between H_1 and H_2 before testing H_3 ?

Recycling between primary endpoints before testing H_3

We may prefer to gain maximum power for tests of the co-primary endpoints before testing the secondary endpoint at all.

This is achieved by recycling error probability from H_1 to H_2 , and vice versa, before allocating any error probability to a test of H_3 .

A graphical representation is:



More complex procedures

As we add more options, and get more creative, we can produce some quite complex procedures.

It is still necessary to check that the familywise type I error rate is protected.

At the same time, we should try to avoid excessive conservatism.

We also want to be able to construct testing procedures that fit with:

A logical sequence for considering hypotheses, e.g., primary endpoint before secondary endpoint,

The relative impact of decisions to reject different hypotheses,

The perceived chance of being able to reject each hypothesis.

General methodology

Two papers, published simultaneously, describe an elegant and understandable way to describe complex multiple testing procedures.

These procedures are closed testing procedures in which the tests of intersection hypotheses are weighted Bonferroni tests.

The papers are:

“A recycling framework for the construction of Bonferroni-based multiple tests” by Burman, Sonesson and Guilbaud, *Statistics in Medicine*, 2009.

and

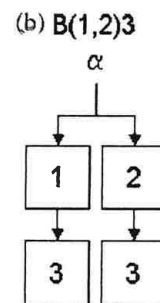
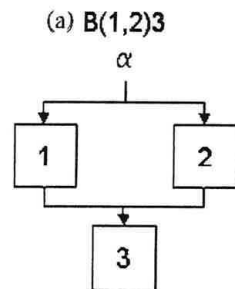
“A graphical approach to sequentially rejective multiple test procedures” by Bretz, Maurer, Brannath and Posch, *Statistics in Medicine*, 2009.

The following diagrams give a flavour of what is possible and the graphical representations of multiple testing procedures used in the two papers .

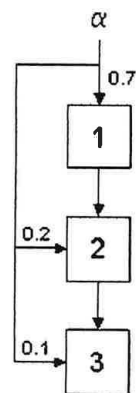
A figure from Burman et al. (2009)

(a) and (b) A parallel gatekeeping procedure (equivalent versions)

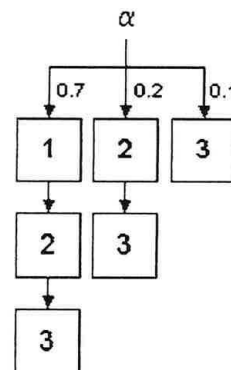
(c) and (d) A fallback procedure (equivalent versions)



(c) $B(123,23,3)[0.7,0.2,0.1]$



(d) $B(123,23,3)[0.7,0.2,0.1]$



A figure from Bretz et al. (2009)

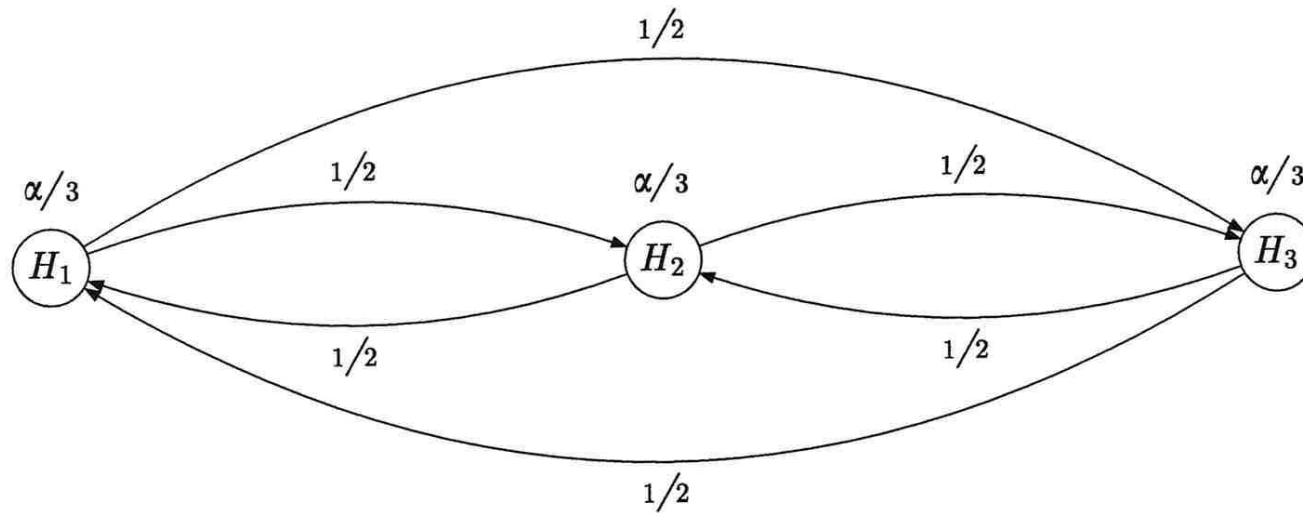


Figure 3. Graphical illustration of the Bonferroni–Holm procedure with $m=3$ hypotheses and initial allocation $\alpha=(\alpha/3, \alpha/3, \alpha/3)$.

10. Multiple testing procedures and group sequential designs

We have just described multiple testing procedures in the context of fixed sample size trial designs.

Here, the sample space is simple and it is straightforward to define a Z -statistic or determine a P-value for each null hypothesis.

We can follow the same principles to test multiple hypotheses when a study is conducted group sequentially — but we shall need to define any P-values with proper attention to the sequential sampling frame.

In particular, the definition of a P-value should not change in response to observed data, either on the endpoint in question or other, correlated endpoints.

These considerations underlie discussion in the paper

“Statistical considerations for testing multiple endpoints in group sequential or adaptive clinical trials” by Hung, Wang and O’Neill, *J. Biopharmaceutical Statistics*, 2007.

Testing a secondary endpoint after a group sequential test

In our earlier example, a trial compares two treatments with regard to their effect on a primary endpoint, then a secondary endpoint is analysed if a positive result is obtained for the primary endpoint.

Denoting the treatment effect on the primary endpoint by θ_1 , a group sequential test is conducted of $H_1: \theta_1 \leq 0$ vs $\theta_1 > 0$.

If H_1 is rejected by the group sequential test, the secondary endpoint, with treatment effect θ_2 , is analysed.

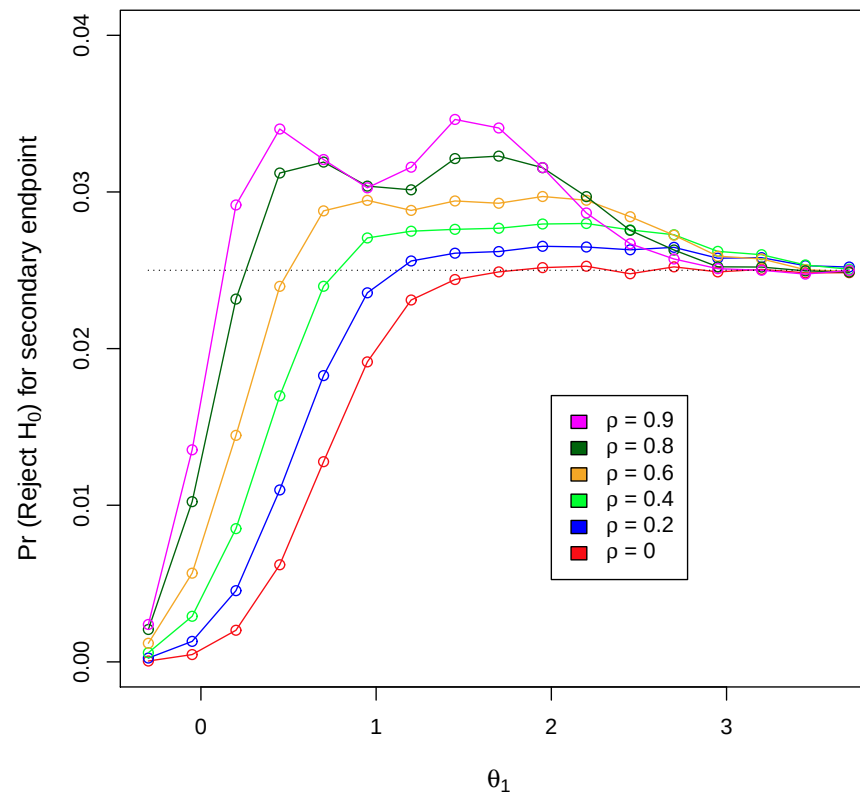
We supposed that investigators chose to conduct a fixed sample size, level α test of $H_2: \theta_2 \leq 0$ against $\theta_2 > 0$ using the data available for the second endpoint.

The investigators claim type I error probability α is passed from the test of H_1 to the test of H_2 , just as in a “gatekeeping” procedure.

Does general theory ensure the familywise type I error rate is protected?

Testing a secondary endpoint after a group sequential test

We have already seen plots of the overall probability of rejecting $H_2: \theta_2 \leq 0$ when $\theta_2 = 0$ which show that familywise error rate is *not* protected at level $\alpha = 0.025$.



For $\rho > 0$, $\Pr\{\text{Reject } H_2\} > \alpha$ for sufficiently high values of θ_1 .

Why the “gatekeeping” argument does not apply

In the proposed design, H_2 is tested if H_1 is rejected.

The test of H_2 is based on the set of measurements of the secondary endpoint at analysis $j = 1, 2, 3$, or 4 , depending on when H_1 is rejected.

Each analysis $j = 1, \dots, 4$ will give a different value for P_2 , $P_2(j)$ say.

The plan is to reject H_2 if $P_2(j^*) \leq \alpha$ when H_1 is rejected at analysis j^* .

If a single value, j^* , were specified and the trial always continued to analysis j^* (so we learn the value of $P_2(j^*)$), we would have

$$P_2(j^*) \sim U(0, 1) \quad \text{under } \theta_2 = 0.$$

Then, rejecting H_2 when $P_2(j^*) \leq \alpha$ would give a level α test.

Instead, the proposal is to reject H_2 when $P_2(J) \leq \alpha$ where J is the random variable denoting the analysis at which the trial terminates.

Why the “gatekeeping” argument does not apply

We have defined the random variable J as the analysis at which the trial stops.

The value taken by J depends on observations on the primary endpoint.

These observations are correlated with those on the secondary endpoint, so there is dependence between J and the values $P_2(1)$, $P_2(2)$, $P_2(3)$ and $P_2(4)$.

The danger is that $P_2(J)$ is more likely to be one of the smaller values in the set $\{P_2(1), P_2(2), P_2(3), P_2(4)\}$, increasing the probability that $P_2(J) \leq \alpha$, and H_2 is rejected, above α .

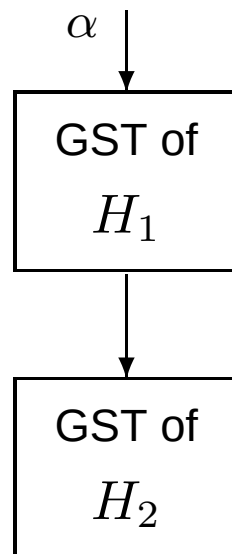
The simulation results for our example show this does indeed happen.

Solution:

We must test carry out a test of H_2 in a way which ***does not change in response to observed data, either on the endpoint in question or on other, correlated endpoints.***

A correct gatekeeping procedure

We need to specify a single test of H_2 which can be applied whenever the trial terminates.



The group sequential test of H_1
determines the stopping time
for the trial

The group sequential test of H_2 is
used for the secondary analysis
if and when H_1 is rejected

The group sequential test of H_2 provides a critical value at each analysis.

If the first test rejects H_1 at analysis J , we compare data on the secondary endpoint to the critical value given by the GST of H_2 at analysis J .

A correct gatekeeping procedure

Let $\{Z_{1,1}, \dots, Z_{1,K}\}$ denote Z -statistics for testing $H_1: \theta_1 \leq 0$ at analyses $1, \dots, K$ when information for θ_1 is $\mathcal{I}_{1,1}, \dots, \mathcal{I}_{1,K}$.

Similarly, let $\{Z_{2,1}, \dots, Z_{2,K}\}$ be Z -statistics for testing $H_2: \theta_2 \leq 0$ at analyses $1, \dots, K$ when information for θ_2 is $\mathcal{I}_{2,1}, \dots, \mathcal{I}_{2,K}$.

The GST of H_1 stops at analysis k to

Reject H_1 if $Z_{1,k} \geq b_k$,

Accept H_1 if $Z_{1,k} < a_k$.

Boundary values are set to control type I error at level α under $\theta_1 = 0$, i.e.,

$$\sum_{k=1}^K \Pr\{Z_{1,1} \in (a_1, b_1), \dots, Z_{1,k-1} \in (a_{k-1}, b_{k-1}), Z_{1,k} > b_k\} = \alpha.$$

A correct gatekeeping procedure

The GST of H_2 rejects H_2 at analysis k if $Z_{2,k} \geq c_k$, where

$$\sum_{k=1}^K \Pr\{Z_{2,1} < c_1, \dots, Z_{2,k-1} < c_{k-1}, Z_{2,k} \geq c_k\} = \alpha.$$

Since the stopping rule for the trial is based on the primary endpoint, the test of H_2 does not need a futility boundary, which would imply early acceptance of H_2 .

In the overall procedure, if the GST of H_1 stops to reject H_1 at analysis k^* , then we also reject H_2 if

$$Z_{2,k^*} \geq c_{k^*}.$$

A gatekeeping procedure using all of $\{Z_{2,1}, \dots, Z_{2,K}\}$ *could* reject H_2 if

$$Z_{2,k} \geq c_k \quad \text{for any } k \in \{1, \dots, K\}.$$

Hence, our overall procedure protects the familywise type I error rate conservatively.

Further options

Conservatism in the overall procedure arises because the test of H_1 may stop at analysis k^* when

$$Z_{2,k^*} < c_{k^*},$$

but

$$Z_{2,k} \geq c_k \text{ for some } k < k^* \text{ or } k > k^*.$$

This suggests options for reducing conservatism and increasing power:

1. Reject H_2 if $Z_{2,k} \geq c_k$ for some $k < k^*$, even though $Z_{2,k^*} < c_{k^*}$.

However, ignoring the most recent data (and the sufficient statistic for θ_2) would cast doubt on the credibility of this decision.

2. Continue the trial in the hope that $Z_{2,k} \geq c_k$ at some future analysis k .

However, if the primary endpoint is also observed for future subjects, is there a risk of “losing” the positive result on the primary endpoint?

Several authors have considered option (2), where a positive result outcome for H_1 is retained, whatever the additional information about θ_1 .

Example: Testing primary and secondary endpoints

A trial compares two treatments with normally distributed responses.

The treatment effect is θ_1 for the primary and θ_2 for the secondary endpoint.

The trial is designed group sequentially with a Pampallona & Tsiatis test of the primary endpoint using $\Delta = 0$, 4 analyses, $\alpha = 0.025$ and power 0.8 at $\theta_1 = 1$.

If $H_1: \theta_1 \leq 0$ is rejected for the primary endpoint, we test the secondary endpoint: when H_1 is rejected at analysis k^* , the test of $H_2: \theta_2 \leq 0$ rejects H_2 if

$$Z_{2,k^*} \geq c_{k^*}.$$

Case A (Pocock):

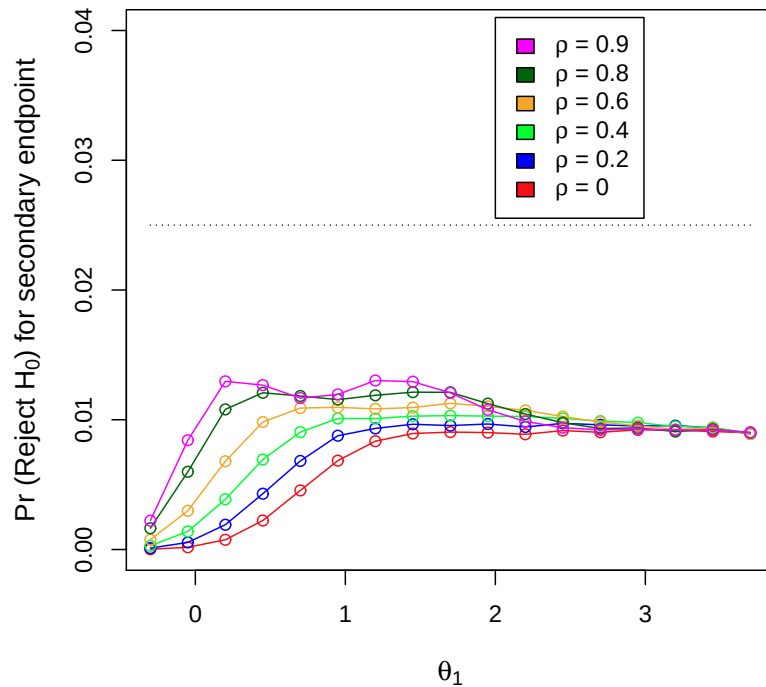
$$c_k = 2.361, \quad k = 1, \dots, 4.$$

Case B (O'Brien & Fleming):

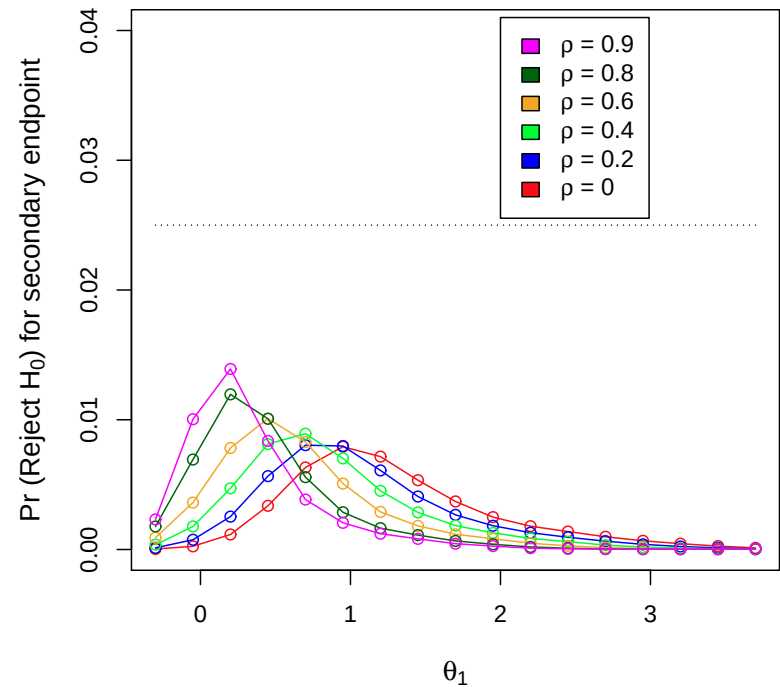
$$c_k = 2.024 \sqrt{\frac{4}{k}}, \quad k = 1, \dots, 4.$$

Type I error probability for testing H_2

A: Pocock boundary for H_2



B: OBF boundary for H_2

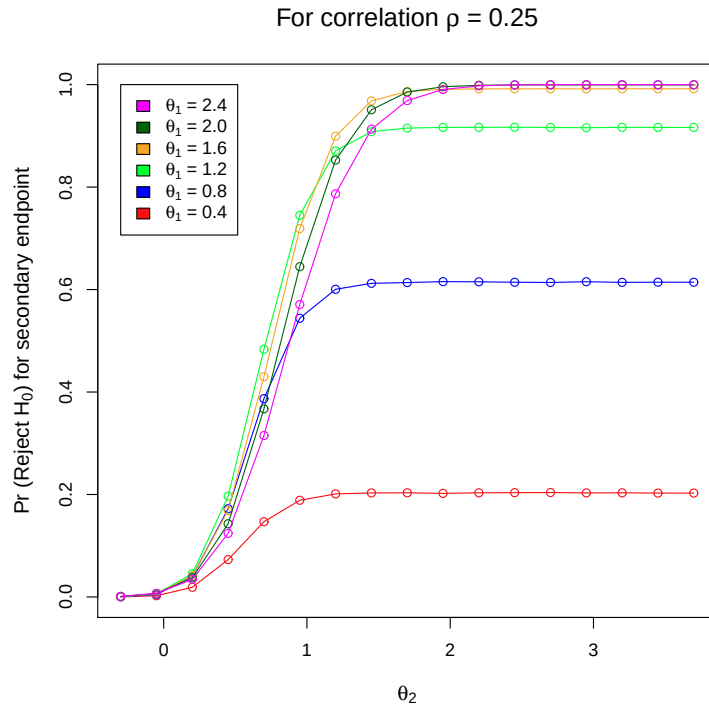


Type I error probabilities are calculated under $\theta_2 = 0$, but they also depend on θ_1 and the correlation, ρ , between the primary and secondary endpoints.

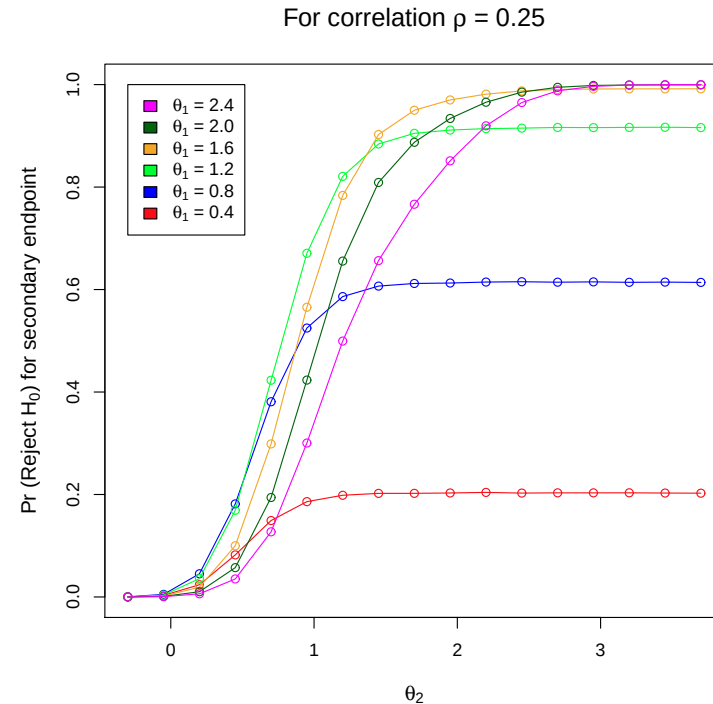
The OBF test of H_2 is particularly conservative under large values of θ_1 .

Power for testing H_2 , $\rho = 0.25$

A: Pocock boundary for H_2



B: OBF boundary for H_2



We have supposed (without real loss of generality) that $\mathcal{I}_{2,k} = 2\mathcal{I}_{1,k}$.

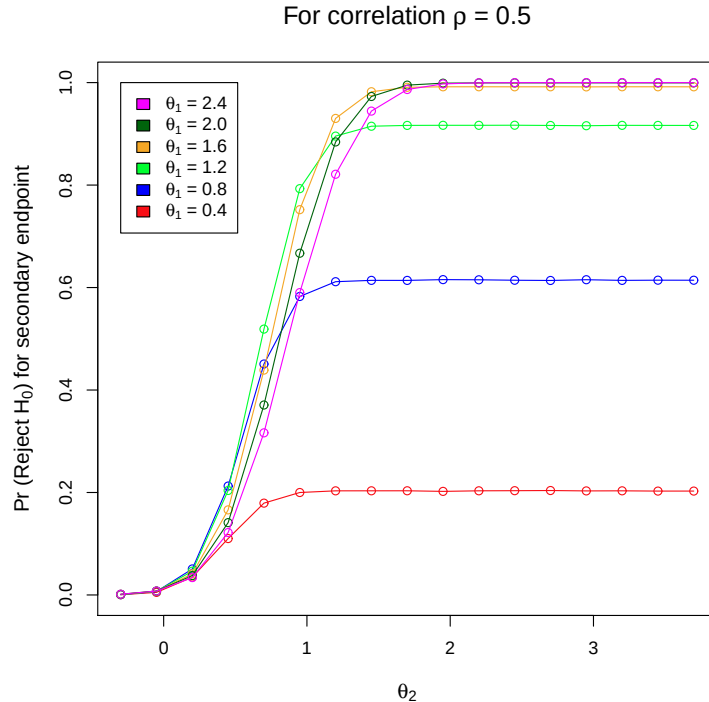
Power is shown as a function of θ_2 for selected values of θ_1 and $\rho = 0.25$.

The value of θ_1 has a large effect on the analysis at which a test of H_2 may occur.

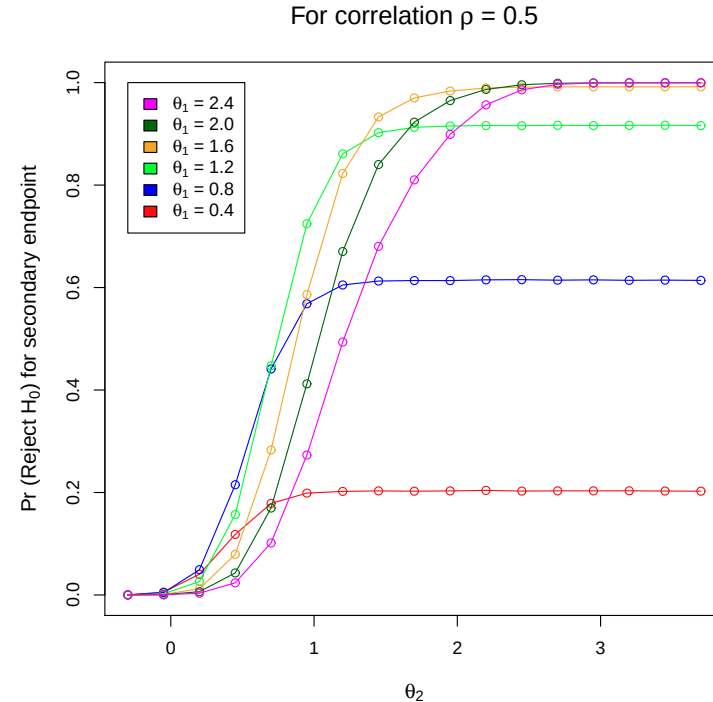
The Pocock boundary for H_2 deals better with the trial's uncertain termination time.

Power for testing H_2 , $\rho = 0.5$

A: Pocock boundary for H_2



B: OBF boundary for H_2



We have supposed (without real loss of generality) that $\mathcal{I}_{2,k} = 2\mathcal{I}_{1,k}$.

Power is shown as a function of θ_2 for selected values of θ_1 and $\rho = 0.5$.

The value of θ_1 has a large effect on the analysis at which a test of H_2 may occur.

The Pocock boundary for H_2 deals better with the trial's uncertain termination time.

11. Further reading on GSTs and multiple hypothesis testing

Tang & Geller (*Biometrics*, 1999) Closed testing procedures for group sequential clinical trials with multiple endpoints.

One treatment vs control with multiple endpoints, or multiple treatments vs control with a single endpoint.

In the closed testing procedure, each intersection hypothesis has its own GST.

Intersection hypotheses are tested systematically, starting with the intersection of all k hypotheses, then intersections of $(k - 1)$ hypotheses, etc.

Glimm, Maurer & Bretz (*Stat. in Med.*, 2010) Hierarchical testing of multiple endpoints in group-sequential trials.

GMB consider hierarchical testing of a secondary endpoint in a group-sequential clinical trial that is mainly driven by a primary endpoint.

The “secondary” endpoint may actually be of prime interest and the primary endpoint only a surrogate to indicate when to test the secondary endpoint.

GSTs and multiple hypothesis testing: further reading

Tamhane, Mehta & Liu (*Biometrics*, 2010) Testing a primary and a secondary endpoint in a group sequential design.

TML reduce the conservatism in Tang & Geller's method for the case of known correlation, ρ , between endpoints.

For given GSTs of H_1 and H_2 and a known value of ρ , they calculate the overall type I error rate for H_2 . They then calibrate the GST for the secondary endpoint so the maximum overall type I error rate for H_2 , over all values of θ_1 , is α .

Tamhane, Wu & Mehta (*Stat. in Med.*, 2012) Adaptive extensions of a two-stage group sequential procedure for testing primary and secondary endpoints (I) unknown correlation between endpoints.

TWM obtain an upper confidence bound, r , for the correlation ρ . They proceed on the basis that $\rho \leq r$, allocating fractions of α to (i) type 1 error for testing H_2 assuming $\rho \leq r$ and (ii) the probability that $\rho > r$.

GSTs and multiple hypothesis testing: further reading

Ye, Liu & Yao (*Statist. in Med.*, 2012) A group sequential Holm procedure with multiple primary endpoints.

In applying the Holm procedure to test h multiple hypotheses, one starts by dividing the familywise type I error probability α between the h hypotheses.

If a hypothesis is rejected, its error probability is re-distributed to the others.

This process continues until no more hypotheses can be rejected.

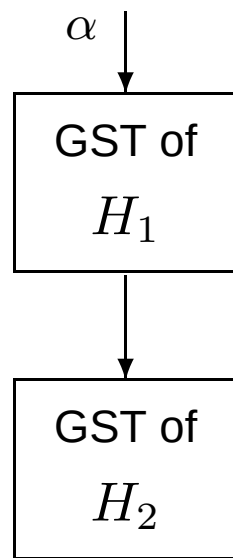
YLY follow this approach with GSTs for each hypothesis — at the appropriate collection of type I error rates.

GSTs and multiple hypothesis testing: further reading

Maurer & Bretz (*Statist. in Biopharm. Research*, 2013) Multiple testing in group sequential trials using graphical approaches.

M&B apply GSTs in multiple testing procedures with a graphical representation.

They give a thorough account of the details of this methodology, including the issue of “concordance” and when a set of GSTs has this property.



The gatekeeping procedure in which each “test” is a group sequential test is a simple example of the approach that Maurer and Bretz describe.

Complex examples start with a more elaborate graph but, again, the key idea is to implement each “test”, at each particular α -level, as a group sequential test.

Recapitulation: GSTs and multiple hypothesis testing

- There are methods available to test multiple hypotheses in a group sequential design AND control the overall type I error probability.
- Closed testing procedures encompass a variety of methods.

Graphical representations (*Statistics in Medicine* papers, 2009) can help investigators to select — and understand — an appropriate procedure.

There are many options to choose from. A suitable choice will depend on the importance to investigators of rejecting each null hypothesis and the likelihood of each null hypothesis being true or false.

- When testing multiple hypotheses in a group sequential trial design, the key point is to use GSTs as the “testing rules” in the multiple testing scheme: if this is not done correctly, FWER may be too high.
- In the above, we have assumed that one positive result may be a satisfactory conclusion. If a positive finding is required on each of several endpoints, different methods are called for.

12. GSTs and co-primary endpoints

Earlier, we mentioned an example of a trial comparing a new treatment against control with respect to two endpoints,

Endpoint 1: Core MACE (*Major Adverse Cardiac Event* — CV-related death, nonfatal stroke, or nonfatal myocardial infarction)

Endpoint 2: Expanded MACE (Core MACE plus hospitalization for unstable angina or coronary revascularization).

One possibility is that approval for the new treatment could be sought based on a positive outcome on at least one endpoint.

In this case, the previously described methods are appropriate.

Suppose instead that a positive outcome is required on *both* endpoints in order for a New Drug Application to be possible.

What are the multiple testing implications?

What can a group sequential design offer in this case?

Co-primary endpoints

Suppose it is required to show a new treatment is effective on both endpoints.

Denote the treatment effects on the two endpoints by θ_1 and θ_2 .

We wish to demonstrate that $\theta_1 > 0$ and $\theta_2 > 0$.

Formally we test the null hypothesis

$$H_0: \theta_1 \leq 0 \text{ or } \theta_2 \leq 0$$

against the alternative

$$H_A: \theta_1 > 0 \text{ and } \theta_2 > 0.$$

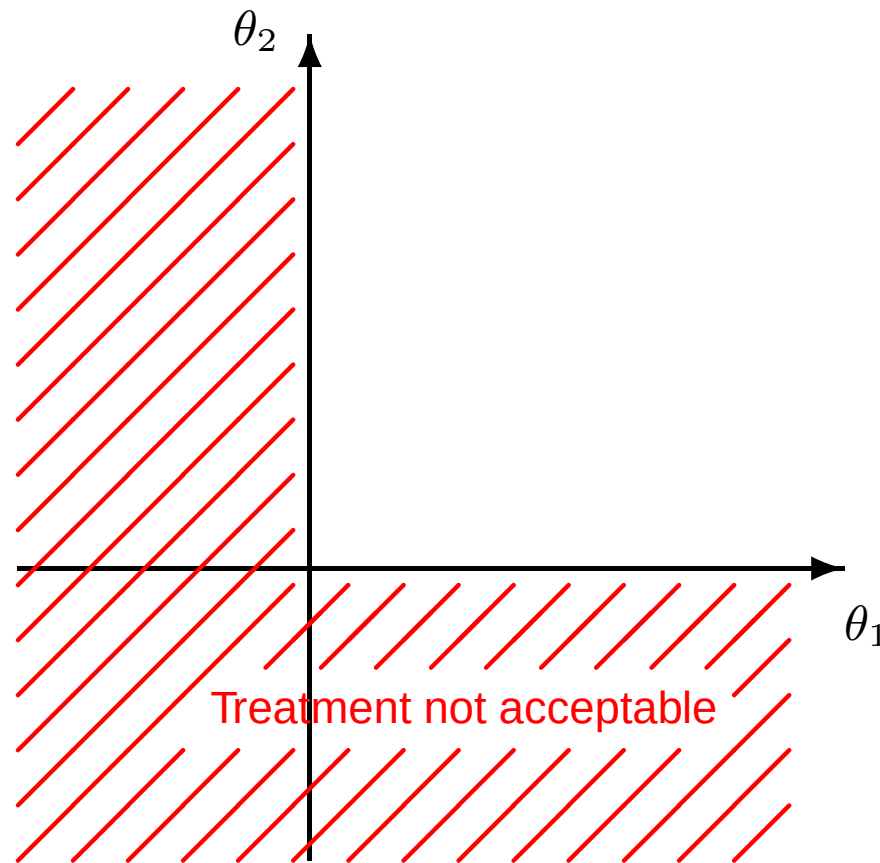
A type I error occurs if the new treatment is claimed to be effective,

i.e., if both $H_1: \theta_1 \leq 0$ and $H_1: \theta_2 \leq 0$ are rejected,

when *either* $\theta_1 \leq 0$ or $\theta_2 \leq 0$.

Co-primary endpoints

The type I error probability must be controlled over all values (θ_1, θ_2) in the null hypothesis $H_0: \theta_1 \leq 0$ or $\theta_2 \leq 0$, as shown below.



The type I error is largest at $(0, \infty)$ or $(\infty, 0)$.

Co-primary endpoints

Suppose a clinical trial is conducted in the hope of showing a treatment effect on both of two co-primary endpoints.

The trial will test

$$H_0: \theta_1 \leq 0 \text{ or } \theta_2 \leq 0$$

against the alternative

$$H_A: \theta_1 > 0 \text{ and } \theta_2 > 0.$$

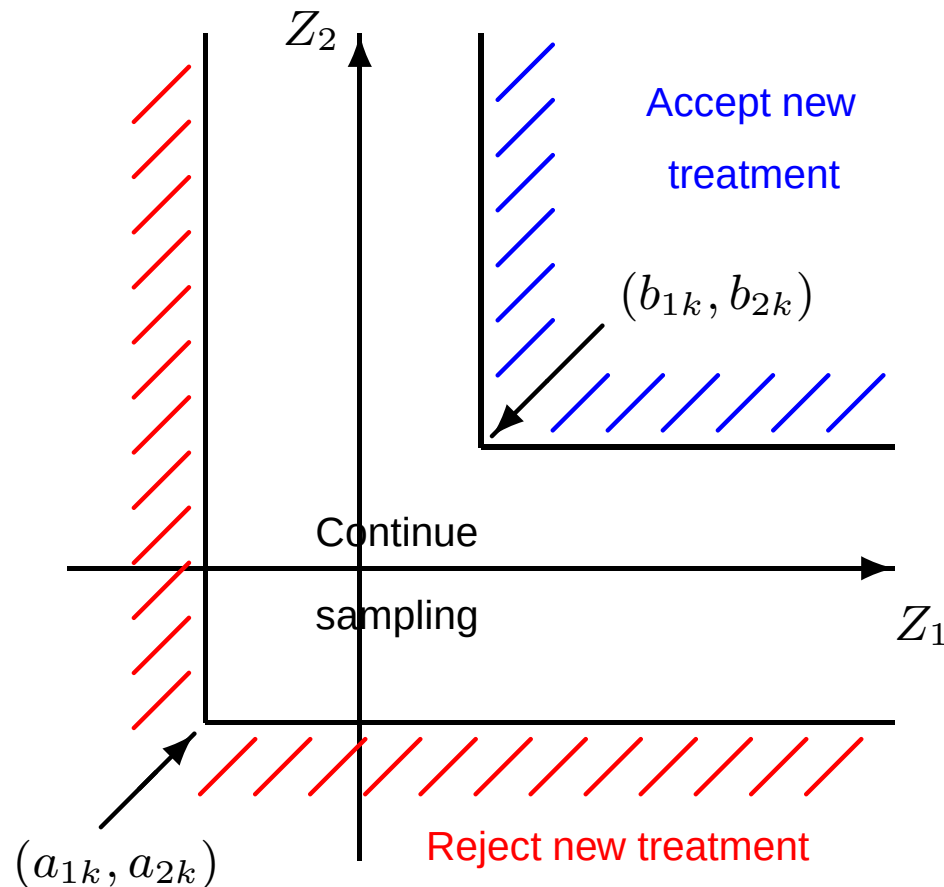
A group sequential design is possible — but this should only stop early for a positive outcome when there is evidence of a treatment effect for both endpoints.

Jennison & Turnbull (*Biometrics*, 1993) proposed such a group sequential design for a trial with efficacy and safety endpoints. (They used a non-inferiority criterion for safety, and so had $\theta_2 \leq -\delta$ rather than $\theta_2 \leq 0$ in their null hypothesis.)

Co-primary endpoints

Jennison & Turnbull's (1993) group sequential designs for a bivariate response have L-shaped boundaries at each analysis k .

The design is set up to achieve power at a specific pair of positive treatment effects.



Session 3. Adaptive clinical trial designs I

- 13. Motivation for adaptive designs
- 14. Combination tests
- 15. Sample size re-estimation for a response variance
- 16. Sample size re-estimation for the estimated treatment effect
- 17. Adaptive methods for a change of null hypothesis
- 18. Enrichment designs

13. Motivation for adaptation in clinical trials

Wall Street Journal, July 2006:

FDA Signals it's Open to Drug Trials that Shift Midcourse

Adaptive designs may allow trials to be adjusted based on early results:

- Route a larger share of patients to the treatment that seems to work best
- Drop treatments that don't seem to be effective
- Add more of the type of patients . . . reacting best to a particular treatment
- Merge two different phases of drug development into one trial

With views from:

Bob O'Neill , FDA

Michael Krams, Wyeth

Paul Gallo, Novartis

Don Berry, M. D. Anderson Cancer Center

Tom Fleming, Univ. Washington

Bruce Turnbull, Cornell University

Adaptation in clinical trials

Adaptation to external factors

Changes in the clinical setting or economic background

Following withdrawal of a competing treatment, a smaller treatment effect is now of clinical interest.

An improved financial position means sponsors can invest more in this trial.

Adaptation to internal factors

Nuisance parameters affecting sample size

In-study estimates of sample variance indicate a greater sample size is needed to achieve the intended power.

Overall failure rates in a survival study are low: higher accrual and longer follow-up are required.

Adaptation in clinical trials

Adaptation to internal factors . . .

Safety outcomes

Higher than expected toxicity implies treatment dosage should be reduced.

A lower rate of adverse events in the experimental treatment suggests it will suffice to demonstrate non-inferiority, rather than superiority.

Sub-group analyses

Treatment benefits a particular sub-group: investigators wish to re-define the target population.

Change of endpoint

An alternative endpoint provides better discrimination between treatment groups: investigators wish to re-define the primary endpoint.

Adaptation in clinical trials

Adaptation to internal factors . . .

Response on primary endpoint

Results on the new treatment are good and it is desirable to reach a conclusion as rapidly as possible.

Responses on the new treatment are not so good: investigators wish to increase sample size for power at a lower effect size than originally planned.

Response-dependent treatment allocation

Interim data suggest one treatment arm could be superior but results are not yet statistically significant. To improve treatment of patients in the trial, weight random allocation in favor of the currently superior treatment arm.

Adaptation in clinical trials

A trial with multiple treatments or dose levels

Eliminate weaker treatments as the study progresses.

Using a dose-response model, optimize treatment allocation in order to learn most efficiently about the best choice of dose level.

Combining Phase IIb (dose finding) and Phase III (confirmatory trial)

Select the best dose level in Phase IIb.

Proceed directly to Phase III and test the treatment at this dose level, eliminating “white space” between phases.

Combine Phase IIb and Phase III data in the final statistical analysis.

Possibly use a more rapidly available surrogate endpoint in Phase IIb.

Optimize the allocation of resource between Phase IIb and Phase III.

FDA (draft) Guidance for Industry, February 2010

Adaptive Design Clinical Trials for Drugs and Biologics

The message from the FDA was largely positive.

The document indicates that all adaptations should be pre-specified and access to interim data should be strictly controlled.

This is summarized in Section XI:

“ . . . there should be comprehensive and prospective, written standard operating procedures (SOPs) that define who will implement the interim analysis and adaptation plan, and all monitoring and related procedures for accomplishing the implementation, providing for the strict control of access to unblinded data (see the DMC guidance)

. . . It is likely that the measures defined by the SOPs will be related to the type of adaptation and the potential for impairing study integrity.”

FDA: Adaptive Design Clinical Trials for Drugs and Biologics

This guidance distinguishes between ***Generally well-understood adaptive designs with valid approaches to implementation***, including:

- V B.) Adaptations to maintain power based on blinded interim analyses,
- V D.) Group sequential tests using unblinded data

and ***Adaptive study designs whose properties are less well understood***, which include:

- VI A.) Adaptations for dose selection studies,
- VI B.) Adaptive randomization based on . . . treatment group responses,
- VI C.) Adaptation of sample size based on interim-effect size estimates,
- VI D.) Adaptation of patient population based on treatment-effect estimates,
- VI E.) Endpoint selection based on interim estimate of treatment effect.

The FDA's next (non-DRAFT?) statement on this topic is still awaited.

FDA (draft) Guidance for Industry, May 2015

Adaptive Designs for Medical Device Clinical Studies

The FDA's recent guidance for Studies of Medical Devices is similar in spirit to that given for Adaptive Trials for Drugs and Biologics.

The phrase “adaptive design” includes group sequential designs with early stopping for efficacy or futility.

The document stresses the importance of:

- The benefits of adaptive designs as a “learning paradigm” and their use to address a small number of uncertainties while maintaining study integrity,

- Pre-planning and the use of pre-specified adaptation rules,

- Protection of the type I error rate,

- Firewalls to prevent information leakage,

- Assessment of the benefits of adaptive design and weighing of these against more complex logistical requirements.

A key property for adaptive and flexible procedures

Protecting the type I error probability

The importance of avoiding inflation of the type I error rate in an adaptive or flexible design cannot be over-emphasized.

- ICH E9 (p. 25)

“The procedures selected should always ensure that the overall probability of type I error is controlled.”

- PhRMA White paper (*J. Biopharmaceutical Statistics*, 2006)

“The key issue in most contexts is preservation of the Type I error rate.”

- Pocock & Hughes (*Controlled Clinical Trials*, 1989)

“Control of Type I error is a vital aid to prevent a flood of false positives into the medical literature.”

For a complex procedure, it may not be straightforward to demonstrate control of the type I error rate over the whole of a multi-dimensional null hypothesis.

14. Combination tests

Suppose we run a clinical trial adaptively in two stages:

The design of Stage 1 is set, then this part of the trial is conducted.

Results from Stage 1 are analysed.

External information can be brought in if appropriate.

Stage 2 design is set, informed by Stage 1 results and external information.

Stage 2 is conducted.

Results from Stage 2 are analysed.

We could continue to further stages, but suppose for now there are just two stages.

How can we test a null hypothesis with proper protection of the type I error rate?

Combination tests

Before the trial commences, define the null hypothesis.

Let θ denote the treatment effect vs control for a specified form of the treatment, patient population, and endpoint.

We shall test $H_0: \theta \leq 0$ against $\theta > 0$, with type I error probability α at $\theta = 0$.

Define one-sided P-values P_1 and P_2 from hypothesis tests of H_0 based on Stage 1 and Stage 2 data, respectively.

Under $\theta = 0$

$$P_1 \sim U(0, 1),$$

Conditionally on all Stage 1 data and the Stage 2 design, $P_2 \sim U(0, 1)$.

Hence, P_1 and P_2 are independent $U(0, 1)$ variates.

Bauer & Köhne's inverse χ^2 combination test

Reference: Bauer & Köhne (*Biometrics*, 1994).

Initial design

Stipulate that Bauer & Köhne's inverse χ^2 combination test will be used.

Define the null hypothesis, H_0 .

Design Stage 1, fixing sample size and test statistic for this stage.

Stage 1

Observe the one-sided P-value, P_1 , based on Stage 1 data.

Design Stage 2 in the light of Stage 1 data.

Stage 2

Observe the one-sided P-value, P_2 , based on Stage 2 data alone.

Key point: Under H_0 , $P_1 \sim U(0, 1)$, $P_2 \sim U(0, 1)$, P_2 is independent of P_1 .

Bauer & Köhne's inverse χ^2 combination test

Bauer & Köhne's test rejects H_0 for low values of $P_1 P_2$.

If $P \sim U(0, 1)$, then

$$-\ln(P) \sim \text{Exp}(1) = \frac{1}{2} \chi_2^2.$$

Thus, under H_0 ,

$$-\ln(P_1 P_2) \sim \frac{1}{2} \chi_4^2.$$

Combining the two P-values in an overall test, we reject H_0 if

$$-\ln(P_1 P_2) > \frac{1}{2} \chi_{4, 1-\alpha}^2.$$

If $\theta < 0$, then P_1 and P_2 are stochastically larger than $U(0, 1)$ random variables and the type I error rate is less than α .

This χ^2 test was originally proposed for combining results of several studies by R. A. Fisher (1932) *Statistical Methods for Research Workers*.

The inverse normal combination test

Initial design

Stipulate the **inverse normal test** will be used to test the null hypothesis, H_0 .

For a two-stage design, specify weights w_1 and w_2 where $w_1^2 + w_2^2 = 1$.

Design Stage 1, fixing sample size and test statistic for this stage.

Stage 1

Observe the one-sided P-value, P_1 , based on Stage 1 data.

Compute $Z_1 = \Phi^{-1}(1 - P_1)$.

Design Stage 2 in the light of Stage 1 data.

Stage 2

Observe the one-sided P-value, P_2 , based on Stage 2 data alone.

Compute $Z_2 = \Phi^{-1}(1 - P_2)$.

Note: Here, Z_2 is the Z -statistic based on data in Stage 2 only.

Key: Under H_0 , $Z_1 \sim N(0, 1)$, $Z_2 \sim N(0, 1)$, Z_1 and Z_2 are independent.

The inverse normal combination test

Overall test

Under H_0 , Z_1 and Z_2 are independent $N(0, 1)$, so

$$w_1 Z_1 + w_2 Z_2 \sim N(0, 1).$$

Hence, for an overall one-sided test with type I error rate α , we reject H_0 if

$$w_1 Z_1 + w_2 Z_2 > \Phi^{-1}(1 - \alpha).$$

If $\theta < 0$, then Z_1 and Z_2 are stochastically smaller than $N(0, 1)$ random variables and the type I error rate is less than α .

An attractive property

If w_1 and w_2 are proportional to the square roots of the Stage 1 and 2 sample sizes (more generally, the square roots of the information for θ), then $w_1 Z_1 + w_2 Z_2$ is the standard Z statistic based on the full data set at the end of Stage 2.

This is a function of the sufficient statistic for θ and the test based on it will have the usual efficiency properties.

15. Sample size re-estimation for a response variance

A combination test can be used to protect the type I error rate when a trial's sample size is changed.

Consider a two-treatment comparison with normal responses of unknown variance.

Observations on treatments A and B, respectively, are

$$X_{Ai} \sim N(\mu_A, \sigma^2) \quad \text{and} \quad X_{Bi} \sim N(\mu_B, \sigma^2).$$

Objective

It is desired to test $H_0: \theta = \mu_A - \mu_B \leq 0$ against $\theta > 0$ with type I error rate α .

The study should achieve power $1 - \beta$ at $\theta = \delta$.

In the case of known variance, the sample size formula

$$n = \frac{\{\Phi^{-1}(1 - \alpha) + \Phi^{-1}(1 - \beta)\}^2 2 \sigma^2}{\delta^2} \quad (1)$$

gives the required value of n , the sample size per treatment.

Sample size re-estimation for a response variance

If variance is unknown, one can follow an adaptive procedure, estimating σ^2 from early data and using this estimate to modify the initial choice of sample size.

Initial design

We specify a two-stage adaptive design, using the inverse χ^2 combination rule to test $H_0: \theta \leq 0$ against $\theta > 0$.

An initial estimate σ_0^2 is used in the sample size formula (1) to obtain a sample size of n_0 per treatment. This would give the desired power if indeed $\sigma^2 = \sigma_0^2$.

Stage 1 is planned with a sample size of $n_1 = n_0/2$ per treatment.

Stage 1

Yields estimates $\hat{\theta}_1$ and $\hat{\sigma}_1^2$, plus the t -statistic t_1 for testing H_0 vs $\theta > 0$.

We convert t_1 to a one-sided P-value, $P_1 = P_{\theta=0}\{T_{2n_1-2} > t_1\}$.

Sample size re-estimation for a response variance

Stage 1 . . .

In re-calculating sample size, the variance estimate $\hat{\sigma}_1^2$ can be substituted in the original calculation, in place of σ_0^2 .

Or, the Stage 2 sample size might be chosen to give conditional power $1 - \beta$, given P_1 , assuming $\theta = \hat{\theta}_1$ and $\sigma^2 = \hat{\sigma}_1^2$.

This defines an additional sample size of n_2 per treatment arm in Stage 2.

Stage 2

We calculate the t -statistic t_2 for testing H_0 vs $\theta > 0$ based on Stage 2 data alone, then convert t_2 to a P-value, $P_2 = P_{\theta=0}\{T_{2n_2-2} > t_2\}$.

The inverse χ^2 combination test rejects H_0 if

$$-\ln(P_1 P_2) > \frac{1}{2} \chi_{4, 1-\alpha}^2$$

and the type I error rate of this test is exactly α .

Sample size re-estimation for a response variance

The above approach adds to the variety of methods for dealing with an unknown parameter, ϕ , affecting the sample size needed to attain a specific power.

Internal pilot:

Wittes & Brittain (*Statistics in Medicine*, 1990) proposed a simple “plug in” of the current estimate $\hat{\phi}$ to update the remaining sample size.

A small inflation of the type I error rate can result from bias in the final $\hat{\phi}$.

Information monitoring:

Mehta & Tsiatis (*Drug Information Journal*, 2001) use a similar “plug in” of estimated information within an error spending group sequential design.

For normal response with unknown σ^2 , the small error rate inflation can be reduced by use of better approximations (Denne & Jennison, *Biometrika*, 2000).

Friede & Miller (*Applied Statistics*, 2012) show that sample size modification based on an unblinded estimate of σ^2 produces much less type I error rate inflation.

Sample size re-estimation for a response variance

Pre-planned combination tests:

The Bauer & Köhne combination test approach attains its type I error rate exactly.

In the case of normal response with unknown σ^2 , the P-values to be combined come from separate t -tests within each stage.

Combination tests with more than two groups:

The combination test approach can be applied in a group sequential framework using a Lehman & Wassmer design, which is constructed as follows.

With Z_i based on Stage i data, $i = 1, \dots, K$, the test statistic

$$Z(k) = (w_1 Z_1 + \dots + w_k Z_k) / (w_1^2 + \dots + w_k^2)^{1/2}$$

at analysis k is compared with critical values a_k and b_k that define a group sequential test with type I error probability α .

Remember: Here, Z_k denotes the Z -statistic based on data in group k alone.

Sample size re-estimation for a response variance

In the Lehmacher & Wassmer test statistic

$$Z(k) = (w_1 Z_1 + \dots + w_k Z_k) / (w_1^2 + \dots + w_k^2)^{1/2},$$

the Z_k are required to be independent, $N(0, 1)$ variates under H_0 .

Let t_k be the observed value of the t -statistic testing $H_0: \theta \leq 0$ vs $\theta > 0$ based on responses in group k alone. Let the degrees of freedom of this t -statistic be ν_k .

We create the P-value

$$P_k = P_{\theta=0}\{T_{\nu_k} > t_k\},$$

where T_{ν_k} is a t -variate with ν_k degrees of freedom, then define

$$Z_k = \Phi^{-1}(1 - P_k).$$

Under H_0 , $P_k \sim U(0, 1)$ and $Z_k \sim N(0, 1)$. Hence, this group sequential combination test controls the type I error rate exactly.

An example of sample size modification

Consider a clinical trial studying treatment of heart failure.

Failure rates on control and experimental treatments are denoted p_c and p_t .

Historical data indicate $p_c \approx 0.25$ and a reduction of 0.05 is hoped for.

Writing $\theta = p_c - p_t$, it is desired to test $H_0: \theta \leq 0$ against $\theta > 0$ with type I error rate $\alpha = 0.025$ and power $1 - \beta = 0.9$ if $\theta = 0.05$.

Initial design

A Bauer & Köhne two-stage design is specified.

With type I error rate and power as specified above, assuming $p_c = 0.25$, a fixed sample test needs 1461 subjects per treatment arm.

Stage 1 is planned with 730 subjects per treatment, with a view to re-assessing requirements for the remainder of the study in the light of their responses.

Sample size modification

Stage 1 with 730 subjects per treatment

Yields $\hat{p}_c = 0.253$ and $\hat{p}_t = 0.219$.

Hence $\hat{\theta} = 0.034$ with standard error 0.0222.

A test of $H_0: \theta \leq 0$ has Z -statistic $0.034/0.0222 = 1.53$ and P-value

$$P_1 = 1 - \Phi(1.53) = 0.0629.$$

The overall test will reject H_0 if $-\ln(P_1 P_2) > 0.5 \chi_{4, 0.975}^2 = 5.57$.

Since $-\ln(0.0629) = 2.77$, results thus far are promising. However, a positive outcome is by no means certain.

Suppose it is learnt that trials of competing treatments have been unsuccessful.

Based on perceived costs and benefits, it is decided to increase the second stage sample size.

Sample size is chosen to give higher probability of a positive outcome under the original alternative, $\theta = 0.05$ — and under smaller effect sizes.

Sample size modification

Planning Stage 2 sample size

p_c	p_t	θ	Stage 2 sample size	Conditional power
0.25	0.22	0.03	750	0.43
			1000	0.51
			1250	0.59
0.25	0.21	0.04	750	0.62
			1000	0.72
			1250	0.80
0.25	0.20	0.05	750	0.78
			1000	0.87
			1250	0.93

Sample size modification

Stage 2

Suppose the chosen sample size is: 1000 patients per treatment arm in Stage 2.

Further, suppose the Stage 2 data (alone) yield $\hat{p}_c = 0.251$ and $\hat{p}_t = 0.221$.

Hence, the Stage 2 data give $\hat{\theta} = 0.030$ with standard error 0.0190.

A test of $H_0: \theta \leq 0$ based on Stage 2 data has Z -statistic $0.030/0.0190 = 1.58$ and, thus, P-value

$$P_2 = 1 - \Phi(1.58) = 0.0570.$$

In the overall test,

$$-\ln(P_1 P_2) = -\ln(0.0629) - \ln(0.0570) = 2.77 + 2.87 = 5.64.$$

This is greater than $\frac{1}{2} \chi_{4, 0.975}^2 = 5.57$, so H_0 is rejected and it is concluded that the new treatment has a lower failure rate.

16. Sample size re-estimation in response to the estimated treatment effect

If sample size is modified in response to an interim estimate of the treatment effect, the resulting design has much in common with a group sequential design.

In each case there is an overall maximum possible sample size but, depending on the observed data, the actual sample size can be smaller than this.

JT have written several papers on adaptive and non-adaptive group sequential designs. (*Statistics in Medicine*, 2003 and 2006, *Biometrika*, 2006). They conclude that familiar group sequential designs can provide almost all the available efficiency gains — whereas some proposals for adaptive designs can be quite inefficient.

More recently, Mehta & Pocock (*Statistics in Medicine*, 2011) proposed their “promising zone” trial design: if interim data fall in the promising zone, additional subjects can be added to increase the (conditional) power. They particularly promote this method to handle a delayed response.

Mehta & Pocock (MP) designs

JT (*Statistics in Medicine*, 2015) have assessed Mehta & Pocock's proposals: they find MP's designs to have rather high expected sample size for their attained power — but they note there are ways in which efficiency can be improved.

Working within Mehta & Pocock's general framework, JT propose sample size rules that optimize specific sample size and power criteria.

The designs produced by JT's approach require modest increases in sample size compared to MP's designs — but these increases occur over a wider range of interim results. Also, the location of the “promising zone” (where sample size is increased) is liable to shift to less favourable interim results.

JT's developments within the MP framework lead, ultimately, to the adaptive version of Hampson and Jennison's Delayed Response GSDs — so we see a convergence of the two approaches.

If fixed group sizes are desired, Hampson and Jennison's non-adaptive DRGSTs can be used and their efficiency is close to the best fully adaptive designs.

17. Adaptive methods for a change of null hypothesis

We have seen methods for modifying sample size when H_0 remains the same.

There may also be reasons to change the null hypothesis:

Change of treatment definition,

Change of endpoint,

Restriction to a sub-group of the patient population,

Switching from a test of superiority to a test of non-inferiority.

When H_0 changes, attention focuses on the **new** null hypothesis. Care is needed to avoid **selection bias** as this hypothesis is data-generated.

Methods must protect the type I error rate when there are multiple hypotheses.

Selected references:

Bauer and Köhne (*Biometrics*, 1994), Bretz, Schmidli et al. (*Biometrical Journal*, 2006), Schmidli, Bretz et al. (*Biometrical Journal*, 2006).

Procedures for testing multiple hypotheses

The familywise error rate

Suppose there are h null hypotheses, $H_i: \theta_i \leq 0$ for $i = 1, \dots, h$.

A procedure's **familywise error rate** under a set of values $(\theta_1, \dots, \theta_h)$ is

$$P\{\text{Reject } H_i \text{ for some } i \text{ with } \theta_i \leq 0\} = P\{\text{Reject any true } H_i\}.$$

The familywise error rate is controlled **strongly** at level α if this error rate is at most α for all possible combinations of θ_i values. Then

$$P\{\text{Reject any true } H_i\} \leq \alpha \quad \text{for all } (\theta_1, \dots, \theta_h).$$

Using such a procedure, the probability of choosing to focus on a parameter θ_{i^*} and then falsely claiming significance for null hypothesis H_{i^*} is at most α .

Closed testing procedure

Marcus et al. (*Biometrika*, 1976) define a ***closed testing procedure*** which provides strong control by combining level α tests of each H_i and of intersections of these hypotheses.

We have null hypotheses $H_i, i = 1, \dots, h$.

For each subset I of $\{1, \dots, h\}$, define the intersection hypothesis

$$H_I = \cap_{i \in I} H_i.$$

Construct a level α test of each intersection hypothesis H_I , i.e., a test which rejects H_I with probability at most α whenever all hypotheses specified in H_I are true.

Closed testing procedure

The simple hypothesis $H_j: \theta_j \leq 0$ is rejected overall if, and only if, H_I is rejected for every set I containing index j .

Closed testing procedure

Proof of strong control of familywise error rate

In the closed testing procedure, overall rejection of the simple hypothesis H_j can only occur if H_I is rejected for every set I containing index j .

Let $\tilde{I}$ be the set of indices of all true hypotheses H_i .

Since $H_{\tilde{I}}$ is true, $P\{\text{Reject } H_{\tilde{I}}\} = \alpha$.

For a familywise error to be committed, $H_{\tilde{I}}$ must be rejected.

Hence, the probability of a familywise error is no greater than α .

Testing an intersection hypothesis

We have a choice of methods to test an intersection hypothesis $H_I = \cap_{i \in I} H_i$.

For each $i \in I$, let P_i denote the 1-sided P-value for testing H_i .

Suppose H_I is the intersection of m simple hypotheses and denote the P-values for the m hypotheses in increasing order by $P_{(k)}$, $k = 1, \dots, m$.

Bonferroni adjustment

The overall P-value for testing H_I is

$$P_I = m P_{(1)}.$$

Simes' method (Biometrika, 1986):

The overall P-value for H_I is

$$P_I = \min_{k=1, \dots, m} (m P_{(k)} / k).$$

Bonferroni and Simes' methods

The Bonferroni adjustment is simple, but conservative.

In the definition of Simes' P-value,

$$P_I = \min_{k=1, \dots, m} (m P_{(k)} / k),$$

the term for $k = 1$ is $mP_{(1)}$, i.e., the Bonferroni adjusted P-value.

Other low P-values can reduce the overall result, e.g., if $P_{(2)}$ is only a little higher than $P_{(1)}$ so $P_{(2)}/2 < P_{(1)}$, then this will reduce P_I .

The Simes method is valid — and still slightly conservative — when the P_i are independent or positively dependent.

Such positive dependence arises in a comparison of m treatments with a common control or in tests for a treatment effect in overlapping sub-populations.

In special cases, such as m comparisons with a control, a precise treatment of the dependence between tests is possible — as we shall see in the next method.

Dunnett's method (JASA, 1955)

Suppose m treatments are compared with a control and, for $i = 1, \dots, m$, the null hypothesis H_i states that treatment i is no better than the control.

We are to test the intersection hypothesis $H_I = \cap_{i \in I} H_i$.

Suppose responses are normal with known variance and the sample sizes on each treatment and the control are equal.

Denote the Z -statistic arising from the test of H_i by Z_i .

In the least favourable case under H_I , each treatment effect is zero, so we have $Z_i \sim N(0, 1)$, $i = 1, \dots, m$, and $\text{Cov}(Z_i, Z_{i'}) = 0.5$ for $i \neq i'$.

Let $Z^* = \max_{i \in I} Z_i$ and let z^* be the value attained by Z^* .

The P-value for testing H_I using Dunnett's test is

$$P\{\max_{i \in I} Z_i > z^*\}$$

under the above multivariate normal joint distribution for $Z_i, i \in I$.

Using combination tests in a closed testing procedure

Suppose we have null hypotheses $H_i, i = 1, \dots, h$, and wish to control the familywise type I error rate at level α using a closed testing procedure.

We need a level α test for each intersection hypothesis

$$H_I = \cap_{i \in I} H_i$$

— a simple hypothesis H_j is a special case where $I = \{j\}$.

Suppose an adaptive trial has two stages. Then, for each H_I , we can apply a combination test across the stages.

Each stage provides a P-value for H_I and we combine these by a pre-specified method, e.g., “an inverse normal combination test with equal weights”.

The definition of the P-value from the second stage must be set before the second stage commences, but it may depend on first stage responses.

18. Enrichment designs

Consider a new treatment developed for a target population, but which is expected to be particularly effective in an identified sub-population.

An “enrichment design” aims to identify differential treatment effects in patient subgroups and adapt the trial’s focus to patients for whom the benefit is greatest.

In a clinical trial with enrichment we

Start by comparing the new treatment against control in the full population.

Examine responses at an interim stage.

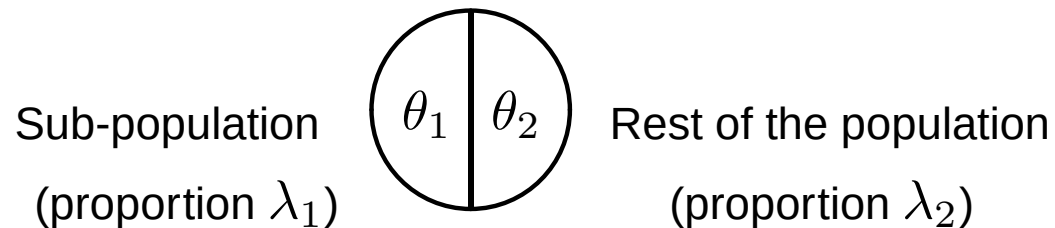
If there is no evidence of treatment effect, stop for futility.

If the new treatment appears effective in the full population, continue as before.

If the new treatment appears to benefit just the subgroup, recruit only from the subgroup and increase the numbers in this subgroup.

Results may support a licence for the full population or just the sub-population.

Enrichment designs



Denote the treatment effect:

In the sub-population by θ_1 ,

In the complement of the sub-population by θ_2 ,

Aggregated over the whole population by $\theta_3 = \lambda_1\theta_1 + \lambda_2\theta_2$.

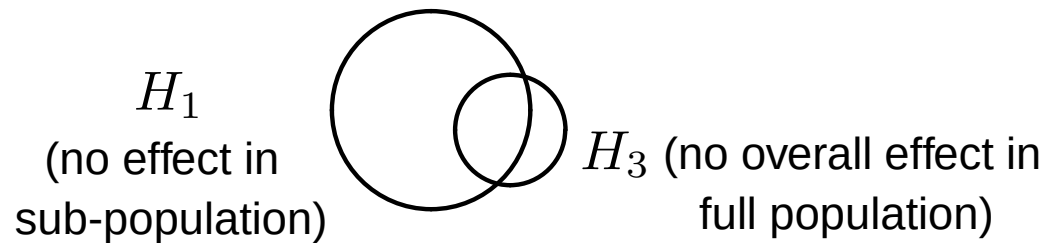
The null hypothesis for the sub-population is $H_1: \theta_1 \leq 0$.

The null hypothesis for the full target population is $H_3: \theta_3 \leq 0$.

The sets H_1 and H_3 are not necessarily nested.

All tests are against one-sided alternatives, $\theta_1 > 0$ or $\theta_3 > 0$.

Enrichment designs



Stage 1

P-value for H_1 from the sub-population data only is $P_{1,1}$.

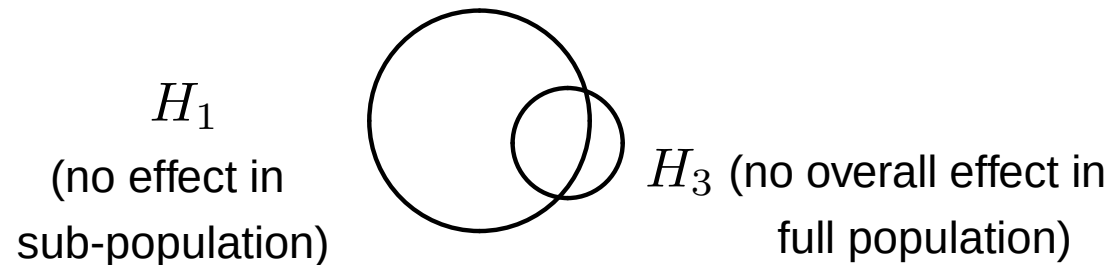
P-value for H_3 from the full data is $P_{1,3}$.

To obtain a P-value from Stage 1 data for $H_{13} = H_1 \cap H_3$, we use Simes' method (equivalently, Hochberg's adjustment, *Biometrika*, 1988):

$$P_{1,13} = \min\{2 \min(P_{1,1}, P_{1,3}), \max(P_{1,1}, P_{1,3})\}.$$

This is conservative since $P_{1,1}$ and $P_{1,3}$ are positively correlated through their overlapping sets of subjects.

Enrichment designs



Case A: Stage 2 continuing with the full population

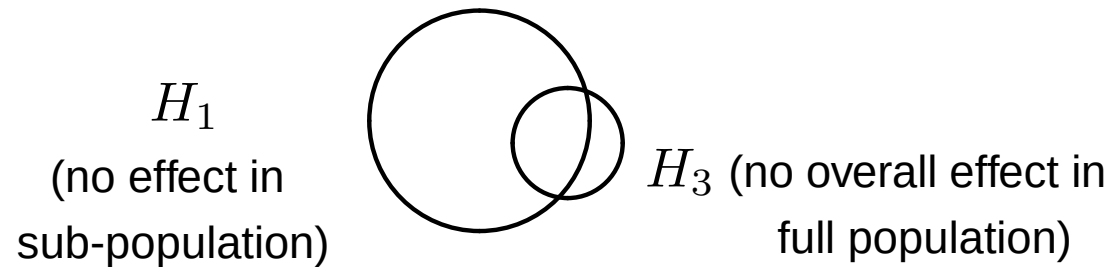
P-value for H_1 from the sub-population data only is $P_{2,1}$.

P-value for H_3 from the full Stage 2 data is $P_{2,3}$.

Simes' method gives the conservative P-value for H_{13} from the Stage 2 data

$$P_{2,13} = \min\{2 \min(P_{2,1}, P_{2,3}), \max(P_{2,1}, P_{2,3})\}.$$

Enrichment designs



Case B: Stage 2 restricted to the sub-population

P-value for H_1 from the sub-population data is $P_{2,1}$.

No P-value is available for testing H_3 in Stage 2,

P-value for H_{13} from Stage 2 data is simply $P_{2,13} = P_{2,1}$.

Enrichment designs

Combining stages

A rule must be pre-specified for combining P-values from the two stages to give tests of H_1 , H_3 and H_{13} .

For example, with P-values p_1 and p_2 from Stages 1 and 2, a weighted inverse normal rule gives

$$Z(p_1, p_2) = w_1 \Phi^{-1}(1 - p_1) + w_2 \Phi^{-1}(1 - p_2).$$

The weights w_1 and w_2 reflect the relative sample sizes planned for Stage 1 and Stage 2 and satisfy $w_1^2 + w_2^2 = 1$.

Then $Z \sim N(0, 1)$ when P_1 and P_2 are independent $U(0, 1)$, and the relevant null hypothesis is rejected at level α if $Z > \Phi^{-1}(1 - \alpha)$.

Global tests of H_1 and H_3 with familywise error rate α are formed from individual level α tests of H_1 , H_3 and H_{13} , each combining P-values from the two stages.

Enrichment designs

Global tests

Case A: Stage 2 continuing with the full population

Global test of H_1 (sub-population effect)

H_1 is rejected at global level α if both H_1 and H_{13} are rejected (in combination tests for Stage 1 and Stage 2 data).

NB This is liable to be a stronger requirement than rejection of elementary hypothesis H_1 due to allowance for testing multiple hypotheses.

Global test of H_3 (full population effect)

H_3 is rejected at global level α if both H_3 and H_{13} are rejected (typically a stronger requirement than rejection of just H_3).

Even when continuing with the full population, it is possible to reject H_1 without rejecting H_3 and so find a treatment effect in the sub-population only.

Enrichment designs

Global tests

Case B: Stage 2 restricted to the sub-population

Global test of H_1 (sub-population effect)

H_1 is rejected at global level α if both H_1 and H_{13} are rejected.

No Stage 2 P-value is available for H_3 but this does not matter as a test of H_3 is not required in this case.

Example 1 (JT, *J. Biopharmaceutical Statistics*, 2007)

We compare treatment vs control in a general population (null hypothesis H_3).

The treatment is expected to be particularly effective in men over 50 years old (null hypothesis H_1).

The familywise type I error rate is set as $\alpha = 0.025$.

Stage 1 and 2 data are combined in an inverse normal test with $w_1 = w_2 = 1/\sqrt{2}$.

Stage 1 results

Older men: $P_{1,1} = 0.02$

Full population: $P_{1,3} = 0.20$

Although the P-value for H_1 is low, it is one of two hypotheses under consideration.

Using Simes' method, the P-value for $H_1 \cap H_3$ from Stage 1 data is

$$P_{1,13} = \min\{2 \min(P_{1,1}, P_{1,3}), \max(P_{1,1}, P_{1,3})\} = 0.04.$$

We decide to continue to recruit from the full population in Stage 2.

Example 1, continued

Stage 2 results

$$\text{Older men: } P_{2,1} = 0.03$$

$$\text{Full population: } P_{2,3} = 0.10$$

$$P_{2,13} = \min\{2 \min(P_{2,1}, P_{2,3}), \max(P_{2,1}, P_{2,3})\} = 0.06.$$

Combining results from the two stages

$$Z_1 = \sqrt{0.5} \Phi^{-1}(1 - 0.02) + \sqrt{0.5} \Phi^{-1}(1 - 0.03) = 2.78, \quad P_1 = 0.003,$$

$$Z_3 = \sqrt{0.5} \Phi^{-1}(1 - 0.20) + \sqrt{0.5} \Phi^{-1}(1 - 0.10) = 1.50, \quad P_3 = 0.067,$$

$$Z_{13} = \sqrt{0.5} \Phi^{-1}(1 - 0.04) + \sqrt{0.5} \Phi^{-1}(1 - 0.06) = 2.34, \quad P_{13} = 0.010.$$

Both H_1 and H_{13} are rejected in individual tests at level $\alpha = 0.025$. Hence, **H_1 is rejected in the global test** controlling the familywise error probability at 0.025.

The adjusted P-value for H_1 is $\max\{P_1, P_{13}\} = 0.010$.

The P-value for the simple H_3 is $0.067 > \alpha$, so **H_3 is not rejected overall.**

Example 1, continued

Now consider a realisation of the trial where Stage 2 is restricted to “Older men”.

Stage 1 results: As before

Stage 2 results

Older men: $P_{2,1} = 0.03$

Full population: $P_{2,3}$ is undefined

$$P_{2,13} = P_{2,1} = 0.03.$$

Combining results from the two stages

$$Z_1 = \sqrt{0.5} \Phi^{-1}(1 - 0.02) + \sqrt{0.5} \Phi^{-1}(1 - 0.03) = 2.78, \quad P_1 = 0.003,$$

$$Z_{13} = \sqrt{0.5} \Phi^{-1}(1 - 0.04) + \sqrt{0.5} \Phi^{-1}(1 - 0.03) = 2.57, \quad P_{13} = 0.005.$$

H_1 is rejected overall as both H_1 and H_{13} are rejected at $\alpha = 0.025$.

The adjusted P-value for H_1 is $\max\{P_1, P_{13}\} = 0.005$.

No combined test for H_3 is available since we have no value for $P_{2,3}$.

Enrichment designs: More sub-populations

The above method extends naturally from the case of just one designated sub-population to more general situations.

In order to do this, the following elements must be specified:

- Sub-populations to be considered (including the full target population),

- Rule for testing an intersection hypothesis within one stage of data,

- Rule for combining results from two stages in an overall hypothesis test .

The study commences with recruitment from the full target population.

At the assigned re-design point, interim data are studied and a decision taken:

- To continue recruiting from the full population,

- Or to restrict recruitment to a specific sub-population,

- Selecting a suitable sample size in either case.

Several sub-populations

In our illustrative example, sub-populations are:

1. The entire population
2. Men only
3. Men over 50
4. Men who are smokers

Within a stage, each null hypothesis will be tested by combining P-values using Simes' method.

We combine P-values from two stages by a weighted inverse normal rule, giving

$$Z(p_1, p_2) = w_1 \Phi^{-1}(1 - p_1) + w_2 \Phi^{-1}(1 - p_2),$$

where $w_1^2 + w_2^2 = 1$. (We shall use $w_1 = w_2 = \sqrt{0.5}$.)

Several sub-populations

Let $\theta_1, \dots, \theta_4$ denote the treatment effects within specified sub-populations.

The elementary null hypotheses are $H_i: \theta_i \leq 0$ for $i = 1, \dots, 4$.

In order to protect the familywise error rate at level α , we test these hypotheses using a Closed Testing Procedure.

In Stage 1 We test each H_i against the alternative $\theta_i > 0$ using the estimate $\hat{\theta}_i$ from subjects within the relevant sub-population, giving P-value $P_{1,i}$.

In Stage 2 It may only be possible to test some of the H_i using Stage 2 data.

For example, if recruitment is restricted to “men only”, we can test H_2, H_3 and H_4 but not H_1 , since θ_1 is a weighted average of effects on both men and women.

Thus, we obtain Stage 2 P-values $P_{2,i}$ for some hypotheses H_i but not for others.

Several sub-populations

Using the Closure Testing Procedure

Formally:

In order to reject $H_{i^*}: \theta_{i^*} \leq 0$, we need to reject each intersection hypothesis H_I with $i^* \in I$ at level α , based on combined Stage 1 and Stage 2 data.

Here, $H_I = \cap_{i \in I} H_i$ states that $\theta_i \leq 0$ for all $i \in I$.

Intuitively:

Some sub-populations will have better than average results due to random variation in patient responses.

In order to avoid an inflated type I error rate, the procedure must make allowance for the multiplicity of null hypotheses being tested.

Under a global null hypothesis, the sub-population with the most favorable results should be viewed as the *best out of n comparisons of an ineffective treatment with the control*, rather than typical results for a single, pre-specified comparison.

Several sub-populations

Testing an intersection hypothesis $H_I: \theta_i \leq 0$ for all $i \in I$

- (a) We need to test an intersection hypothesis.
- (b) We need to combine data from two stages.

Take problem (b) first

A weighted inverse normal combination test is specified.

Letting $P_{1,I}$ and $P_{2,I}$ denote P-values for testing H_I from Stages 1 and 2, we calculate

$$Z(P_{1,I}, P_{2,I}) = w_1 \Phi^{-1}(1 - P_{1,I}) + w_2 \Phi^{-1}(1 - P_{2,I}),$$

using the pre-specified w_1 and w_2 . Then, we reject H_I if

$$Z(P_{1,I}, P_{2,I}) > \Phi^{-1}(1 - \alpha).$$

Testing an intersection hypothesis

(a) Testing H_I in Stage 1

Suppose m hypotheses are involved in $H_I = \cap_{i \in I} H_i$.

We calculate a P-value $P_{1,i}$ from Stage 1 data for each $H_i \in H_I$ and apply Simes' method to test the intersection hypothesis H_I .

Let $P_{1,(i)}$, $i = 1, \dots, m$, denote the m P-values in increasing order.

Then the P-value for testing H_I is

$$P_{1,I} = \min_{k=1, \dots, m} (m P_{1,(k)} / k).$$

(b) Testing H_I in Stage 2

Applying the same procedure to the Stage 2 P-values $P_{2,i}$ gives the Stage 2 P-value $P_{2,I}$ for H_I .

If some sub-populations are dropped in Stage 2, consider only those H_i involved in H_I for which a P-value $P_{2,i}$ is available, reducing the value of m accordingly.

Several sub-populations

Simes' test: Example

Suppose $P_{1,1} = 0.2$, $P_{1,2} = 0.04$, $P_{1,3} = 0.05$, $P_{1,4} = 0.03$.

In conducting a global test of H_{i^*} , we consider all sets I containing I^* .

Thus for $i^* = 4$, we need P-values for

$$I = \{4\}, \quad I = \{1, 4\}, \quad I = \{2, 4\}, \quad I = \{3, 4\}, \quad I = \{1, 2, 4\},$$

$$I = \{1, 3, 4\}, \quad I = \{2, 3, 4\}, \quad I = \{1, 2, 3, 4\}.$$

As an example of testing H_I , take the case $I = \{1, 3, 4\}$.

The ordered P-values are $P_{1,(1)} = 0.03$, $P_{1,(2)} = 0.05$, $P_{1,(3)} = 0.2$. Thus,

$$P_{1,I} = \min_{k=1,\dots,3} (3 P_{1,(k)} / k) = 3 \times 0.05 / 2 = 0.075.$$

Several sub-populations

(a) Testing H_I in Stage 2

In Stage 2, we have P-values, $P_{2,i}$, for some of the H_i : $\theta_i \leq 0$, depending on the section of the population from which recruitment took place in this stage.

Let $\tilde{I}$ denote the set of indices $i \in I$ for which we have a P-value $P_{2,i}$ and suppose there are $\tilde{m}$ such indices.

We can apply Simes' method on the reduced set $\tilde{I}$, as long as it is non-empty, yielding the P-value for testing H_I

$$P_{2,I} = \min_{k=1,\dots,\tilde{m}} (\tilde{m} P_{2,(k)} / k),$$

where $P_{2,(k)}$, $k = 1, \dots, \tilde{m}$, are the $\tilde{m}$ available P-values in increasing order.

(If the set $\tilde{I}$ is empty, its P-value will not be needed.)

Several sub-populations

Restrictions on sub-populations that can be tested in Stage 2

If the study continues to recruit from the full population in Stage 2, a P-value can be calculated for each sub-population, so all combination tests are feasible.

If recruitment is restricted, elementary tests are only possible for sub-populations which are contained completely in the new recruitment pool.

If, in our example, recruitment is restricted to “Men only”, the sub-populations “Men over 50” and “Men who are smokers” are still sampled fully so we can test H_3 and H_4 as well as H_2 . We cannot test H_1 since θ_1 is a weighted average of effects on both men and women and Stage 2 provides no information about women.

Consequently, we can test H_2 , H_3 and H_4 at the global level. For example, in testing H_2 , the relevant sets I all contain $i = 2$, so there is at least one element in the reduced set $\tilde{I}$.

Example 2 (JT, *J. Biopharm. Statist.*, 2007)

In the preceding example, suppose we observe

Stage 1 results

Full population: $P_{1,1} = 0.20$

All men: $P_{1,2} = 0.10$

Men over 50 years: $P_{1,3} = 0.03$

Men who smoke: $P_{1,4} = 0.03$.

After restricting recruitment in Stage 2 to “Men only”, we obtain

Stage 2 results

All men: $P_{2,2} = 0.11$

Men over 50 years: $P_{2,3} = 0.08$

Men who smoke: $P_{2,4} = 0.03$.

Example 2, continued

At the end of Stage 2, we have the P-values:

	Stage 1	Stage 2
H_1 : Entire population	0.20	—
H_2 : All men	0.10	0.11
H_3 : Men over 50 years	0.03	0.08
H_4 : Men who smoke	0.03	0.03

We wish to combine P-values over stages to test H_2 , H_3 and H_4 .

We use a closed testing procedure, protecting familywise error rate for the four original hypotheses (including H_1) at level $\alpha = 0.025$.

Within a stage, we combine P-values for an intersection hypothesis by Simes' test.

P-values are combined over stages by a weighted inverse normal test with equal weights, $w_1 = w_2 = \sqrt{0.5}$.

Tests of intersection hypotheses

	P-values		Combined	
	Stage 1	Stage 2	Z	P
H_1	.20	—	—	—
H_2	.10	.11	1.77	.038
H_3	.03	.08	2.32	.010
H_4	.03	.03	2.66	.004
H_{12}	.20	.11*	1.46	.072
H_{13}	.06	.08*	2.09	.018
H_{14}	.06	.03*	2.43	.008
H_{23}	.06	.11	1.97	.025
H_{24}	.06	.06	2.20	.014
H_{34}	.03	.06	2.43	.008
H_{123}	.09	.11*	1.82	.035
H_{124}	.09	.06*	2.05	.020
H_{134}	.045	.06*	2.30	.011
H_{234}	.045	.09	2.15	.016
H_{1234}	.06	.09*	2.05	.020

* Value for $P_{2, \{1\} \cup I}$ in Stage 2 equals $P_{2, I}$ for $I \subseteq \{2, 3, 4\}$.

Adjusted combined P-values

In order to reject H_i at global level $\alpha = 0.025$, each intersection hypothesis H_I with $i \in I$ must be rejected at this level.

The adjusted P-value $\tilde{P}_i$ for testing H_i with protection of familywise error rate is the maximum P-value (combined over Stages 1 and 2) for all H_I with $i \in I$.

$$\begin{aligned} H_2: \text{ All men} \quad \tilde{P}_2 &= \max\{P_2, P_{12}, P_{23}, P_{24}, P_{123}, \\ &\quad P_{124}, P_{234}, P_{1234}\} \\ &= 0.072 \end{aligned}$$

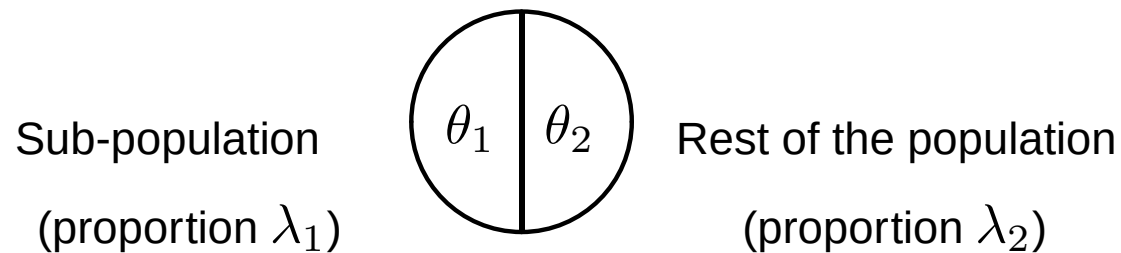
$$H_3: \text{ Men over 50 years} \quad \tilde{P}_3 = \max\{P_3, \dots, P_{1234}\} = 0.035$$

$$H_4: \text{ Men who smoke} \quad \tilde{P}_4 = \max\{P_4, \dots, P_{1234}\} = 0.020$$

We can reject H_4 at global significance level $\alpha = 0.025$ and quote an adjusted P-value of $\tilde{P}_4 = 0.020$.

Assessing the benefits of enrichment designs

Consider our original problem where there is special interest in one sub-population. A trial is to investigate whether a new treatment is beneficial to the full population or, failing that, to the sub-population.



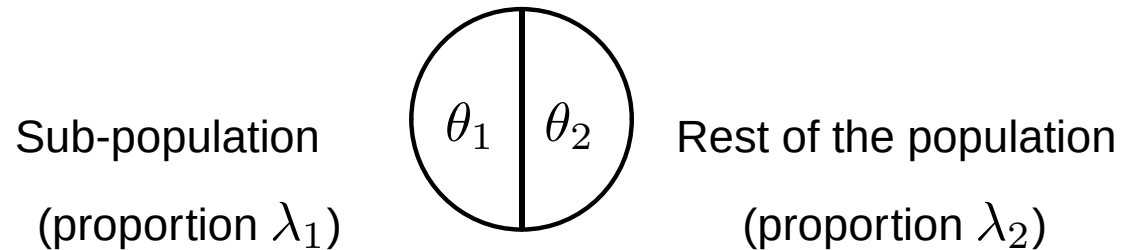
The treatment effect is θ_1 in the sub-population, θ_2 in its complement, and the average effect in the full population is $\theta_3 = \lambda_1\theta_1 + \lambda_2\theta_2$.

We wish to test:

The null hypothesis for the full population, $H_3: \theta_3 \leq 0$ vs $\theta_3 > 0$,

The null hypothesis for the sub-population, $H_1: \theta_1 \leq 0$ vs $\theta_1 > 0$.

The benefits of enrichment



First, consider a design testing just for a whole population effect,

$$\theta_3 = \lambda_1 \theta_1 + \lambda_2 \theta_2.$$

The design has two analyses and one-sided type I error probability 0.025.

Sample size is set to achieve power 0.9 at $\theta_3 = 20$.

Data in each stage are summarised by a Z -value:

	<i>Stage 1</i>	<i>Stage 2</i>	<i>Overall</i>
$H_3: \theta_3 \leq 0$	$Z_{1,3}$	$Z_{2,3}$	$Z_3 = \frac{1}{\sqrt{2}} Z_{1,3} + \frac{1}{\sqrt{2}} Z_{2,3}$

The benefits of enrichment

Two stage design testing for a whole population effect, θ_3 .

	Stage 1	Stage 2	Overall
$H_3: \theta_3 \leq 0$	$Z_{1,3}$	$Z_{2,3}$	$Z_3 = \frac{1}{\sqrt{2}} Z_{1,3} + \frac{1}{\sqrt{2}} Z_{2,3}$

Decision rules:

If $Z_{1,3} < 0$	Stop at Stage 1, Accept H_3
If $Z_{1,3} \geq 0$	Continue to Stage 2, then
If $Z_3 < 1.96$	Accept H_3
If $Z_3 \geq 1.96$	Reject H_3

As the futility boundary is treated as non-binding, this design is slightly conservative.

The benefits of enrichment

Assume the sub-population comprises half the total population, so $\lambda_1 = \lambda_2 = 0.5$.

Properties of design for the whole population effect, θ_3 :

θ_1	θ_2	θ_3	<i>Power for</i> <i>$H_3: \theta_3 \leq 0$</i>
20	20	20	0.90
10	10	10	0.37
20	0	10	0.37

Is it feasible to identify at Stage 1 that θ_3 is low but θ_1 may be higher, so one might switch resources to test the sub-population?

The benefits of enrichment

We wish to be able to consider two null hypotheses:

$H_3: \theta_3 \leq 0$ Treatment is not effective in the whole population,

$H_1: \theta_1 \leq 0$ Treatment is not effective in the sub-population.

Since $\theta_3 = 0.5 \theta_1 + 0.5 \theta_2$, either of H_1 and H_3 may be true on its own.

In applying a ***closed testing procedure***, we also test the intersection hypothesis

$$H_{13}: \theta_1 \leq 0 \text{ and } \theta_3 \leq 0.$$

Then to reject H_1 overall, while protecting the family-wise type I error rate, we need to reject both H_1 and H_{13} in individual tests at significance level α .

Similarly, we can reject H_3 overall if both H_3 and H_{13} are rejected in level α tests.

An adaptive design

At Stage 1, if $\hat{\theta}_3 < 0$, stop to accept $H_3: \theta_3 \leq 0$.

If $\hat{\theta}_3 > 0$ and the trial continues:

If $\hat{\theta}_2 < 0$ and $\hat{\theta}_1 > \hat{\theta}_2 + 8$ Restrict to sub-population 1 and test H_1 only,
needing to reject H_1 and H_{13} .

Else, Continue with full population and test H_3 ,
needing to reject H_3 and H_{13} .

The same *total* sample size for Stage 2 is retained in both cases, increasing the numbers for the sub-population when enrichment occurs.

An adaptive design

Each null hypothesis, H_1 , H_3 and H_{13} , is tested in a 2-stage group sequential test.

Let H_i be one of these hypotheses and Z_1 and Z_2 the Z -statistics for H_i based on data collected in Stages 1 and 2.

The 2-stage group sequential test rejects H_i after Stage 2 if

$$\frac{1}{\sqrt{2}}Z_1 + \frac{1}{\sqrt{2}}Z_2 \geq 1.96.$$

Defining standardised test statistics

We need to specify standardised statistics for testing each of the hypotheses H_1 , H_3 and H_{13} in Stages 1 and 2.

For testing H_1 , we can define $Z_{1,1}$ and $Z_{2,1}$ from estimates $\hat{\theta}_{1,1}$ and $\hat{\theta}_{2,1}$ of θ_1 based on the data collected in each stage.

Similarly, for H_3 , we define $Z_{1,3}$ and $Z_{2,3}$ from estimates $\hat{\theta}_{1,3}$ and $\hat{\theta}_{2,3}$ of θ_3 based on the data from each stage.

An adaptive design

Testing H_{13} in Stage 1

If we wish to maximise power to find an effect in the full population, we can set $Z_{1,13}$, the test statistic for H_{13} , to be the $Z_{1,3}$ used to test H_3 .

Alternatively, we can set $Z_{1,13}$ to be the $Z_{1,1}$ for testing H_1 . This will improve power to find an effect in the sub-population if only this effect is present.

We have found that a good compromise is to take an average of these statistics,

$$Z_{1,13}^* = (Z_{1,3} + Z_{1,1})/\sqrt{(2 + \sqrt{2})},$$

where the factor $1/\sqrt{(2 + \sqrt{2})}$ is chosen to make the variance of $Z_{1,13}^*$ equal to 1 — and this avoids the conservatism of Simes' test.

Testing H_{13} in Stage 2

We set $Z_{2,13} = Z_{2,3}$ if the full population continues into Stage 2.

If the trial is restricted to the sub-population, we set $Z_{2,13} = Z_{2,1}$.

An adaptive design: Summary of Z -statistics

When continuing with the full population:

	Stage 1	Stage 2
H_1	$Z_{1,1}$	$Z_{2,1}$
H_3	$Z_{1,3}$	$Z_{2,3}$
H_{13}	$Z_{1,13}^*$	$Z_{2,3}$

When switching to the sub-population:

	Stage 1	Stage 2
H_1	$Z_{1,1}$	$Z_{2,1}$
H_{13}	$Z_{1,13}^*$	$Z_{2,1}$

Here $Z_{i,1}$ and $Z_{i,3}$ are based on $\hat{\theta}_1$ and $\hat{\theta}_3$ from responses in Stage i .

The Stage 1 statistic for testing H_{13} is $Z_{1,13}^* = (Z_{1,3} + Z_{1,1})/\sqrt{(2 + \sqrt{2})}$.

Note the common test statistic for H_{13} in Stage 1 — which must be pre-specified.

Simulation results: Power of non-adaptive and adaptive designs

				<i>Non-adaptive</i>	<i>Adaptive</i>		
	θ_1	θ_2	θ_3	<i>Full popⁿ</i>	<i>Sub-popⁿ only</i>	<i>Full popⁿ</i>	<i>Total</i>
1.	30	0	15	0.68	0.47	0.41	0.88
2.	20	0	10	0.37	0.33	0.25	0.58
3.	20	20	20	0.90	0.04	0.83	0.87
4.	20	10	15	0.68	0.15	0.57	0.72

Cases 1 & 2: Testing focuses (correctly) on H_1 , but it is still possible to find an effect (wrongly) for the full population. Overall power is increased.

Case 3: Restricting to the sub-population reduces power for finding an effect in the full population.

Case 4: Adaptation improves overall power a little.

The benefits of enrichment

In creating an enrichment design, we can favour specific goals when defining:

- The rule for switching to the sub-population,

- The test of the intersection hypothesis.

However, we cannot eliminate the probability of making an error in these decisions.

This is to be expected. The standard error of the interim estimates $\hat{\theta}_1$ and $\hat{\theta}_2$ is 12.3 — much higher than the differences between θ_1 and θ_2 that interest us.

Although restricting attention to a sub-population can help improve power, higher overall sample size is needed for accurate sub-population inference.

Conclusions: Enrichment designs

- With the advent of biomarkers and personalised medicine, there is a need for methods which allow adaptation to focus on a subgroup of patients during a clinical trial.
- Such methods are feasible using combination tests and a closed testing procedure.
- Restricting attention to a sub-population can be effective in improving power.
- However, higher overall sample size is needed for more accurate sub-population inference.
- Open problems remain regarding more complex trial scenarios and inference on termination.

Session 4. Adaptive clinical trial designs II

- 19. Seamless Phase IIb / Phase III designs
- 20. Example of a Phase IIb / Phase III trial
- 21. Efficiency gains from combining Phase IIb and Phase III
- 22. A phase III trial with treatment selection and a survival endpoint
- 23. A closed testing procedure with combination tests for survival data
- 24. Assessing the benefits of an adaptive design with treatment selection

19. Seamless Phase IIb / Phase III designs

The roles of Phase IIb and Phase III trials

Phase IIb

Several dose levels of a treatment are compared against a control, to select a dose (or variant of a treatment) and confirm there is improvement against the control.

Phase III

The trial is run as a confirmatory study to demonstrate superiority against control of the treatment selected in Phase IIb.

Stages:

Write Phase IIb protocol, seek ethical and regulatory approval, (FDA, IRBs, ...)

Run Phase IIb, analyse data, reach conclusions.

Write Phase III protocol, seek ethical and regulatory approval, (FDA, IRBs, ...)

Run Phase III, analyse data, reach final conclusion.

Planning the Phase III trial after Phase IIb is complete

Planning the Phase III trial after Phase IIb allows investigators to make use of information gained in Phase IIb.

They may decide to modify:

Treatment definition,

Target population,

Primary endpoint,

Sample size.

Positive results in Phase IIb will help recruitment for participation in Phase III.

But, planning and gaining approval for the Phase III trial can be time-consuming.

If the final outcome is positive, the sooner this conclusion is reached, the better.

Seamless Phase IIb / Phase III trials

Requirements

A single protocol for the combined Phase IIb and Phase III trials.

Rules for a committee managing the trials to follow as they:

- Decide whether to proceed to Phase III,

- Select the dose level for Phase III, considering efficacy and safety data,

- Respond to information from Phase IIb, e.g., using estimated treatment effect and/or response variance to set Phase III sample size.

NB Expect everyone else to be blinded to the Phase IIb results.

Potential benefits

- Eliminating the “white space” between phases,

- Gaining efficiency from using Phase IIb data in the Phase III analysis.

(1) Applying a closed testing procedure with combination tests

Reference: Bretz, Schmidli et al. (*Biometrical Journal*, 2006).

Let θ_i , $i = 1, \dots, k$, denote the effect size of dose level i vs the control treatment.

Define H_i : $\theta_i \leq 0$, for $i = 1, \dots, k$.

Procedure

Select dose level i^* to advance to Phase III.

Test H_{i^*} : $\theta_{i^*} \leq 0$ at significance level α , with allowance for having selected H_{i^*} from k null hypotheses based on the results for dose i^* .

Formally

We shall define a procedure controlling the **familywise error rate**.

Then, for all possible sets of treatment effects $\{\theta_i\}$

$$P\{\text{Reject any true } H_i\} \leq \alpha.$$

Applying a closed testing procedure

Phase IIb

Observe estimated treatment effects $\hat{\theta}_{1,i}$, $i = 1, \dots, k$.

Select dose level i^* to go forward to Phase III.

Dose level i^* will have a high estimate $\hat{\theta}_{1,i^*}$ and good safety outcomes.

Phase III

Test dose level i^* against control.

In order to reject $H_{i^*}: \theta_{i^*} \leq 0$, we need to reject each intersection hypothesis H_I with $i^* \in I$ at level α .

Here, $H_I = \cap_{i \in I} H_i$ states that $\theta_i \leq 0$ for all $i \in I$.

Applying a closed testing procedure

Formally:

In order to reject $H_{i^*}: \theta_{i^*} \leq 0$, we need to reject each intersection hypothesis H_I with $i^* \in I$ at level α , **based on combined Phase IIb and Phase III data**.

Intuitively:

Dose i^* is chosen for the good results observed at this dose in Phase IIb.

We must adjust for this choice in order to avoid **selection bias** when adding the Phase IIb data on dose level i^* to the final analysis after Phase III.

Under a global null hypothesis of no treatment effect at any dose, the Phase IIb data on dose i^* should be viewed as *possibly the best results out of k ineffective doses*, rather than typical results at a single, pre-specified dose.

Testing an intersection hypothesis $H_I: \theta_i \leq 0$ for all $i \in I$

- a) We need to test an intersection hypothesis.
- b) We need to combine data from two stages, Phase IIb and Phase III.

Take problem (b) first

Denote the P-value for testing H_I in Phase IIb by $P_{1,I}$.

Denote the P-value for testing H_I in Phase III by $P_{2,I}$.

An inverse χ^2 combination test rejects H_I if

$$-\log(P_{1,I} P_{2,I}) > \frac{1}{2} \chi_{4, 1-\alpha}^2.$$

An inverse normal combination test rejects H_I if

$$w_1 \Phi^{-1}(1 - P_1) + w_2 \Phi^{-1}(1 - P_2) > \Phi^{-1}(1 - \alpha).$$

where w_1 and w_2 are pre-specified and $w_1^2 + w_2^2 = 1$.

Testing an intersection hypothesis $H_I: \theta_i \leq 0$ for all $i \in I$

(a) *Testing H_I is most complex in Phase IIb*

For each $i \in I$, let $P_{1,i}$ denote the 1-sided P-value for testing $H_i: \theta_i \leq 0$.

If H_I is the intersection of m simple hypotheses, denote the P-values for the m hypotheses in increasing order by $P_{1,(j)}$, $j = 1, \dots, m$.

Using Bonferroni adjustment

The overall P-value for testing H_I is

$$P_{1,I} = m P_{1,(1)}.$$

If we use Simes' method (Biometrika, 1986):

The overall P-value for H_I is

$$P_{1,I} = \min_{j=1,\dots,m} (m P_{1,(j)} / j).$$

Remarks on Simes' method

Using Simes' method:

We are interested in testing intersection hypotheses H_I where I contains i^* , the index of the dose level selected for Phase III.

The P-value for testing H_I , where I has m elements, is

$$P_{1,I} = \min_{j=1,\dots,m} (m P_{1,(j)} / j).$$

If treatment i^* has the highest $\hat{\theta}_{1,i}$ and smallest P-value of all k treatments, we have $P_{1,(1)} = P_{1,i^*}$ in any set I containing i^* .

The term $mP_{1,(j)}/j$ with $j = 1$ is mP_{1,i^*} , which is also the Bonferroni adjusted version of P_{1,i^*} .

Simes' method "borrows strength" from other dose levels, allowing their P-values to reduce the overall result: if a second dose level performs well, $P_{1,(2)}/2$ may be smaller than P_{1,i^*} , reducing $P_{1,I}$.

Dunnett's method (JASA, 1955)

Consider $H_I = \cap_{i \in I} H_i$, the intersection of m simple hypotheses. Each H_i concerns the comparison of one dose level against the control.

Suppose responses are normal with known variance and sample sizes on each treatment and the control are equal.

Denote the Z -statistic arising from the test of H_i by Z_i .

In the case that each treatment effect is zero, the Z_i are multivariate normal, $Z_i \sim N(0, 1)$, $i = 1, \dots, m$, and $\text{Cov}(Z_i, Z_{i'}) = 0.5$ for $i \neq i'$.

Let $Z^* = \max_{i \in I} Z_i$ and let z^* be the value attained by Z^* .

The P-value for testing H_I using Dunnett's test is

$$P\{\max_{i \in I} Z_i > z^*\}$$

under the above multivariate normal joint distribution for Z_i , $i \in I$.

Testing an intersection hypothesis $H_I: \theta_i \leq 0$ for all $i \in I$

Returning to (a) now consider testing of H_I in Phase III

In order to reject $H_{i^*}: \theta_{i^*} \leq 0$ overall, we need to reject each H_I with $i^* \in I$.

Only dose level i^* is studied in Phase III, so a test of such an H_I using Phase III data is based on $\hat{\theta}_{2,i^*}$ — and there is just one such test.

Hence, all H_I of interest have a common P-value in Phase III of $P_{2,I} = P_{2,i^*}$.

With $P_{2,I} = P_{2,i^*}$, the inverse χ^2 combination test of H_I is a function of $\log(P_{1,I} P_{2,I})$, while the inverse normal combination test depends on

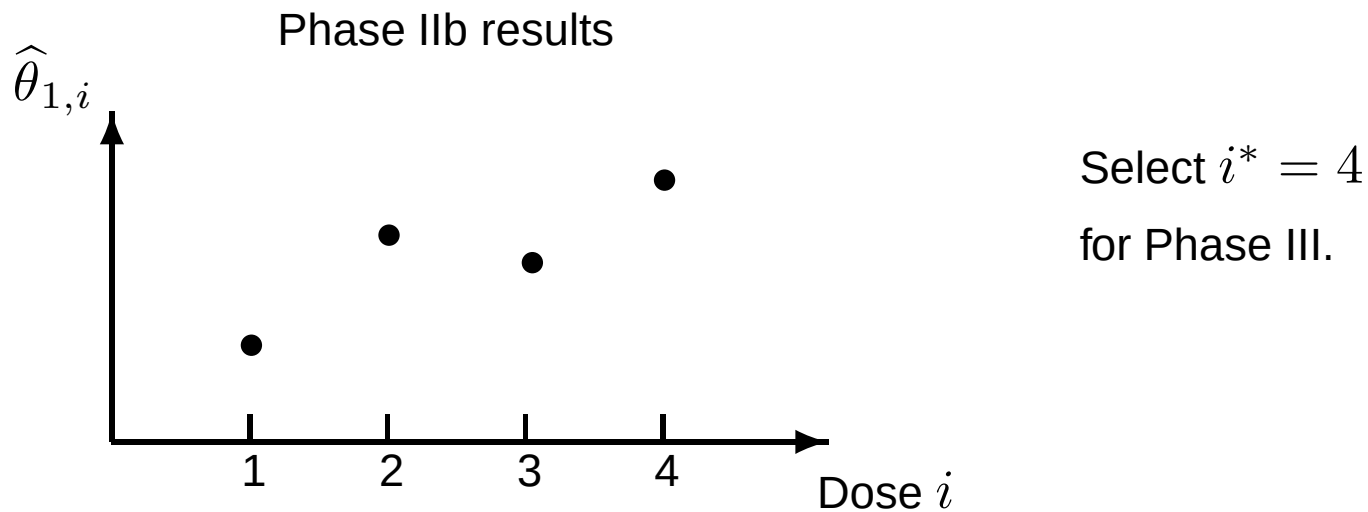
$$w_1 \Phi^{-1}(1 - P_{1,I}) + w_2 \Phi^{-1}(1 - P_{2,I}).$$

Rejection of H_{i^*} depends on the highest value of $P_{1,I}$ appearing in these formulae.

Thus, the key statistic from Phase IIb is:

$$\max_I P_{1,I} \text{ over sets } I \text{ containing } i^*.$$

Example of Simes' test



Suppose $P_{1,1} = 0.2$, $P_{1,2} = 0.04$, $P_{1,3} = 0.05$, $P_{1,4} = 0.03$.

For sets I containing $i^* = 4$, maximum $P_{1,I}$ comes from $I = \{1, 3, 4\}$.

Ordered P-values are $P_{1,(1)} = 0.03$, $P_{1,(2)} = 0.05$, $P_{1,(3)} = 0.2$:

$$P_{1,I} = \min_{j=1,\dots,3} (3 P_{1,(j)} / j) = 3 \times 0.05 / 2 = 0.075.$$

More details of Simes' test Example

Recall $P_{1,1} = 0.2$, $P_{1,2} = 0.04$, $P_{1,3} = 0.05$, $P_{1,4} = 0.03$.

Single element sets I containing $i^* = 4$

There is just one P-value $P_{1,4} = 0.03$, so $P_{1,I} = 0.03$.

Two-element sets I containing $i^* = 4$

Consider $I = \{1, 4\}$ with ordered P-values $P_{1,(1)} = 0.03$, $P_{1,(2)} = 0.2$:

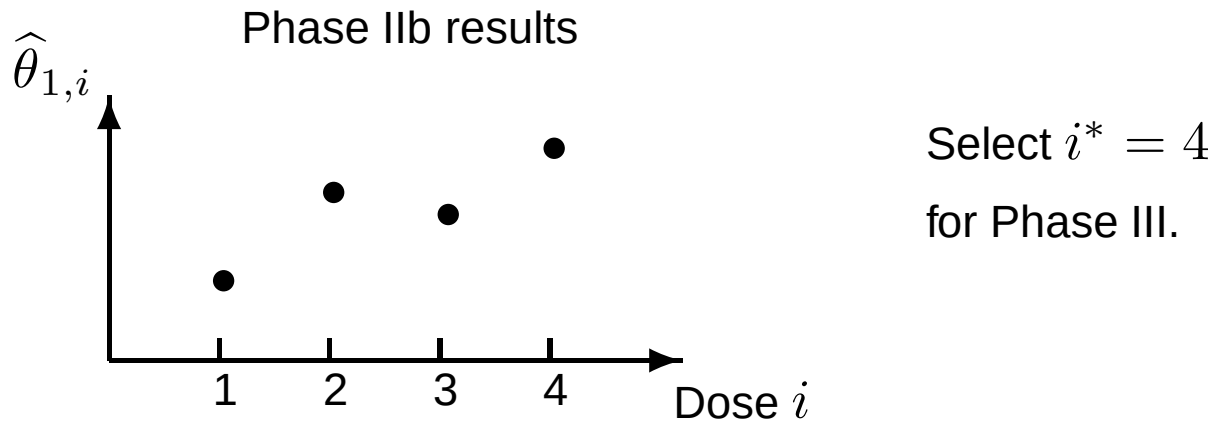
$$P_{1,I} = \min_{j=1,2} (2 P_{1,(j)} / j) = 2 \times 0.03 = 0.06.$$

Four-element sets I containing $i^* = 4$

Ordered P-values for $I = \{1, 2, 3, 4\}$ are $P_{1,(1)} = 0.03$, $P_{1,(2)} = 0.04$, $P_{1,(3)} = 0.05$, $P_{1,(4)} = 0.2$:

$$P_{1,I} = \min_{j=1,\dots,4} (4 P_{1,(j)} / j) = 4 \times 0.05 / 3 = 0.067.$$

Example of Dunnett's test



Suppose $P_{1,1} = 0.2$, $P_{1,2} = 0.04$, $P_{1,3} = 0.05$, $P_{1,4} = 0.03$.

For sets I containing $i^* = 4$, the maximum $P_{1,I}$ comes from $I = \{1, 2, 3, 4\}$.

The lowest P-value is $P_{1,4} = 0.03$, for which $\Phi^{-1}(1 - 0.03) = 1.881$.

The Dunnett test has P-value

$$P_{1,I} = \Pr\left\{\max_{i=1,\dots,4} (Z_i) > 1.881\right\} = 0.0918,$$

calculated with $Z_i \sim N(0, 1)$, $i = 1, \dots, 4$, and $\text{Cov}(Z_i, Z_{i'}) = 0.5$ for $i \neq i'$.

Summary of combination of Phase IIb and Phase III data

In Phase IIb: Select treatment i^* and carry forward $P_1^* = \max_{I:i^* \in I} P_{1,I}$.

In Phase III:

Test treatment i^* against control and find P_{2,i^*} .

We use P_{2,i^*} for all tests of intersection hypotheses involving treatment i^* .

Overall: Combine the P-values P_1^* and P_{2,i^*} to see if $H_{i^*}: \theta_{i^*} \leq 0$ is rejected.

Flexibility:

Treatment i^* can be selected for efficacy, safety, or other factors
— not necessarily the treatment with maximum $\hat{\theta}_{1,i}$.

Efficiency:

Use of Phase IIb data in the final analysis should increase power
or reduce the Phase III sample size needed for a given power.

(2) The method of Thall, Simon and Ellenberg (*Biometrika*, 1988)

Phase IIb

Take m_1 observations per treatment and control.

Denote estimated effect of treatment i against control by $\hat{\theta}_{1,i}$ and let the maximum of these be $\hat{\theta}_{1,i^*}$.

If $\hat{\theta}_{1,i^*} < C_1$, stop and accept $H_0: \theta_1 \leq 0, \dots, \theta_k \leq 0$,

if $\hat{\theta}_{1,i^*} \geq C_1$, select treatment i^* and proceed to Phase III.

Phase III

Take m_2 observations on treatment i^* and the control.

Combine data in $T_{i^*} = (m_1 \hat{\theta}_{1,i^*} + m_2 \hat{\theta}_{2,i^*}) / (m_1 + m_2)$.

If $T_{i^*} < C_2$, accept H_0 ,

if $T_{i^*} \geq C_2$, reject H_0 and conclude $\theta_{i^*} > 0$.

Thall, Simon and Ellenberg (TSE)

NB The statistic T_{i^*} is the simple estimate of treatment θ_{i^*} obtained by pooling Phase IIb and Phase III data.

Type I error and power requirements imply values for m_1 , m_2 , C_1 and C_2 .

Type I error

Treatment i^* is said to be “chosen” if

Treatment i^* is selected at the end of Phase IIb, and

H_0 is rejected in favour of $\theta_{i^*} > 0$ in the final analysis.

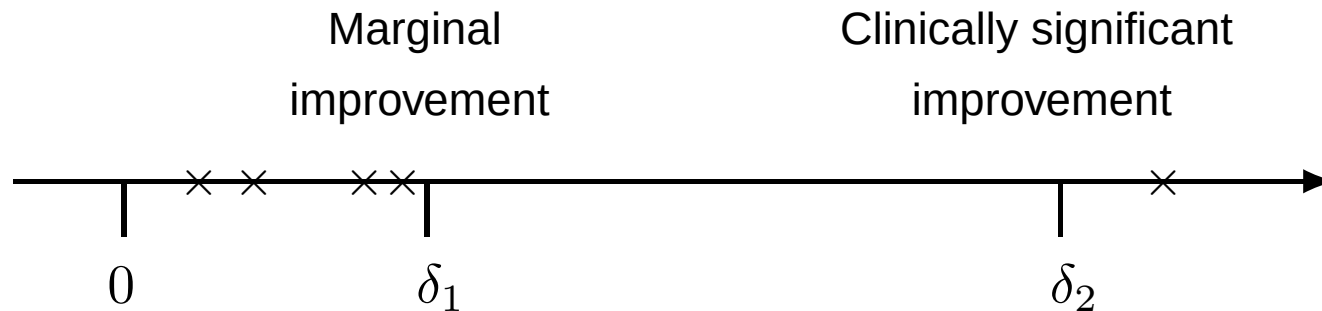
TSE define the type I error rate as the maximum value of

$$Pr\{\text{Any experimental treatment is “chosen”}\}$$

under $H_0: \theta_1 \leq 0, \dots, \theta_k \leq 0$, which occurs when $\theta_1 = \dots \theta_k = 0$.

Thall, Simon and Ellenberg

Power



Power depends on the full vector $\boldsymbol{\theta} = (\theta_1, \dots, \theta_k)$.

Any treatment with $\theta_i \geq \delta_2$ is said to be “acceptable”.

Consider cases of $\boldsymbol{\theta}$ where:

At least one treatment is acceptable,

No θ_i lies in the interval (δ_1, δ_2) .

The power function is $Pr_{\boldsymbol{\theta}} \{ \text{An acceptable choice is made} \}$, i.e., the probability an acceptable treatment is selected and the corresponding H_i is then rejected.

Thall, Simon and Ellenberg

Power

TSE show power is minimized in cases described above under the *least favourable configuration*:

$$\theta_1 = \dots = \theta_{k-1} = \delta_1 \quad \text{and} \quad \theta_k = \delta_2.$$

They call this configuration θ^* and set their power condition at $\theta = \theta^*$.

Numerical integration under H_0 and $\theta = \theta^*$ enables parameters m_1, m_2, C_1 and C_2 to be found satisfying type I error and power conditions.

Optimisation

Tests minimizing expected sample size averaged over these two cases are found by searching feasible parameter combinations.

Thall, Simon and Ellenberg

The TSE method has familywise error rate α

TSE set their type I error probability α when $\theta_1 = \dots = \theta_k = 0$.

Family wise error rate concerns the probability of “choosing” a treatment with $\theta_i \leq 0$ under any vector $\boldsymbol{\theta}$, including cases where $\theta_j > 0$ for some indices j .

Starting from the case $\theta_1 = \dots = \theta_k = 0$, one can show that moving some effects to values $\theta_i < 0$ decreases the familywise error probability.

Then, increasing other values to $\theta_j > 0$ reduces the error rate as there are fewer treatments with $\theta_i \leq 0$ and the chance of selecting one of these is decreased.

We conclude that the TSE method protects the familywise error rate at level α .

Thall, Simon and Ellenberg

The TSE method is a Closed Testing Procedure

Define a test of H_I for I containing i^* as follows:

If stopping at Phase IIb, accept H_I .

When proceeding to Phase III, define

$$\begin{aligned} T_I &= \frac{m_1 \max_{i \in I} \hat{\theta}_{1,i} + m_2 \hat{\theta}_{2,i^*}}{m_1 + m_2} \\ &= \frac{m_1 \hat{\theta}_{1,i^*} + m_2 \hat{\theta}_{2,i^*}}{m_1 + m_2} = T_{i^*} \end{aligned}$$

and reject H_I for $T_I > c_I$, accept H_I otherwise.

The value c_I for a level α test of H_I is highest when $I = \{1, \dots, k\} = I_k$, say. Rejecting H_{I_k} implies rejecting every other H_I with $i^* \in I$.

Equating C_2 and c_{I_k} , we see that “choosing” i^* is equivalent to rejecting H_{i^*} according to the Closure Principle with these tests of intersection hypotheses H_I .

Thall, Simon and Ellenberg

Remarks on the TSE method

The single summary of Phase IIb data is $\hat{\theta}_{1,i^*}$. This is combined with Phase III data in the overall estimate of θ_{i^*} using the final sufficient statistic for θ_{i^*} .

The critical value for the final test statistic allows for selecting treatment i^* as the best performer in Phase IIb.

The null distribution taken for $\hat{\theta}_{1,i^*}$ is that of the maximum of n estimated effects when the true effects are all zero.

Adding flexibility:

The TSE method can be used, as defined, when a treatment i^* is selected with $\hat{\theta}_{1,i^*}$ less than the highest $\hat{\theta}_{1,i}$. The type I error rate is then met conservatively.

Alternatively, the above definitions of T_I , etc., can be extended to create a less conservative procedure via the Closure Principle.

20. Example of a Phase IIb / Phase III trial

Reference: JT (J. Biopharmaceutical Statistics, 2007)

Scenario:

- Disease: Asthma.
- Endpoint: Asthma quality of life score (AQLS) at 6 weeks.
- Four treatment groups and a control group.
- Responses are approximately normal with s.d. $\sigma/\sqrt{2}$, where $\sigma = 5$ units.
- **Stage 1:**
 - 100 observations per group.
 - If the highest mean is less than the control, the experiment is abandoned as futile and H_0 is accepted.
 - Otherwise, the treatment with the highest mean is selected for Stage 2.
- **Stage 2:**
 - 500 observations on the control and on the treatment selected in Stage 1.

Example

You are a new statistician just hired to a small pharma company.

You are presented with data from a Phase IIb trial and a Phase III trial for asthma treatments, conducted as described above.

You are to analyze these data and, if results are positive, present these in an NDA submission.

(The statistician who designed the trials has just moved to a competitor.)

Example: The results

Phase IIb results

	Control	Trt 1	Trt 2	Trt 3	Trt 4
n	100	100	100	100	100
P (1-sided)		0.20	0.04	0.05	0.03
Z		0.84	1.75	1.64	1.88

Treatment $i^* = 4$ was selected to go forward to Phase III.

Phase III results

	Control	Trt 4
n	500	500
P (1-sided)		0.04
Z		1.75

Should treatment 4 be recommended at significance level $\alpha = 0.025$?

Example: What did the protocol or SAP specify?

You find one of the following:

1. **Conventional**

Separate Phase IIb and Phase III trials were conducted with the final decision to be made on the Phase III data alone.

2. **Bauer and Köhne**

Combination of Phase IIb and Phase III results using

- (a) inverse χ^2 or (b) weighted inverse normal combination test,
- (i) Simes' test or (ii) Dunnett's test for intersection hypotheses.

3. **Thall, Simon and Ellenberg**

The final Z -statistic is based on the simple estimate of treatment θ_{i^*} obtained by pooling Phase IIb and Phase III data — but the critical value accounts for the selection of treatment i^* .

Results: Conventional protocol

- The Phase III P-value of 0.04 is greater than 0.025.
- The null hypothesis is not rejected.
- The result of the trial is negative.

Results: Bauer and Köhne

Stage 1:

First, we test intersection hypotheses in Stage 1.

We need P_1 , the largest P-value $P_{1,I}$ for a set I containing the index of the selected treatment, $i^* = 4$.

(i) Simes' method

$$P_1 = \max_{I: 4 \in I} \{P_{1,I}\} = 0.075.$$

The maximum is from $I = \{1, 3, 4\}$ and equals 0.075, as explained earlier.

(ii) Dunnett's method

$$P_1 = \max_{I: 4 \in I} \{P_{1,I}\} = 0.0918.$$

The maximum $P_{1,I}$ comes from $I = \{1, 2, 3, 4\}$.

Results: Bauer and Köhne

Stage 2:

Intersection hypotheses of interest in Stage 2 are those containing $i^* = 4$.

All of these yield the same P-value, $P_2 = P_{2,I} = 0.04$.

(a) Combining P_1 and P_2 by the inverse χ^2 method

(i) With $P_1 = 0.075$ from Simes' test, the combination statistic is

$$-2 \log(0.075 \times 0.04) = 11.62$$

with significance level $Pr\{\chi_4^2 > 11.6\} = 0.0204$.

(ii) With $P_1 = 0.0918$ from Dunnett's test, the combination statistic is

$$-2 \log(0.0918 \times 0.04) = 11.21$$

with significance level $Pr\{\chi_4^2 > 11.21\} = 0.0243$.

Thus, both tests find treatment 4 effective at overall level $\alpha = 0.025$.

Results: Bauer and Köhne

Stage 2:

Intersection hypotheses of interest in Stage 2 yield $P_2 = P_{2,I} = 0.04$.

(b) Combining P_1 and P_2 by the weighted inverse normal method

Weights for each stage are proportional to the square root of sample size.

(i) With $P_1 = 0.075$ from Simes' test, the combination statistic is

$$\sqrt{100/600} \Phi^{-1}(1 - 0.075) + \sqrt{500/600} \Phi^{-1}(1 - 0.04) = 2.186$$

with significance level $1 - \Phi(2.186) = 0.0144$.

(ii) With $P_1 = 0.0918$ from Dunnett's test, the combination statistic is

$$\sqrt{100/600} \Phi^{-1}(1 - 0.0918) + \sqrt{500/600} \Phi^{-1}(1 - 0.04) = 2.141$$

with significance level $1 - \Phi(2.141) = 0.0161$.

Again, both tests find treatment 4 effective at overall level $\alpha = 0.025$.

Results: TSE

- **Stage 1**

$$Z_{1,i^*} = 1.88, \quad \hat{\theta}_{1,i^*} = \sigma \cdot Z_{1,i^*} / \sqrt{n_1} = 0.940.$$

- **Stage 2**

$$Z_2 = 1.75, \quad \hat{\theta}_{2,i^*} = \sigma \cdot Z_2 / \sqrt{n_2} = 0.391.$$

- **Combining**

The TSE statistic is

$$T_{i^*} = \frac{100 \hat{\theta}_{1,i^*} + 500 \hat{\theta}_{2,i^*}}{600} = \frac{\sigma}{\sqrt{600}} Z = 0.483,$$

where

$$Z = \sqrt{\frac{100}{600}} \cdot Z_{1,i^*} + \sqrt{\frac{500}{600}} \cdot Z_2 = 2.365.$$

Results: TSE

The TSE statistic is $T_{i^*} = 0.483$ or, equivalently, $Z = 2.365$.

The critical value C_2 for T_{i^*} is determined by the requirement

$$Pr\{\hat{\theta}_{1,i^*} > C_1 = 0 \text{ and } T_{i^*} > C_2\} = \alpha = 0.025$$

when $\theta_1 = \dots = \theta_k = 0$.

This gives $C_2 = 0.449$ or an equivalent critical value of 2.20 for Z .

Since $T_{i^*} > 0.449$ and, in keeping with this, $Z > 2.20$, the null hypothesis of no effect for treatment i^* is rejected in the overall test.

Thus, the trial has a positive outcome and a recommendation can be made in support of treatment 4.

21. Efficiency gains from combining Phase IIb and Phase III

Seamless Phase IIb / III designs carry a high organisational cost, so it is important that they should provide substantial benefits.

It is therefore of interest to compare:

Separate Phase IIb and Phase III trials

versus

Combined design with Phase IIb data used at the end of Phase III.

Since the vector of treatment effects is k -dimensional, there are many scenarios to consider.

There are also many options for conducting the final analysis.

Efficiency gains from combining Phase IIb and Phase III

Jennison & Turnbull (*J. Biopharmaceutical Statistics*, 2007) considered situations when the vector of treatment effects is of the form $\theta = (0, 0, 0, \delta)$.

We shall show results for:

The “conventional” method, using only Phase III data in the final hypothesis test, Inverse normal and inverse χ^2 combination tests paired with Simes’ and Dunnett rules for testing intersection hypotheses,

TSE: The method proposed by Thall, Simon & Ellenberg (*Biometrika*, 1988).

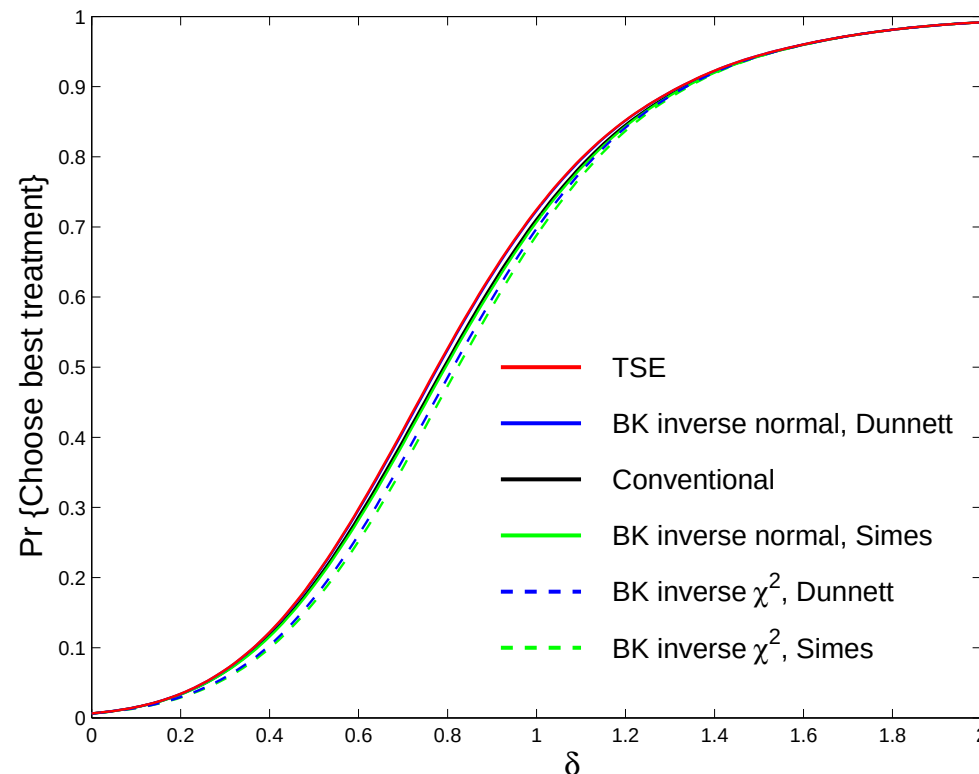
The TSE method compares the pooled estimate of θ_{i^*} with a critical value that takes account of the treatment selection in Phase IIb.

It can be viewed as a special case of the group sequential design of Stallard & Todd (*Statistics in Medicine*, 2003).

The TSE method can also be formulated as a closed testing procedure.

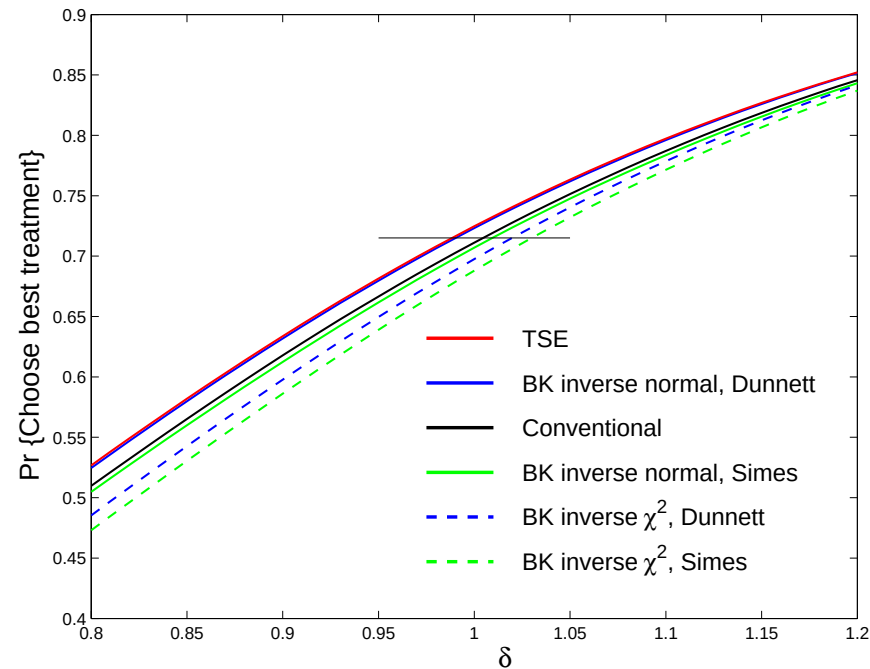
Power functions of the six selection and testing procedures

Power of six 2-stage selection/testing procedures when $\theta = (0, 0, 0, \delta)$,
i.e., three treatments are ineffective and the other has effect size δ :



NB The sample size distribution is the same for all six methods.

Power functions of the six selection and testing procedures



The Thall, Simon & Ellenberg design has slightly higher power than the inverse normal combination test using a Dunnett rule.

The Conventional procedure — with no data combination — is a little worse than the TSE design, and superior to three versions of the BK combination test !!

Differences in power equate to differences in sample size of around 4% to 8%.

Discussion of six power functions

- The variations in performance between procedures are surprising.
- While one BK method improves on the Conventional design, the others fail to gain any advantage from using Phase IIb data in the final hypothesis test.
Given the practical difficulties involved in a seamless Phase IIb / III design, a positive benefit is crucial.
- The TSE method and the BK design with weighted inverse normal combination test and Dunnett's test of intersection hypotheses perform well for mean vectors of the form $\theta = (0, 0, 0, \delta)$.

Are these methods also the best choices for other configurations of treatment means?

Is it possible to quantify the value of including Phase IIb data in the final analysis, e.g., in terms of an equivalent number of Phase III observations?

Assessing the benefits of seamless designs

Reference: Hampson & Jennison, *Statistics in Medicine*, 2015.

Hampson & Jennison create a decision theoretic formulation of the testing problem at the end of a Phase IIb / Phase III trial and search for optimal decision rules in a variety of scenarios.

They conclude that good all round performance is achieved by

The TSE method,

An inverse normal combination test paired with Dunnett's test for intersection hypotheses.

Using Simes' rule, rather than the Dunnett test, can lead to inefficiency when some treatments have low effect sizes.

Conclusions about the benefits of seamless designs

The value of Phase IIb data: A rule of thumb

Hampson & Jennison compare the power obtained by the seamless designs with that of a “conventional” design with a higher Phase III sample size.

Let m_1 be the number of subjects treated at each dose in Phase IIb.

To attain the power of a seamless design, a conventional design needs γm_1 more Phase III observations on each of the selected treatment arm and the control arm, where γ varies between 0.2 and 1.0, depending on the vector of treatment effects.

In the most plausible scenarios, the Phase IIb data on the selected treatment and control provide about half their face value.

In some cases, this may be seen as too small an advantage to justify the extra effort in running a seamless Phase IIb / Phase III trial.

In other cases, e.g., a rare disease where patients are very difficult to find, data combination may still be an attractive proposition.

Further topics for Phase IIb / Phase III trials

Other variations on the preceding methods are possible

More than one treatment may be carried forward to Phase III.

Sequential monitoring could result in early elimination of inferior treatments in Phase IIb, or an early Phase III decision, either positive or negative.

Phase IIb could use adaptive allocation of patients to doses.

Dose response modelling

In Phase IIb dose selection, it is only natural to model the response curve.

It is not straightforward to combine such modelling with hypothesis tests at individual doses — but see the “MCP MOD” methodology of Bretz, Pinheiro & Branson (*Biometrics*, 2005)

A DIA working group has been looking into the joint planning of Phase IIb and Phase III trials from an over-arching perspective; see Antonijevic et al., Ch. 6 of *Optimization of Pharmaceutical R&D Programs and Portfolios* (2015).

22. A phase III trial with treatment selection and a survival endpoint

Consider a trial of cancer treatments comparing

Experimental Treatment 1: Intensive dosing,

Experimental Treatment 2: Slower dosing,

Control treatment

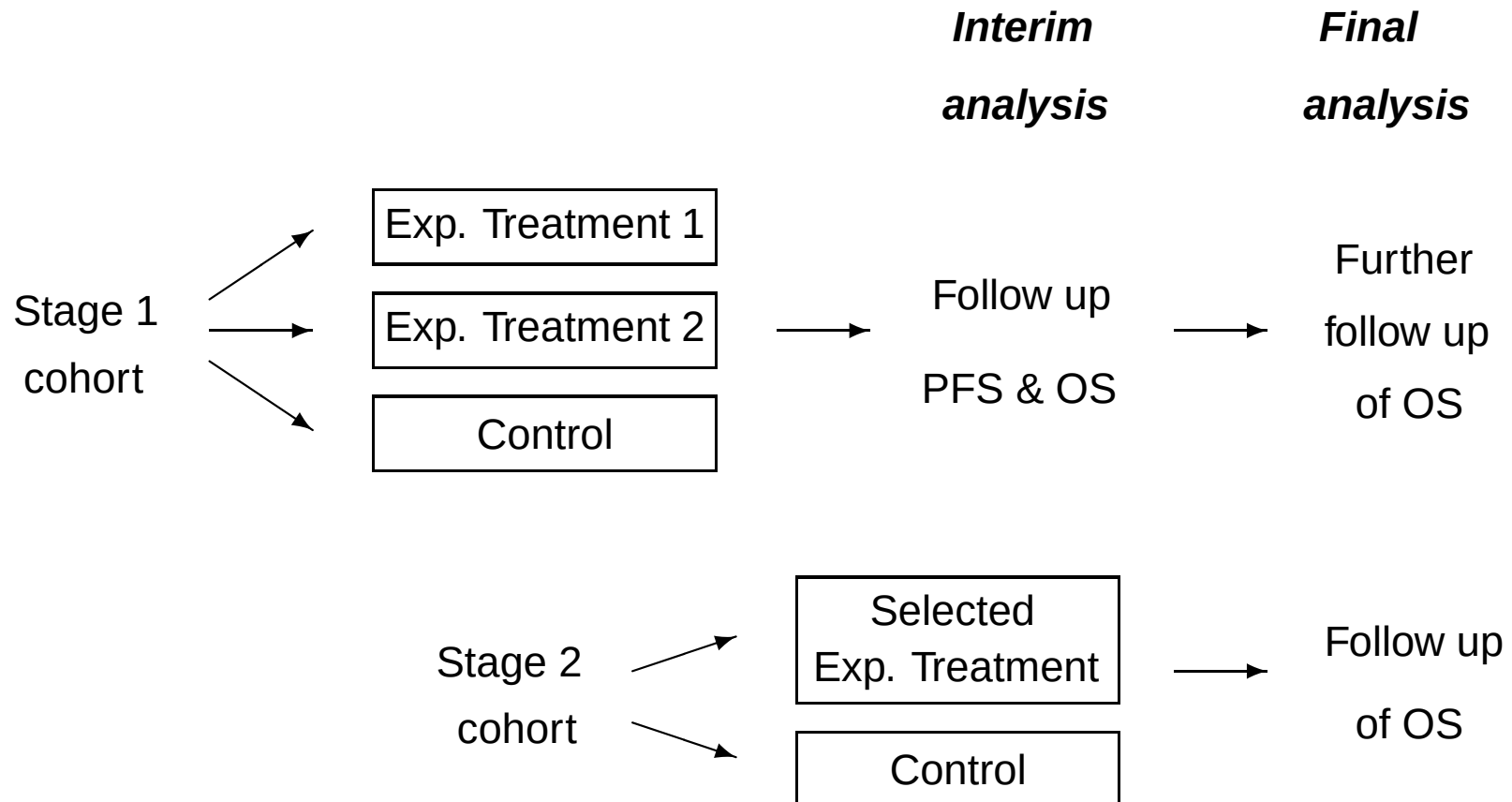
The primary endpoint is Overall Survival (OS).

Information on OS, Progression Free Survival (PFS) and safety will be used at an interim analysis to choose between the two experimental treatments.

Note that PFS is useful here as it is more rapidly observed.

After the interim analysis, patients will only be recruited to the selected treatment and the control.

Overall plan of the trial



At the final analysis, we test the null hypothesis that OS on the selected treatment is no better than OS on the control.

Points to consider

1. Protecting the type I error rate in an adaptive design

A closed testing procedure

Combination tests

2. Properties of log-rank statistics

3. Applying a combination test to survival data

4. Analysing an adaptive trial

Method 1 — and why it may inflate the type I error rate

Method 2 (Jenkins, Stone & Jennison, *Pharmaceutical Statistics*, 2011)

5. Properties of the proposed adaptive design

23. A closed testing procedure with combination tests for survival data

We shall assume a proportional hazards model with

λ_1 = Hazard ratio, Control vs Exp. Treatment 1

λ_2 = Hazard ratio, Control vs Exp. Treatment 2

$$\theta_1 = \log(\lambda_1), \quad \theta_2 = \log(\lambda_2).$$

We test null hypotheses

$H_{0,1}: \theta_1 \leq 0$ vs $\theta_1 > 0$ (*Exp. Treatment 1 superior to control*),

$H_{0,2}: \theta_2 \leq 0$ vs $\theta_2 > 0$ (*Exp. Treatment 2 superior to control*).

In order to control the “familywise error rate”, we require

$$Pr\{\text{Reject any true null hypothesis}\} \leq \alpha.$$

A closed testing procedure

Define level α tests of

$$H_{0,1}: \theta_1 \leq 0,$$

$$H_{0,2}: \theta_2 \leq 0$$

and of the intersection hypothesis

$$H_{0,12} = H_{0,1} \cap H_{0,2}: \theta_1 \leq 0 \text{ and } \theta_2 \leq 0.$$

Then:

*Reject $H_{0,1}$ **overall** if the above tests reject $H_{0,1}$ and $H_{0,12}$,*

*Reject $H_{0,2}$ **overall** if the above tests reject $H_{0,2}$ and $H_{0,12}$.*

The requirement to reject $H_{0,12}$ compensates for testing multiple hypotheses and the “selection bias” in choosing the treatment to focus on in Stage 2.

Combining data across stages

Consider testing a generic null hypothesis $H_0: \theta \leq 0$ against $\theta > 0$.

With data gathered in two stages, suppose Stage 1 data produce Z_1 where

$$Z_1 \sim N(0, 1) \text{ (or stochastically smaller) under } H_0.$$

After possible adaptations, Stage 2 data produce Z_2 with *conditional* distribution

$$Z_2 \sim N(0, 1) \text{ (or stochastically smaller) under } H_0.$$

Weighted inverse normal combination test

With pre-specified weights w_1 and w_2 satisfying $w_1^2 + w_2^2 = 1$,

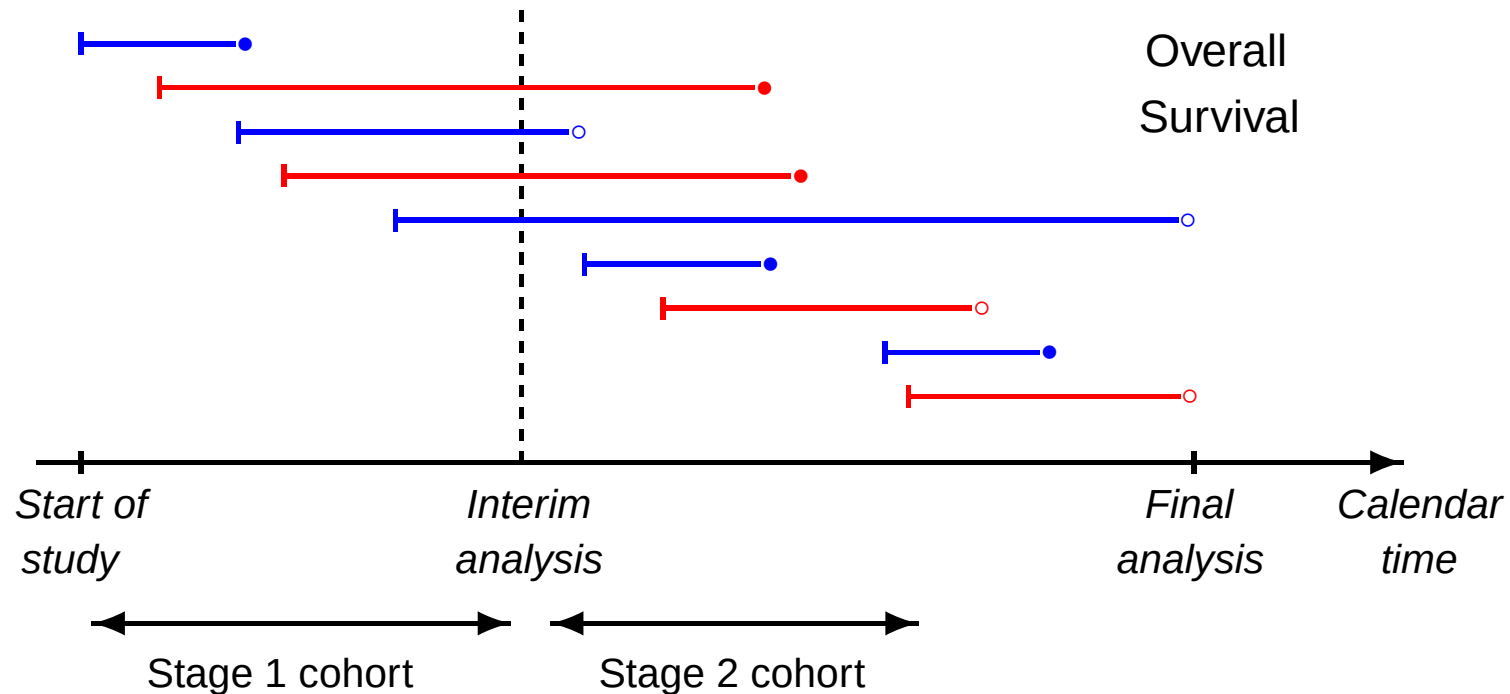
$$Z = w_1 Z_1 + w_2 Z_2 \sim N(0, 1) \text{ (or stochastically smaller) under } H_0.$$

So, for a level α test, we reject H_0 if $Z > \Phi^{-1}(1 - \alpha)$.

Or, the test can be defined in terms of $P_1 = 1 - \Phi(Z_1)$ and $P_2 = 1 - \Phi(Z_2)$.

Properties of log-rank tests

For now, consider Experimental Treatment 1 vs Control.



Key:

- Subjects randomised to Exp. Treatment 1
- Subjects randomised to Control
- Death observed
- Censored observation.

Properties of log-rank tests

Comparing Experimental Treatment 1 vs Control, define

S_1 = Unstandardised log-rank statistic an interim analysis,

$\mathcal{I}_1$ = Information for θ_1 at interim analysis $\approx$ (Number of deaths)/4

S_2 = Unstandardised log-rank statistic an final analysis,

$\mathcal{I}_2$ = Information for θ_1 at final analysis $\approx$ (Number of deaths)/4

Here, “Number of deaths” refers to Experimental Treatment 1 and Control arms only.

Then, approximately,

$$S_1 \sim N(\mathcal{I}_1 \theta_1, \mathcal{I}_1),$$

$$S_2 - S_1 \sim N(\{\mathcal{I}_2 - \mathcal{I}_1\} \theta_1, \{\mathcal{I}_2 - \mathcal{I}_1\})$$

and S_1 and $(S_2 - S_1)$ are **independent** — the “independent increments” property (Tsiatis, *Biometrika*, 1981). NB This result holds for staggered entry.

A combination test for survival data

We can create Z statistics

Based on data at the interim analysis:

$$Z_1 = \frac{S_1}{\sqrt{\mathcal{I}_1}},$$

Based on data accrued between the interim and final analyses:

$$Z_2 = \frac{S_2 - S_1}{\sqrt{\mathcal{I}_2 - \mathcal{I}_1}}.$$

If $\theta_1 = 0$, then $Z_1 \sim N(0, 1)$ and $Z_2 \sim N(0, 1)$ are independent.

If $\theta_1 < 0$, Z_1 and Z_2 are stochastically smaller than this.

So we can use $Z = w_1 Z_1 + w_2 Z_2$ in a combination test of $H_{0,1}: \theta_1 \leq 0$.

A combination test for survival data

In the above, it is crucial that

$$Z_2 = \frac{S_2 - S_1}{\sqrt{\mathcal{I}_2 - \mathcal{I}_1}} \sim N(0, 1) \quad \text{under } \theta_1 = 0,$$

regardless of decisions taken at the interim analysis.

For this to be true, the conduct of the second part of the trial should not depend on the prognosis of Stage 1 patients at the interim analysis.

Bauer & Posch (*Statistics in Medicine*, 2004) note that this condition may fail to hold

Suppose, for example,

- PFS at the interim analysis is better for patients on Exp. Treatment 1 than Control, implying better prospects for OS on the Exp. Treatment 1 arm.
- Stage 2 cohort size is reduced and Stage 1 patients are followed up longer.

The change to Stage 2 cohort size increases the contribution of Stage 1 patients to Z_2 , biasing it upwards — we shall need to avoid such potential biases.

Analysing an adaptive survival trial

Recall, we wish to apply a Closed Testing Procedure based on level α tests of

$$H_{0,1}: \theta_1 \leq 0,$$

$$H_{0,2}: \theta_2 \leq 0,$$

$$H_{0,12}: \theta_1 \leq 0 \text{ and } \theta_2 \leq 0.$$

Combination tests for these hypotheses are formed from:

	<i>Stage 1 data</i>	<i>Stage 2 data</i>
$H_{0,1}$	$Z_{1,1}$	$Z_{2,1}$
$H_{0,2}$	$Z_{1,2}$	$Z_{2,2}$
$H_{0,12}$	$Z_{1,12}$	$Z_{2,12}$

The question is how we should define $Z_{1,1}$, $Z_{2,1}$, etc.

Analysing an adaptive survival trial: Method 1

A natural choice is to:

Base $Z_{1,1}$, $Z_{1,2}$ and $Z_{1,12}$ on data available at the interim analysis,

Base $Z_{2,1}$, $Z_{2,2}$ and $Z_{2,12}$ on the additional information accruing between interim and final analyses.

We shall take $Z_{1,1}$ and $Z_{1,2}$ to be standardised log-rank statistics, and $Z_{2,1}$ and $Z_{2,2}$ to be standardised increments in these logrank statistics between analyses.

For $Z_{1,12}$ we could compare the pooled Exp Tr 1 and Exp Tr 2 patients with the Control group. However, for “consonance”, it is probably better to combine $Z_{1,1}$ and $Z_{1,2}$ through, say, Simes’ test or Dunnett’s test.

If we select Experimental Treatment 1 at the interim analysis, we no longer wish to test $H_{0,2}$ — we do not need $Z_{2,2}$ and we can set $Z_{2,12} = Z_{2,1}$.

Similarly, if we select Experimental Treatment 2, we no longer need $Z_{2,1}$ and we can set $Z_{2,12} = Z_{2,2}$.

Method 1, continued

Stage 1 statistics are calculated at the interim analysis:

$Z_{1,1}$ from log-rank test of Exp Tr 1 vs Control

$Z_{1,2}$ from log-rank test of Exp Tr 2 vs Control

$Z_{1,12}$ from pooled log-rank test, or a Simes or Dunnett test.

If Exp. Treatment 1 is selected at the interim analysis, Stage 2 statistics are

$Z_{2,1}$ from increment in log-rank statistic testing Exp Tr 1 vs Control,
combining Stage 1 and Stage 2 cohorts

$Z_{2,12} = Z_{2,1}.$

If Exp. Treatment 2 is selected, Stage 2 statistics are

$Z_{2,2}$ from increment in log-rank statistic testing Exp Tr 2 vs Control,
combining Stage 1 and Stage 2 cohorts

$Z_{2,12} = Z_{2,2}.$

Method 1: What can go wrong?

The first stage statistics are fine.

Suppose Experimental Treatment 1 is selected at the interim analysis.

Then, $Z_{2,1}$ is the increment in the log-rank statistic testing Exp Tr 1 vs Control, combining Stage 1 and Stage 2 cohorts.

$Z_{2,1}$ The issues raised earlier should be considered — might Stage 2 be modified in the light of interim data in a way that biases $Z_{2,1}$?
Regulators are likely to worry about such possibilities !

$Z_{2,12}$ Setting $Z_{2,12} = Z_{2,1}$ *will* cause bias.

Exp Tr 1 is selected when subjects on this arm have good PFS, so the Exp Tr 1 patients who continue to be followed for OS in Stage 2 are liable to have good prognoses.

This method is almost certain to inflate the overall type I error rate !!

Method 2: Jenkins, Stone & Jennison (2011)

In constructing a combination test, Method 1 separates data into the parts accrued before and after the interim analysis:

	Z_1	Z_2
Stage 1 cohort	Overall survival (during Stage 1)	Overall survival (during Stage 2)
Stage 2 cohort		Overall survival (during Stage 2)

Instead, we divide the data into the parts arising from the two cohorts:

Stage 1 cohort	Overall survival (during Stage 1)	Overall survival (during Stage 2)	Z_1
Stage 2 cohort		Overall survival (during Stage 2)	Z_2

Method 2

All patients in the Stage 1 cohort are followed for overall survival up to a fixed time OR up to a fixed number of observed events.

The “Stage 1” statistics are based on the final OS data for the Stage 1 cohort

$Z_{1,1}$ from log-rank test of Exp Tr 1 vs Control

$Z_{1,2}$ from log-rank test of Exp Tr 2 vs Control

$Z_{1,12}$ from pooled log-rank test, or a Simes or Dunnett test.

The “Stage 2” statistics are based on OS data for the Stage 2 cohort

If Exp. Treatment 1 is selected:

$Z_{2,1}$ from log-rank test of Exp Tr 1 vs Control, $Z_{2,12} = Z_{2,1}$

If Exp. Treatment 2 is selected:

$Z_{2,2}$ from log-rank test of Exp Tr 2 vs Control, $Z_{2,12} = Z_{2,2}$.

Method 2

Notes

Jenkins, Stone & Jennison (2011) introduced “Method 2” in a design where a choice is made between testing for an effect in the full population or a sub-population.

If the length of follow up of the Stage 1 cohort for OS can be influenced by interim information about the likely survival of continuing patients, error rate inflation could result (as noted by Bauer & Posch, 2004).

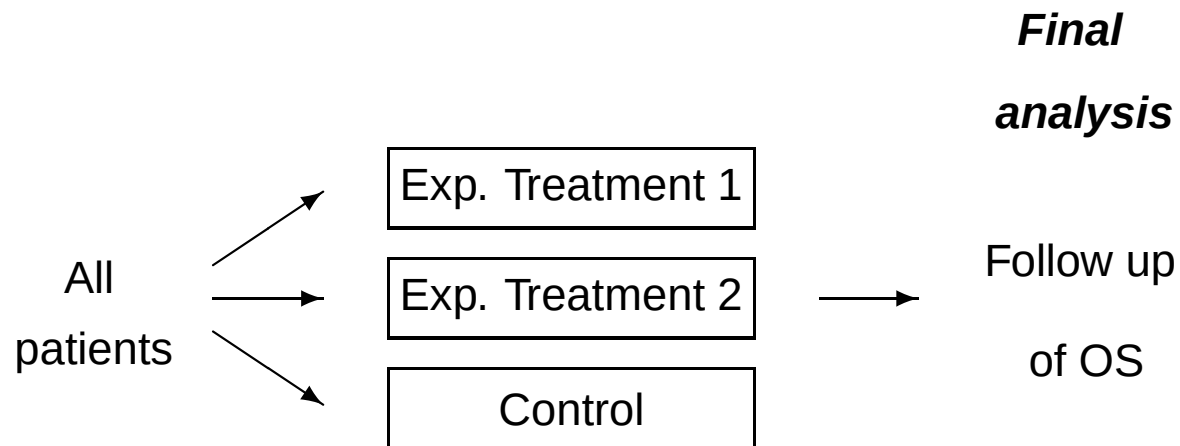
Hence, we stipulate the amount of follow up and require that this is not changed.

Some adaptive designs allow an early decision based on summaries of “Stage 1” data at an interim analysis.

Our statistics $Z_{1,1}$, $Z_{1,2}$ and $Z_{1,12}$ are not known at the time of the interim analysis, so we cannot apply formal stopping rules defined in terms of these — but with a lack of OS data at the interim analysis, that is not a serious limitation.

24. The benefits of an adaptive design with treatment selection

We shall make comparisons with a non-adaptive trial design in which patients are randomised to both experimental treatments and control *throughout* recruitment.



Here, a closed testing procedure will be used to control the familywise error rate.

When the total numbers of patients and lengths of follow-up are the same in adaptive and non-adaptive designs,

Does the adaptive design provide higher power?

Are there other advantages?

Assessing the adaptive design: Model assumptions

Overall Survival

Log hazard ratio

Exp. Treatment 1 vs control θ_1

Exp. Treatment 2 vs control θ_2

Logrank statistics are correlated because of the common control arm.

Progression Free Survival (for treatment selection)

Log hazard ratio

Exp. Treatment 1 vs control ψ_1

Exp. Treatment 2 vs control ψ_2

We suppose correlation between logrank statistics for OS and PFS = ρ .

Proportional hazards models for both endpoints are not essential (or reasonable?)

— the implications for the joint distribution of logrank statistics are what matter.

Assessing the adaptive design: Model assumptions

Log hazard ratios for OS: θ_1, θ_2 .

Log hazard ratios for PFS: ψ_1, ψ_2 .

We suppose

$$\psi_1 = \gamma \times \theta_1 \quad \text{and} \quad \psi_2 = \gamma \times \theta_2$$

Final number of OS events for Stage 1 cohort = 300 (over 3 treatment arms)

Number of OS events for Stage 2 cohort = 300 (over 2 or 3 treatment arms)

Number of PFS events at interim analysis = $\lambda \times 300$.

From large sample theory, the standardised logrank statistic based on d observed events is, approximately,

$$N(\theta\sqrt{d/4}, 1)$$

when the log hazard ratio is θ .

Testing the intersection hypothesis

We have null hypotheses

$$H_{0,1}: \theta_1 \leq 0 \quad \text{and} \quad H_{0,2}: \theta_2 \leq 0.$$

In the closed testing procedure we must also test the intersection hypothesis

$$H_{0,12} = H_{0,1} \cap H_{0,2}: \theta_1 \leq 0 \quad \text{and} \quad \theta_2 \leq 0.$$

We can test $H_{0,12}$ by pooling the Exp. Treatment 1 and Exp. Treatment 2 patients and carrying out a logrank test vs the Control group.

Alternatively we could use a **Simes** test or a **Dunnnett** test.

Suppose P_1 and P_2 are the P-values for logrank tests of Exp. Treatment 1 vs control and Exp. Treatment 2 vs Control.

The corresponding normal deviates are

$$Z_1 = \Phi^{-1}(1 - P_1) \quad \text{and} \quad Z_2 = \Phi^{-1}(1 - P_2).$$

Testing the intersection hypothesis

Simes' test

Given observed values p_1 and p_2 of P_1 and P_2 , Simes' test of $H_{0,12}$ yields the P-value

$$\min (2 \min(p_1, p_2), \max(p_1, p_2)).$$

Simes' test protects type I error conservatively when P_1 and P_2 are independent or positively associated.

Dunnett's test for comparisons with a common control

If z_1 and z_2 are the observed values of Z_1 and Z_2 , the Dunnett test of $H_{0,12}$ yields the P-value

$$P(\max(Z_1, Z_2) \geq \max(z_1, z_2))$$

where (Z_1, Z_2) is bivariate normal with $Z_1 \sim N(0, 1)$, $Z_2 \sim N(0, 1)$ and $\text{Corr}(Z_1, Z_2) = 0.5$.

Comparing tests of the intersection hypothesis

Setting $\psi_1 = \theta_1$, $\psi_2 = \theta_2$ and $\rho = 0.6$, we simulated logrank statistics from their large sample distributions under the adaptive design.

We noted

$$P(1) = P(\text{Select Treatment 1 and Reject } H_{0,1} \text{ overall})$$

$$P(2) = P(\text{Select Treatment 2 and Reject } H_{0,2} \text{ overall})$$

$$E(\text{Gain}) = \theta_1 \times P(1) + \theta_2 \times P(2).$$

Here “Gain” represents a possible utility, in which the value of a positive outcome is proportional to the effect size of the recommended treatment.

In the **adaptive trial design**, we have compared use of

the test based on pooled data, Simes' test, and Dunnett's test

for determining the Stage 1 P-value $P_{1,12}$ and, hence, $Z_{1,12} = \Phi^{-1}(1 - P_{1,12})$.

Comparing tests of the intersection hypothesis

We compare intersection tests to produce $Z_{1,12}$ in an adaptive trial design with

$$\psi_1 = \theta_1, \quad \psi_2 = \theta_2, \quad \lambda = 1, \quad \rho = 0.6, \quad \alpha = 0.025.$$

θ_1	θ_2	$P(1)$			$E(\text{Gain})$		
		Pooled	Simes	Dunnett	Pooled	Simes	Dunnett
0.3	0.0	0.77	0.85	0.86	0.232	0.254	0.259
0.3	0.1	0.78	0.81	0.82	0.238	0.245	0.247
0.3	0.2	0.68	0.68	0.69	0.238	0.237	0.238
0.3	0.25	0.58	0.58	0.58	0.250	0.249	0.249
0.3	0.295	0.48	0.47	0.47	0.275	0.274	0.274

All simulation results are based on 1,000,000 replicates.

The Dunnett test is most effective — it has good power and, unlike the pooled test, is well aligned (consonant) with individual tests of $H_{0,1}$ and $H_{0,2}$.

Comparing adaptive and non-adaptive trial designs

We shall compare designs using a Dunnett test for the intersection hypothesis.

For the non-adaptive design

The intersection hypothesis is tested at the final analysis as part of the closed testing procedure.

As for the adaptive design, we found the Dunnett test to give the best power.

If both $H_{0,1}$ and $H_{0,2}$ are rejected, we suppose the treatment with the higher observed effect size will be “chosen” for registration and marketing.

Accordingly, we define

$$P(1) = P(\hat{\theta}_1 > \hat{\theta}_2 \text{ and } H_{0,1} \text{ is rejected overall})$$

$$P(2) = P(\hat{\theta}_2 > \hat{\theta}_1 \text{ and } H_{0,2} \text{ is rejected overall})$$

$$E(\text{Gain}) = \theta_1 \times P(1) + \theta_2 \times P(2).$$

Comparing adaptive and non-adaptive trial designs

We compare designs using a Dunnett test for the intersection hypothesis, with

$$\psi_1 = \theta_1, \quad \psi_2 = \theta_2, \quad \lambda = 1, \quad \rho = 0.6, \quad \alpha = 0.025.$$

		Non-adaptive			Adaptive		
θ_1	θ_2	$P(1)$	$P(2)$	$E(\text{Gain})$	$P(1)$	$P(2)$	$E(\text{Gain})$
0.3	0.0	0.78	0.00	0.235	0.86	0.00	0.259
0.3	0.1	0.78	0.01	0.234	0.82	0.02	0.247
0.3	0.2	0.70	0.11	0.234	0.69	0.16	0.238
0.3	0.25	0.60	0.26	0.244	0.58	0.30	0.249
0.3	0.295	0.47	0.43	0.267	0.47	0.44	0.274

The adaptive design has higher $P(1)$ when θ_1 is substantially greater than θ_2 .

When θ_1 and θ_2 are closer, the adaptive design still has the higher $E(\text{Gain})$.

Comparing adaptive and non-adaptive trial designs

The adaptive design can only be effective if there is appropriate information to select the correct treatment at the interim analysis.

This requires that

Treatment effects on PFS are reliable indicators of treatment effects on OS,

Sufficient information on PFS is available at the time of the interim analysis.

For the case $\theta_1 = 0.3$, $\theta_2 = 0.1$, we have investigated varying the parameters γ and λ where

$$\psi_1 = \gamma \times \theta_1 \quad \text{and} \quad \psi_2 = \gamma \times \theta_2$$

Final number of OS events for Stage 1 cohort = 300 (over 3 treatment arms)

Number of OS events for Stage 2 cohort = 300 (over 2 or 3 treatment arms)

Number of PFS events at interim analysis = $\lambda \times 300$.

NB It is quite plausible that γ should be greater than 1.

Comparing adaptive and non-adaptive trial designs

We compare designs with $\theta_1 = 0.3$, $\theta_2 = 0.1$, $\rho = 0.6$, $\alpha = 0.025$,

PFS log hazard ratios: $\psi_1 = \gamma \theta_1$, $\psi_2 = \gamma \theta_2$,

Number of PFS events at interim analysis = $\lambda \times 300$.

		Non-adaptive			Adaptive		
γ	λ	$P(1)$	$P(2)$	$E(\text{Gain})$	$P(1)$	$P(2)$	$E(\text{Gain})$
1.5	1.2				0.88	0.00	0.264
1.2	1.0				0.85	0.01	0.256
1.0	1.0	0.78	0.01	0.234	0.82	0.02	0.247
0.9	0.9	for all γ and λ			0.78	0.03	0.238
0.8	0.8	(PFS is not used)			0.74	0.04	0.225
0.7	0.7				0.68	0.05	0.208

Adaptation works well if there is enough PFS information for treatment selection.

Conclusions about the benefits of an adaptive design

1. The adaptive design offers the chance to select the better treatment and focus on this treatment in the second stage of the trial.
2. Overall, the adaptation is beneficial as long as there is sufficient information to make a reliable treatment selection decision.

3. Other evidence may be used in reaching this decision:

Safety data

Pharmacokinetic data

Overall survival

4. In addition to reaching a final decision, the adaptive trial compares the two forms of treatment: the conclusions from this comparison may be useful in other settings.

Related work

1. Irle & Schäfer (*JASA*, 2012) propose similar adaptive designs for survival data.

Changes to the design and critical values for test statistics are made, preserving the conditional probability of rejecting a null hypothesis.

As the “Conditional Probability of Rejection” principle is related to combination tests, the method has much in common with that of Jenkins, Stone & Jennison (2011).

Irle & Schäfer’s method imposes the same requirement of a fixed length of follow-up for “Cohort 1” patients.

Even with this condition in place, determining the conditional probability of a future event is problematic, since the final information level (in a log-rank statistic, say) is not known at the time this probability is calculated.

We recommend our combination test approach as simpler to explain and easier to implement.

Related work

2. Friede et al. (*Statistics in Medicine*, 2011) consider a seamless phase II/III trial design with treatment selection based on short-term and long-term responses.

In a study of treatments for multiple sclerosis, several experimental treatments are compared to a control. When the treatment selection decision is made, only a short-term response is available for some subjects but these will go on to provide a long-term response later.

Although the primary endpoint is not a time-to-event response, similar issues arise. When patients on the selected treatment are followed up, results are likely to be biased towards showing a positive treatment effect, given the short-term response data on which the treatment selection decision was based.

These authors follow a similar approach to Jenkins, Stone & Jennison (2011) and apply a combination test to the long-term response data from the *cohorts* of patients admitted before and after the interim decision point.

Discussion

1. Adaptive designs can enable sample size re-assessment, treatment selection or decisions about the population in which a treatment effect is to be sought.
2. When there are multiple null hypotheses, Closed Testing Procedure can be employed and Combination Tests used to carry out each level α hypothesis test with data from two (or more) stages.

3. For survival data, the “independent increments” property of the log-rank statistic can fail, and other biases can arise, if adaptations are based on data that are informative about the later survival of continuing patients.

This problem can be avoided by defining the elements of a combination test in terms of the complete survival data for separate cohorts of patients.

4. It is most important to assess the efficiency of complex trial designs and determine whether they offer worthwhile improvements over simpler, conventional designs.